

PART 4: THE SELECTION–LEVEL–BRANDOLINI HYBRID ROUTE (RIGOROUS VERSION, DIMENSION-FREE MAIN CHAIN)

1. PURPOSE AND STATUS

This note records the hybrid Selection–Level–Brandolini route in one place, with every step proved. The main chain is *dimension-free*: each building block holds for arbitrary $n \geq 2$, and the route delivers $\mathcal{A}_F(U)^2 \leq C \delta_T(U)$ with the sharp exponent in every dimension. The explicit constants are written out for $n = 2$ (where they are most transparent); the general- n specialisation is catalogued in §2.6. The proof combines:

1. the quantitative selection reduction and its $C^{1,\gamma}$ regularity output (Alt–Caffarelli, no Schauder bootstrap);
2. a level extraction at the $\sqrt{\delta_T}$ boundary-layer scale;
3. a quantitative $D_H \rightarrow L^\infty$ -Serrin upgrade on the stopped level (using γ -Hölder interpolation);
4. inheritance of the selected $C^{1,\gamma}$ regularity by the inset;
5. the Brandolini–Nitsch–Salani–Trombetti Serrin stability theorem in its $C^{1,\gamma}$ form (proved here as Theorem 13.1), giving a concentric annular core;
6. a dimension-free *radial-graph promotion* (Lemma 6.1): the $C^{1,\gamma}$ regularity and the Brandolini inclusion together force $\partial \tilde{E}$ to be a $C^{1,\gamma}$ radial graph from a barycentre, with no pocket-filling needed;
7. a barycentre-recentring step;
8. the direct small-amplitude radial-graph stability theorem.

Step 6 is the load-bearing dimension-free part of the route: in any dimension $n \geq 2$, the chain delivers the sharp exponent $\mathcal{A}_F(U)^2 \lesssim \delta_T(U)$ once a dimension-free radial-graph theorem (G_n) is supplied (the planar case is proved in §12; the general- n dimension-free derivation is cited in §12.7). For $n = 2$ we additionally record, in §7, the elementary polar-slicing “hole-filling” alternative to Step 6: a planar radial-hull pocket lemma that gives an $O(\delta_T)$ -volume radialisation by the sharp planar isoperimetric inequality $|Q| \leq P(Q)^2/(4\pi)$ followed by super-additivity. The planar alternative is cute and self-contained; it does not extend to $n \geq 3$ at the sharp scaling because in higher dimensions the analogous quadratic isoperimetric bound is replaced by a $P(Q)^{n/(n-1)}$ bound, which yields only $v = O(\delta_T^{n/(2(n-1))}) \neq O(\delta_T)$. The radial-graph route of Step 6 sidesteps this obstruction entirely.

The route does *not* use Brasco–De Philippis–Velichkov’s quantitative stability theorem as a black box. It does use, with explicit statements:

- (I1) the planar volume-variable level identity (Plan 2/Plan 3);
- (I2) the $C^{1,\gamma}$ Brandolini–Nitsch–Salani–Trombetti Serrin stability theorem;
- (I3) the direct radial-graph theorem proved in the Plan 3 graph note.

The selected-minimizer regularity package $\mathcal{S}$ and the quantitative selection principle (P) — previously imported as Plan 1 black boxes — are now *proved in full* in §11 below. The only deep ingredient required for that proof is the $C^{1,\gamma}$ Alt–Caffarelli free-boundary regularity for selected quasi-minimisers; everything in (S1)–(S6) beyond (S3) is then derived from Lieberman boundary-Schauder estimates for the torsion equation on $C^{1,\gamma}$ domains and the Hopf lemma, in a quantitative form. *No further Schauder bootstrap to $C^{2,\alpha}$ is required*, because the BNST theorem (B) is itself proved here at the $C^{1,\gamma}$ regularity level (§13).

Everything else is proved below in full. In particular, all asymmetry quantities are kept in their Fraenkel form $\mathcal{A}_F$, so that the final estimate is sharp in the exponent without any auxiliary smooth-asymmetry detour.

2. CONVENTIONS, THE SELECTION CLASS, AND IMPORTED INPUTS

2.1. Torsion convention. Throughout this note, the torsion function u_D of a bounded open set $D \subset \mathbb{R}^2$ is the unique $H_0^1(D)$ solution of

$$(1) \quad -\Delta u_D = 1 \quad \text{in } D, \quad u_D = 0 \quad \text{on } \partial D.$$

Equivalently, $\Delta u_D = -1$ and $u_D \geq 0$ in D . We set

$$T(D) := \int_D u_D \, dx.$$

The (unnormalised) Saint-Venant deficit of D at fixed area is

$$\delta_T(D) := T(B_D) - T(D),$$

where B_D is any disk with $|B_D| = |D|$. Under the convention (1), the explicit torsion of a planar disk is

$$(2) \quad T(B_r) = \frac{\pi r^4}{8}, \quad T(B_A) = \frac{A^2}{8\pi} \quad (A = |B_A|).$$

On the disk B_1 of radius 1 (area π) the gradient satisfies $|\nabla u_{B_1}|(x) = |x|/2$, and in particular $|\nabla u_{B_1}| \equiv 1/2$ on ∂B_1 .

We normalize the comparison ball to area π , so the area- π disk has radius 1. Constants change harmlessly under any other normalisation by an explicit power of the dilation factor.

The *Fraenkel asymmetry* of a bounded measurable set $E \subset \mathbb{R}^2$ with $|E| > 0$ is

$$\mathcal{A}_F(E) := \inf_{z \in \mathbb{R}^2} \frac{|E \Delta B(z)|}{|E|},$$

where $B(z)$ is the disk of area $|E|$ centred at z . This is the asymmetry we control: the corollary in §10 is stated for $\mathcal{A}_F$.

2.2. The selection class. Selected minimisers of the BDV variational problem live in a class with explicit uniform regularity. Fix once and for all the structural constants

$$R_0, K_{1,\gamma}, \rho_0, \Lambda_\gamma, q_-, q_+, r_0, c_0, C_0 > 0, \quad 0 < \gamma < 1.$$

Definition 2.1 (Selection class $\mathcal{S}$). The selection class $\mathcal{S} = \mathcal{S}(R_0, K_{1,\gamma}, \rho_0, \Lambda_\gamma, q_\pm, r_0, c_0, C_0, \gamma)$ is the family of all open sets $U \subset \mathbb{R}^2$ satisfying:

- (S1) U is bounded, connected, and simply connected;
- (S2) $|U| = \pi$ and $U \subset B_{R_0}$;
- (S3) ∂U is a closed $C^{1,\gamma}$ Jordan curve with $C^{1,\gamma}$ atlas pair $(K_{1,\gamma}, \rho_0)$ (i.e. every $x_0 \in \partial U$ admits a chart $\psi : B_{\rho_0}(x_0) \rightarrow \mathbb{R}^2$ with $\|\psi\|_{C^{1,\gamma}}, \|\psi^{-1}\|_{C^{1,\gamma}} \leq K_{1,\gamma}$ straightening ∂U to $\partial \mathbb{R}_+^2$);
- (S4) the torsion function $u = u_U$ satisfies $\|u\|_{C^{1,\gamma}(\overline{U})} \leq M_{1,\gamma}$ and the gradient Hölder bound $\|Du\|_{C^{0,\gamma}(\overline{U})} \leq \Lambda_\gamma$;
- (S5) the Hopf bound $q_- \leq |\nabla u| \leq q_+$ holds on ∂U ;
- (S6) the linear-growth pair holds in the collar:

$$c_0 \operatorname{dist}(x, \partial U) \leq u(x) \leq C_0 \operatorname{dist}(x, \partial U) \quad \text{for } x \in \mathcal{N}_{r_0}(\partial U) \cap U.$$

Here $M_{1,\gamma} = M_{1,\gamma}(R_0, K_{1,\gamma}, \rho_0)$ is the universal constant from boundary Lieberman–Schauder estimates for $-\Delta u = 1$ on $C^{1,\gamma}$ domains, depending only on $(R_0, K_{1,\gamma}, \rho_0)$; we absorb it into the structural constants.

The constants in $\mathcal{S}$ are fixed throughout; every constant generated below is allowed to depend on them. We write

$$C \text{ or } C(\mathcal{S})$$

for a constant depending only on $\mathcal{S}$.

Remark 2.2. *Properties (S1)–(S6) constitute the regularity package that the Alt–Caffarelli step of the BDV selection attaches to a selected minimiser; crucially, only the $C^{1,\gamma}$ free-boundary regularity (R4) of Theorem 11.4 is required to verify (S3), and (S4) follows from boundary Lieberman–Schauder estimates for $-\Delta u = 1$ on $C^{1,\gamma}$ domains. No further $C^{2,\alpha}$ bootstrap is needed, since the BNST input (B) of §2.3 is the $C^{1,\gamma}$ version (proved in §13). We use $\mathcal{S}$ as a black box throughout §§3–8. §9 imports the quantitative selection principle (P) that delivers $U \in \mathcal{S}$ from a small-asymmetry near-counterexample. The two BDV-conditional inputs of this note are thus (i) the $C^{1,\gamma}$ regularity class $\mathcal{S}$ (Definition 2.1), derived in §11 from the Alt–Caffarelli regularity package (R1)–(R4), and (ii) the selection principle (P) of §9.*

2.3. The Brandolini–Nitsch–Salani–Trombetti theorem under $C^{1,\gamma}$ regularity. We use the following $C^{1,\gamma}$ version of the planar Brandolini–Nitsch–Salani–Trombetti theorem [5]. The original BNST result is stated for $C^{2,\alpha}$ domains; we strengthen it to $C^{1,\gamma}$ in §13, where it is proved in full as Theorem 13.1. This strengthening allows the manuscript to bypass any Schauder bootstrap of the boundary: the $C^{1,\gamma}$ regularity required below is delivered directly by the Alt–Caffarelli step (R4) of the BDV regularity package (Theorem 11.4).

(B) Let $\gamma \in (0, 1]$ and let the *atlas pair* $(K_{1,\gamma}, \rho_0)$ consist of the $C^{1,\gamma}$ chart norm $K_{1,\gamma}$ and a uniform chart radius $\rho_0 > 0$. There exist $\eta_B \in (0, 1/4)$ and constants $C_{\text{Br}} > 0$ and $\mu = 1/34$ (the planar value of $\mu = 1/(2(4n+9)(n-1))$ at $n = 2$), depending polynomially on $(K_{1,\gamma}, 1/\rho_0, R_0, \Lambda_\gamma)$ and on $1/\gamma$, such that the following holds. If $D \subset \mathbb{R}^2$ is bounded, connected, with diameter $\leq R_0$ and $C^{1,\gamma}$ atlas pair bounded by $(K_{1,\gamma}, \rho_0)$, and if $w \in C^{1,\gamma}(\overline{D}) \cap C^\infty(D)$ solves

$$\Delta w = 2 \text{ in } D, \quad w = 0 \text{ on } \partial D, \quad \||Dw| - 1\|_{L^\infty(\partial D)} \leq \eta \leq \eta_B,$$

with $\|Dw\|_{C^{0,\gamma}(\overline{D})} \leq \Lambda_\gamma$, then there exists $z \in \mathbb{R}^2$ and $r_{\text{in}}, r_{\text{out}} > 0$ with

$$(3) \quad B_{r_{\text{in}}}(z) \subset D \subset B_{r_{\text{out}}}(z), \quad r_{\text{out}} - r_{\text{in}} \leq C_{\text{Br}} \eta^\mu, \quad |r_{\text{in}} - \rho_D| + |r_{\text{out}} - \rho_D| \leq C_{\text{Br}} \eta^\mu,$$

where $\rho_D := \sqrt{|D|/\pi}$ is the radius of the disk of area $|D|$.

Two observations: (i) No *smallness* of the $C^{1,\gamma}$ atlas norm is required, only finiteness; $K_{1,\gamma}$ and $1/\rho_0$ enter polynomially in C_{Br} , and the only smallness parameter is η . (ii) The exponent $\mu = 1/34$ is unchanged from the $C^{2,\alpha}$ version: the rate comes from the L^1 multi-ball argument of BNST (their Theorem 5; see §13.2 step (B1)), which uses no boundary regularity above Lipschitz. Only the constant C_{Br} is affected by the $C^{2,\alpha} \rightarrow C^{1,\gamma}$ relaxation.

2.4. Imported planar level identity. We use the volume-variable form of the level-set deficit identity (Plan 2/Plan 3, planar specialisation):

(L) For every bounded open $U \subset \mathbb{R}^2$ of finite perimeter, with torsion function $u = u_U$ and inverse distribution function $v(s) := \inf\{t > 0 : |\{u > t\}| \leq s\}$, and with

$$E_t := \{u > t\}, \quad f_t := |\nabla u| \upharpoonright_{\partial E_t}, \quad P_t := \mathcal{H}^1(\partial E_t), \quad A_t := |E_t|,$$

the level deficits

$$(4) \quad D_H(t) := \left(\int_{\partial E_t} f_t d\mathcal{H}^1 \right) \left(\int_{\partial E_t} \frac{1}{f_t} d\mathcal{H}^1 \right) - P_t^2, \quad D_I(t) := P_t^2 - 4\pi A_t,$$

are non-negative and satisfy

$$(5) \quad \int_0^{|U|} (D_H(v(s)) + D_I(v(s))) ds = 8\pi \delta_T(U).$$

The constant 8π is the planar specialisation of the general identity, which has the dimensional factor $n^2 \omega_n^{2/n} s^{1-2/n}$; in $n = 2$ this equals $4\pi \cdot 1$ and the equality with the deficit is obtained after multiplication by 2 and by $|U|$ (cf. Plan 2, agent1).

2.5. Imported radial-graph theorem. We state the radial-graph theorem in dimension-free form (G_n) , valid for every $n \geq 2$; the planar case ($n = 2$) is the form used in the main chain of §8, and is proved in full as Theorem 12.8 in §12. The dimension- n version is proved as Theorem 12.10 in §12.9.

(G_n) Let $n \geq 2$. There exist $\eta_G^{(n)}, c_G^{(n)}, C_G^{(n)} > 0$ such that the following holds. If $\varphi \in C^0(S^{n-1})$ and

$$G = \{z + r\omega : 0 \leq r < 1 + \varphi(\omega), \omega \in S^{n-1}\},$$

satisfies

$$\|\varphi\|_{L^\infty(S^{n-1})} \leq \eta_G^{(n)}, \quad |G| = \omega_n, \quad \int_G (x - z) dx = 0$$

(i.e. the barycentre of G is at the radial centre z), then

$$(6) \quad \mathcal{A}_F(G)^2 \leq C_G^{(n)} \delta_T(G).$$

At $n = 2$ this is the planar (G) used in §8; we then drop the superscript and write η_G, c_G, C_G . Admissible explicit constants are $\eta_G = 1/13, c_G = 15/1024, C_G < 48$ (planar), or $c_G^{(n)} \geq 1/(16n^2)$ (general).

Both the smallness of $\|\varphi\|_{L^\infty}$ and the barycentre condition are needed in the proof of (G_n) , the latter to suppress the P_1 -mode in the spherical-harmonic expansion of φ . We carry out the barycentre recentring in Lemma 5.4 below.

2.6. General- n specialisation. The detailed write-up of §§3–9 is carried out at $n = 2$, where every constant is explicit. The same chain extends verbatim to general $n \geq 2$ via the following substitutions:

- Ambient: $\mathbb{R}^2 \rightarrow \mathbb{R}^n, S^1 \rightarrow S^{n-1}, \mathcal{H}^1 \rightarrow \mathcal{H}^{n-1}$, Jordan curve $\rightarrow$ closed embedded $C^{1,\gamma}$ hypersurface.
- Normalisations: $|U| = \pi \rightarrow |U| = \omega_n$ (volume of the unit ball); area of optimal disk $\pi \rightarrow \omega_n$; torsion of the unit ball $T(B_1) = \pi/8 \rightarrow T(B_1) = \omega_n/(n(n+2))$; gradient of the torsion function on ∂B_1 is $1/n$ (so $|Dw| = 1$ on ∂B_1 corresponds to $w = -2u_0$ with $u_0 = (1 - |x|^2)/(2n)$).
- Level identity (5): the constant $8\pi = 4\pi \cdot 2$ becomes $C_{\text{lev}}(n) := n^2 \omega_n^{2/n} (|U|^{2-2/n} n / (2(n-1)))$ — concretely, the integral identity $\int_0^{|U|} (D_H(v) + D_I(v)) ds = C_{\text{lev}}(n) \delta_T(U)$ with $C_{\text{lev}}(n)$ as above and $D_I(t) = P_t^2 - n^2 \omega_n^{2/n} A_t^{2-2/n}$ (the n -D isoperimetric deficit); cf. Plan 2/Plan 3. At $n = 2$, $C_{\text{lev}}(2) = 8\pi$ and $n^2 \omega_n^{2/n} = 4\pi$ recover the planar form.
- BNST exponent: $\mu = 1/34 \rightarrow \mu = 1/(2(4n+9)(n-1))$ (see Theorem 13.1 for the $C^{1,\gamma}$ form, which is stated for $\mathbb{R}^2$ but whose proof — §13 — uses only Lipschitz boundary in steps (B1) and (B3), and the Hölder-modulus replacement of $D^2 w$ in (B2) is dimension-free).
- $D_H \rightarrow L^\infty$ Serrin interpolation (Lemma 4.3): exponent $\gamma/(1+2\gamma) \rightarrow \gamma/(2\gamma+n-1)$ (the denominator counts the surface dimension, with $n-1$ in place of 1), with constant depending on the lower density of ∂E_t and the chart radius (both controlled by $\mathcal{S}$).
- Radial-graph promotion (Lemma 6.1): already stated and proved for arbitrary $n \geq 2$.
- Radial-graph theorem: planar (G) (Theorem 12.8) is replaced by (G_n) (Theorem 12.10 of §12.9), with $c_G^{(n)} = 1/(16n^2)$ and $C_G^{(n)} = 8n^2(C_n/\omega_n)^2$.
- Hole-filling (§7): *not* available in general n , since the planar isoperimetric bound $|Q| \leq P(Q)^2/(4\pi)$ is replaced by $|Q| \leq C_n P(Q)^{n/(n-1)}$, which yields only $v = O(\delta^{n/(2(n-1))})$, strictly weaker than $O(\delta)$ for $n \geq 3$. The radial-graph route of §6 is the dimension-free replacement.

With these substitutions, the selected-domain bound $\mathcal{A}_F(U)^2 \leq C_\star(\mathcal{S}, n) \delta_T(U)$ of Theorem 8.1 holds for every $n \geq 2$ with the same sharp exponent. In the rest of this article we write all explicit constants for $n = 2$ (where they are most transparent); the reader substitutes the entries above for general n .

Remark 2.3. *The regularity class $\mathcal{S}$ (Definition 2.1) and the quantitative selection principle (P) of §9 are proved in §11 (see in particular Theorem 11.11 for (S) and Theorem 11.12 for (P)); the rest of §§3–10 uses them as already established hypotheses. The only “deep” input required for the proof of (S) is $C^{1,\gamma}$ Alt–Caffarelli free-boundary regularity of a selected quasi-minimiser (Brasco–De Philippis–Velichkov 2015, Theorem 4.18), invoked as a single black box and recorded as (R4) in Theorem 11.4. (S1), (S2), (S4), (S5), (S6) are then derived from (S3) = (R4) by boundary Lieberman–Schauder estimates for the torsion equation on $C^{1,\gamma}$ domains and the Hopf lemma, in a quantitative form, with no smallness hypothesis on the $C^{1,\gamma}$ atlas norm.*

3. LEVEL-SET IDENTITIES AND THE STOPPED LEVEL

Lemma 3.1 (Good level in a prescribed layer). *Let $U \in \mathcal{S}$, set $\delta := \delta_T(U)$, and fix $0 < \lambda \leq |U|/2 = \pi/2$. There is a regular value $\hat{t} > 0$ of u such that*

$$(7) \quad \frac{\lambda}{2} \leq |U \setminus E_{\hat{t}}| \leq \lambda,$$

and

$$(8) \quad D_H(\hat{t}) + D_I(\hat{t}) \leq \frac{16\pi}{\lambda} \delta.$$

Proof. In the volume variable $s = |E_t|$, the layer $|U| - \lambda \leq s \leq |U| - \lambda/2$ has length $\lambda/2$. By (5) and non-negativity of D_H, D_I ,

$$\int_{|U|-\lambda}^{|U|-\lambda/2} (D_H(v(s)) + D_I(v(s))) ds \leq 8\pi\delta,$$

so for some s in this layer $D_H(v(s)) + D_I(v(s)) \leq (8\pi\delta)/(\lambda/2) = 16\pi\delta/\lambda$. Sard’s theorem applied to u gives that regular values of u are full-measure in the volume variable (the singular set of values is mapped under $t \mapsto |E_t|$ to a Lebesgue-null subset of $[0, |U|]$). We may therefore choose s so that $\hat{t} := v(s)$ is a regular value, which gives (7) and (8). $\square$

Lemma 3.2 (Boundary-layer height). *Let $U \in \mathcal{S}$. There are constants $c_h, C_h > 0$ depending on $\mathcal{S}$ such that, for every regular value t with $0 < t < r_0 c_0/2$,*

$$(9) \quad c_h t \leq |U \setminus E_t| \leq C_h t.$$

Consequently, when $\hat{t}$ is the level from Lemma 3.1 with $\lambda < \lambda_0 := \min(\pi/2, r_0 c_0 C_h/2)$,

$$(10) \quad \frac{\lambda}{2C_h} \leq \hat{t} \leq \frac{\lambda}{c_h}.$$

Proof. By (S6), in the collar of width r_0 ,

$$\{\text{dist}(\cdot, \partial U) \leq t/C_0\} \cap U \subset \{0 < u \leq t\} \subset \{\text{dist}(\cdot, \partial U) \leq t/c_0\} \cap U.$$

The chosen range $t < r_0 c_0/2$ makes $t/c_0 < r_0/2$, so both inclusions lie strictly inside the collar. Since $\partial U \in C^{1,\gamma}$ with atlas pair $(K_{1,\gamma}, \rho_0)$ (S3), the normal vector $\nu_U : \partial U \rightarrow S^1$ is γ -Hölder. By the standard tubular-neighbourhood formula for $C^{1,\gamma}$ hypersurfaces (cf. [7, Theorem 3.1]), normal coordinates $\Phi : (y, s) \mapsto y - s\nu_U(y)$ define a bi-Lipschitz parametrisation of the collar $\mathcal{N}_{r'_0}(\partial U) \cap U$ for some $r'_0 = r'_0(K_{1,\gamma}, \rho_0) > 0$, with Jacobian satisfying $|J(y, s) - 1| \leq c(K_{1,\gamma}) s^\gamma$ (a γ -Hölder modulus of ν_U). In particular

$$1 - c(K_{1,\gamma}) r^\gamma \leq J(y, s) \leq 1 + c(K_{1,\gamma}) r^\gamma \quad \text{for } s \in [0, r],$$

giving

$$|\{\text{dist}(\cdot, \partial U) \leq r\} \cap U| = \int_{\partial U} \int_0^r J(y, s) ds d\mathcal{H}^1(y) = r \mathcal{H}^1(\partial U) + O(r^{1+\gamma}),$$

with constants in $O(\cdot)$ controlled by $K_{1,\gamma}$. Since $U \in \mathcal{S}$ is bounded, $|U| = \pi$ and the boundary has uniform $C^{1,\gamma}$ atlas, the perimeter satisfies $P(U) \in [P_-, P_+]$ with $P_\pm$ depending only on $\mathcal{S}$ (lower bound from the planar isoperimetric inequality $P \geq 2\pi$; upper bound from the $C^{1,\gamma}$ chart covering of $\partial U \subset B_{R_0}$). Substituting $r = t/c_0$ and $r = t/C_0$ yields (9) with $c_h = P_-/(2C_0)$ and $C_h = 2P_+/c_0$ (the higher-order term $O(r^{1+\gamma}) = O(t^{1+\gamma})$ being absorbed into the leading r -linear term for $t \leq t_*(K_{1,\gamma})$ small). The statement (10) is immediate from (7). $\square$

Lemma 3.3 (Monotone deficit transfer to the stopped level). *Let $U \in \mathcal{S}$, set $\delta := \delta_T(U)$, and let $E := E_{\hat{t}}$ be a regular level with $\lambda := |U \setminus E| \leq \pi/2$. Then*

$$(11) \quad \delta_T(E) \leq \delta,$$

where $\delta_T(E) = T(B_{|E|}) - T(E)$ is computed without area renormalisation. After rescaling E by $s := \sqrt{\pi/|E|}$ to obtain $\tilde{E} := sE$ with $|\tilde{E}| = \pi$,

$$(12) \quad \delta_T(\tilde{E}) = s^4 \delta_T(E) \leq \left(\frac{\pi}{\pi - \lambda}\right)^2 \delta \leq 4\delta.$$

Proof. Set $\hat{t} = v(|E|)$ so $E = \{u > \hat{t}\}$ and the function $u_E := u - \hat{t}$ is the torsion function of E ($-\Delta u_E = 1$ in E , $u_E = 0$ on ∂E). For $\tau > 0$ and $x \in E$, $u_E(x) > \tau \iff u(x) > \hat{t} + \tau$, so

$$\{u_E > \tau\} = \{u > \hat{t} + \tau\} \quad (\tau > 0),$$

and on the level $\{u_E = \tau\}$ we have $|\nabla u_E| = |\nabla u|$ and the same perimeter and area. Consequently, with the obvious notation $D_H^E(\tau) = D_H^U(\hat{t} + \tau)$ and $D_I^E(\tau) = D_I^U(\hat{t} + \tau)$.

Now apply the identity (5) to E :

$$8\pi \delta_T(E) = \int_0^{|E|} (D_H^E(v_E(\sigma)) + D_I^E(v_E(\sigma))) d\sigma.$$

In the volume variable $\sigma = |\{u_E > \tau\}| = |\{u > \hat{t} + \tau\}|$, the substitution $\sigma \mapsto s = \sigma$ (over the range $[0, |E|]$, with the same area variable) makes the integrand on the right equal to $D_H^U(v(s)) + D_I^U(v(s))$ for $s \in [0, |E|]$. Hence

$$8\pi \delta_T(E) = \int_0^{|E|} (D_H^U(v(s)) + D_I^U(v(s))) ds \leq \int_0^{|U|} (D_H^U(v(s)) + D_I^U(v(s))) ds = 8\pi \delta,$$

using non-negativity of the integrand on $(|E|, |U|)$. This is (11).

For (12), recall that torsion satisfies $T(sE) = s^4 T(E)$ (because $u_{sE}(y) = s^2 u_E(y/s)$ solves $-\Delta = 1$ in sE), and $B_{|\tilde{E}|}$ is the scaled $B_{|E|}$, so $\delta_T(\tilde{E}) = s^4 \delta_T(E)$. With $s^2 = \pi/(\pi - \lambda)$ and $\lambda \leq \pi/2$, the factor $s^4 \leq 4$. $\square$

4. FROM D_H TO POINTWISE SERRIN

Lemma 4.1 (Exact variance identity). *Let $U \in \mathcal{S}$ and let t be a regular value of u . Set $\bar{f}_t := A_t/P_t = |E_t|/P_t$. Then*

$$(13) \quad \int_{\partial E_t} \frac{(|\nabla u| - \bar{f}_t)^2}{|\nabla u|} d\mathcal{H}^1 = \frac{A_t}{P_t^2} D_H(t).$$

Proof. By the divergence theorem applied to ∇u on E_t ,

$$\int_{\partial E_t} |\nabla u| d\mathcal{H}^1 = - \int_{E_t} \Delta u dx = |E_t| = A_t.$$

Hence, expanding the square and using $\int_{\partial E_t} (\bar{f}_t)^2 / |\nabla u| = \bar{f}_t^2 \int 1/|\nabla u|$,

$$\begin{aligned} \int_{\partial E_t} \frac{(|\nabla u| - \bar{f}_t)^2}{|\nabla u|} &= \int_{\partial E_t} |\nabla u| d\mathcal{H}^1 - 2\bar{f}_t P_t + \bar{f}_t^2 \int_{\partial E_t} \frac{d\mathcal{H}^1}{|\nabla u|} \\ &= A_t - 2\bar{f}_t P_t + \frac{A_t^2}{P_t^2} \int_{\partial E_t} \frac{d\mathcal{H}^1}{|\nabla u|} \\ &= \frac{A_t}{P_t^2} \left[A_t \int_{\partial E_t} \frac{d\mathcal{H}^1}{|\nabla u|} - P_t^2 \right] = \frac{A_t}{P_t^2} D_H(t), \end{aligned}$$

where we used $2\bar{f}_t P_t = 2A_t$ in the third line. $\square$

Lemma 4.2 (L^2 Serrin on the good level). *Let $\hat{t}$ be the level produced by Lemma 3.1 with $\lambda \leq \lambda_0$. Then*

$$(14) \quad \int_{\partial E_{\hat{t}}} (|\nabla u| - \bar{f}_{\hat{t}})^2 d\mathcal{H}^1 \leq C_{L^2} \frac{\delta}{\lambda},$$

with C_{L^2} depending only on $\mathcal{S}$.

Proof. By (10) we have $\hat{t} \leq \lambda/c_h \leq \lambda_0/c_h < r_0/2$, so the good level lies in the selected collar. By (S5) and (S4), $|\nabla u| \leq q_+ + \Lambda_\gamma r_0^\gamma$ in the collar (from the Hölder modulus of Du). By Lemma 4.1 and $A_{\hat{t}} \leq |U| = \pi$,

$$\int_{\partial E_{\hat{t}}} (|\nabla u| - \bar{f}_{\hat{t}})^2 d\mathcal{H}^1 \leq (q_+ + \Lambda_\gamma r_0^\gamma) \int_{\partial E_{\hat{t}}} \frac{(|\nabla u| - \bar{f}_{\hat{t}})^2}{|\nabla u|} d\mathcal{H}^1 \leq \frac{(q_+ + \Lambda_\gamma r_0^\gamma) \pi}{P_-^2} D_H(\hat{t}).$$

Using $D_H(\hat{t}) \leq 16\pi\delta/\lambda$ from (8) finishes the proof with $C_{L^2} = 16\pi^2(q_+ + \Lambda_\gamma r_0^\gamma)/P_-^2$. $\square$

Lemma 4.3 (One-dimensional interpolation on a level curve, Hölder version). *Let $\Sigma \subset \mathbb{R}^2$ be a compact C^1 curve satisfying the uniform lower length density*

$$(15) \quad \mathcal{H}^1(\Sigma \cap B_r(x)) \geq c_\Sigma r \quad (x \in \Sigma, 0 < r \leq r_\Sigma).$$

Let $\gamma \in (0, 1]$. If $g : \Sigma \rightarrow \mathbb{R}$ is γ -Hölder continuous with seminorm $[g]_{C^{0,\gamma}(\Sigma)} \leq L_\gamma$ and $\|g\|_{L^2(\Sigma)}^2 \leq B$, then

$$(16) \quad \|g\|_{L^\infty(\Sigma)} \leq C_{\text{int}} \left[B^{\gamma/(1+2\gamma)} L_\gamma^{1/(1+2\gamma)} + B^{1/2} \right], \quad C_{\text{int}} = C_{\text{int}}(c_\Sigma, r_\Sigma, \gamma) > 0.$$

Proof. Set $A := \|g\|_{L^\infty(\Sigma)}$ and pick $x_0 \in \Sigma$ with $|g(x_0)| \geq A/2$. Assume $A > 0$ (otherwise the bound is trivial).

Case 1: $L_\gamma > 0$ and $r := (A/(4L_\gamma))^{1/\gamma} \leq r_\Sigma$. For $y \in \Sigma \cap B_r(x_0)$, $|g(y)| \geq |g(x_0)| - L_\gamma|y - x_0|^\gamma \geq A/2 - L_\gamma r^\gamma = A/2 - A/4 = A/4$, so

$$B \geq \int_{\Sigma \cap B_r(x_0)} g^2 d\mathcal{H}^1 \geq \frac{A^2}{16} c_\Sigma r = \frac{c_\Sigma A^2}{16} \left(\frac{A}{4L_\gamma} \right)^{1/\gamma} = \frac{c_\Sigma A^{2+1/\gamma}}{16 \cdot 4^{1/\gamma} L_\gamma^{1/\gamma}},$$

giving $A^{2+1/\gamma} \leq (16 \cdot 4^{1/\gamma}/c_\Sigma) B L_\gamma^{1/\gamma}$, i.e., $A \leq C_1(\gamma, c_\Sigma) B^{\gamma/(1+2\gamma)} L_\gamma^{1/(1+2\gamma)}$, with $C_1(\gamma, c_\Sigma) := (16 \cdot 4^{1/\gamma}/c_\Sigma)^{\gamma/(1+2\gamma)}$.

Case 2: $L_\gamma = 0$, or $L_\gamma > 0$ with $(A/(4L_\gamma))^{1/\gamma} > r_\Sigma$. In either case $|g| \geq A/2 - L_\gamma r_\Sigma^\gamma \geq A/4$ on $\Sigma \cap B_{r_\Sigma}(x_0)$ (the case $L_\gamma = 0$ being trivial, the second case using $L_\gamma r_\Sigma^\gamma < A/4$ by hypothesis), so

$$B \geq \frac{A^2}{16} c_\Sigma r_\Sigma, \quad A \leq \left(\frac{16}{c_\Sigma r_\Sigma} \right)^{1/2} B^{1/2}.$$

Taking the maximum of the two cases yields (16) with $C_{\text{int}} = \max(C_1(\gamma, c_\Sigma), (16/(c_\Sigma r_\Sigma))^{1/2})$. For $\gamma = 1$ the exponent $\gamma/(1+2\gamma) = 1/3$ and $1/(1+2\gamma) = 1/3$ recover the Lipschitz case $A \leq C(BL)^{1/3} + CB^{1/2}$. $\square$

Lemma 4.4 (Pointwise Serrin on the good level). *Under the hypotheses of Lemma 4.2, with the inset regularity constants from Lemma 5.1 below,*

$$(17) \quad \||\nabla u| - \bar{f}_{\hat{t}}\|_{L^\infty(\partial E_{\hat{t}})} \leq C_{L^\infty} \left[\left(\frac{\delta}{\lambda} \right)^{\gamma/(1+2\gamma)} + \left(\frac{\delta}{\lambda} \right)^{1/2} \right],$$

with C_{L^∞} depending only on $\mathcal{S}$.

Proof. By (S4), the function $g := |\nabla u| - \bar{f}_{\hat{t}}$ is γ -Hölder on $\partial E_{\hat{t}}$ with seminorm $[g]_{C^{0,\gamma}(\partial E_{\hat{t}})} \leq \Lambda_\gamma$ (since $\bar{f}_{\hat{t}}$ is constant); by Lemma 5.1, $\partial E_{\hat{t}}$ satisfies the lower length density (15) with constants c_Σ, r_Σ depending only on $\mathcal{S}$. Apply Lemma 4.3 with $B = C_{L^2} \delta/\lambda$ from Lemma 4.2 and $L_\gamma = \Lambda_\gamma$. The first term in (16) becomes $C_{\text{int}} \cdot C_{L^2}^{\gamma/(1+2\gamma)} \Lambda_\gamma^{1/(1+2\gamma)} \cdot (\delta/\lambda)^{\gamma/(1+2\gamma)}$; the second is $C_{\text{int}} \cdot C_{L^2}^{1/2} (\delta/\lambda)^{1/2}$. Both fit into (17) with $C_{L^\infty} := C_{\text{int}} \max(C_{L^2}^{\gamma/(1+2\gamma)} \Lambda_\gamma^{1/(1+2\gamma)}, C_{L^2}^{1/2})$. For $\gamma = 1$ the exponent $\gamma/(1+2\gamma) = 1/3$ recovers the Lipschitz case. $\square$

5. INSET REGULARITY, RECENTRING, AND BRANDOLINI

Lemma 5.1 (Inset regularity). *Let $U \in \mathcal{S}$. There are $t_0, \rho_0^*, K_0^*, d_0 > 0$ depending only on $\mathcal{S}$ such that every regular value $t \in (0, t_0)$ yields an inset E_t which is open, connected, simply connected, of diameter $\leq d_0$, and bounded by a closed $C^{1,\gamma}$ Jordan curve with atlas pair (K_0^*, ρ_0^*) .*

Proof. By (S5) and the Hölder continuity of ∇u on the collar (from (S4): $\|Du\|_{C^{0,\gamma}} \leq \Lambda_\gamma$), after shrinking the collar radius once and for all to $r'_0 \in (0, r_0]$ depending only on $\mathcal{S}$, we have $|\nabla u| \geq q_-/2$ throughout $\mathcal{N}_{r'_0}(\partial U) \cap U$.

Introduce the normal coordinate map

$$\Phi(y, s) := y - s \nu_U(y), \quad y \in \partial U, \quad s \in [0, r'_0].$$

By (S3) ($C^{1,\gamma}$ atlas of ∂U), $\nu_U \in C^{0,\gamma}(\partial U)$, so Φ is a $C^{1,\gamma}$ map onto the collar. Its differential at $s = 0$ is the identity-plus-perturbation on $\partial U \times \mathbb{R}$, so for r'_0 small enough, Φ is a $C^{1,\gamma}$ diffeomorphism with uniformly bounded norm and inverse norm. The composition $F(y, s) := u(\Phi(y, s))$ is $C^{1,\gamma}$ with norm bounded by $M_{1,\gamma}$ from (S4) and $\partial_s F(y, 0) = |\nabla u(y)| \geq q_-$. After further shrinking the collar to $r''_0 \leq r'_0$ (depending only on $M_{1,\gamma}, q_-, \gamma$) we have $|\partial_s F| \geq q_-/2$ throughout.

Set $t_0 := \min(q_- r''_0/2, r_0 c_0/2)$. For $0 < t < t_0$, the quantitative implicit function theorem for $C^{1,\gamma}$ maps gives a unique $C^{1,\gamma}$ function $s_t : \partial U \rightarrow (0, r''_0)$ with $\|s_t\|_{C^{1,\gamma}} \leq K_0^*$ and

$$\partial E_t = \{y - s_t(y) \nu_U(y) : y \in \partial U\}.$$

This makes ∂E_t a closed $C^{1,\gamma}$ Jordan curve with atlas pair (K_0^*, ρ_0^*) depending only on $\mathcal{S}$, and makes E_t a Jordan domain (open, connected, simply connected) by the normal-graph structure. The diameter satisfies $\text{diam}(E_t) \leq \text{diam}(U) \leq 2R_0 =: d_0$. $\square$

Lemma 5.2 (Lower length density of ∂E_t). *Under the hypotheses of Lemma 5.1, the curve $\Sigma := \partial E_t$ satisfies (15) with explicit constants $c_\Sigma := 1/(2(1 + K_0^*))$ and $r_\Sigma := \rho_0^*/2$.*

Proof. By Lemma 5.1, ∂E_t admits at every point a $C^{1,\gamma}$ chart of size ρ_0^* with chart norm bounded by K_0^* . Since $C^{1,\gamma} \subset C^{0,1}$ (Lipschitz), the chart is bi-Lipschitz with constant $\leq 1 + K_0^*$. For any disk $B_\rho(x_0)$ centred on Σ with $\rho \leq \rho_0^*/2$, the chart pulls $B_\rho(x_0) \cap \Sigma$ back to a subset of $\partial \mathbb{R}_+^2$ containing an interval of length $\geq \rho/(1 + K_0^*)$; pushing forward (Lipschitz with constant $\leq 1 + K_0^*$ in the opposite direction) gives the curve segment of $\mathcal{H}^1$ -measure $\geq \rho/((1 + K_0^*)^2)$. Hence $\mathcal{H}^1(B_\rho(x_0) \cap \Sigma) \geq \rho/(2(1 + K_0^*))$ for the slightly more conservative constant $c_\Sigma = 1/(2(1 + K_0^*))$. $\square$

Lemma 5.3 (Brandolini annular core for the rescaled inset). *Let $U \in \mathcal{S}$, set $\delta := \delta_T(U)$, and let $K \geq 1$ be a constant to be chosen later. Apply Lemma 3.1 with $\lambda = K\sqrt{\delta}$, assuming $\delta \leq \delta_1(K, \mathcal{S})$ so that $\lambda < \lambda_0$. Let $E := E_t$ and let $\tilde{E} := sE$ with $s = \sqrt{\pi/|E|}$ so $|\tilde{E}| = \pi$. There exists $\delta_2(K, \mathcal{S}) \in (0, \delta_1]$ such that, for $\delta \leq \delta_2$,*

$$(18) \quad \eta(K, \delta) := 2C_{L^\infty} \left(\frac{\sqrt{\delta}}{K} \right)^{\gamma/(1+2\gamma)} + C_h^* \sqrt{\delta} \leq \eta_B,$$

and there exist $z \in \mathbb{R}^2$, $r_{\text{in}}, r_{\text{out}} > 0$ such that

$$(19) \quad B_{r_{\text{in}}}(z) \subset \tilde{E} \subset B_{r_{\text{out}}}(z), \quad r_{\text{out}} - r_{\text{in}} + |r_{\text{in}} - 1| + |r_{\text{out}} - 1| \leq 3C_{\text{Br}} \eta(K, \delta)^\mu.$$

Here $C_h^* = C_h^*(\mathcal{S})$ is a constant depending only on $\mathcal{S}$.

Proof. Step 1: Pointwise Serrin on $\tilde{E}$. By Lemma 4.4 applied with $\lambda = K\sqrt{\delta}$,

$$\| |\nabla u| - \bar{f}_t \|_{L^\infty(\partial E)} \leq C_{L^\infty} \left[\left(\frac{\sqrt{\delta}}{K} \right)^{\gamma/(1+2\gamma)} + \left(\frac{\sqrt{\delta}}{K} \right)^{1/2} \right] \leq 2C_{L^\infty} \left(\frac{\sqrt{\delta}}{K} \right)^{\gamma/(1+2\gamma)}$$

for $\delta \leq \delta_1$ small (the $1/2$ -power being dominated by the $\gamma/(1+2\gamma)$ -power for small $\sqrt{\delta}/K$, since $\gamma/(1+2\gamma) \leq 1/3 < 1/2$).

Now consider $\tilde{u}(y) := s^2 u(y/s)$ on $\tilde{E}$; then $-\Delta \tilde{u} = 1$ in $\tilde{E}$, $\tilde{u} = 0$ on $\partial \tilde{E}$, and $|\nabla \tilde{u}|(y) = s |\nabla u|(y/s)$. Set $\tilde{f} := |\tilde{E}|/P(\tilde{E}) = \pi/(sP(E)) = s\tilde{f}_t$. Then

$$\| |\nabla \tilde{u}| - \tilde{f} \|_{L^\infty(\partial \tilde{E})} = s \| |\nabla u| - \tilde{f}_t \|_{L^\infty(\partial E)} \leq 2s C_{L^\infty} \left(\frac{\sqrt{\delta}}{K} \right)^{\gamma/(1+2\gamma)}.$$

With $s = \sqrt{\pi/|E|} = 1 + O(\sqrt{\delta})$, the factor $s \leq 2$ for δ small.

Step 2: $\tilde{f}$ close to $1/2$. From $D_I(\tilde{t}) \leq 16\pi\delta/\lambda = 16\pi\sqrt{\delta}/K$ we get, by the planar isoperimetric inequality applied to $\tilde{E}$,

$$P(\tilde{E})^2 = s^2 P(E)^2 \leq s^2 (4\pi|E| + 16\pi\sqrt{\delta}/K) = 4\pi^2 + \frac{16\pi^2\sqrt{\delta}}{K|E|} \cdot 1 \leq 4\pi^2 + \frac{32\pi\sqrt{\delta}}{K},$$

so $P(\tilde{E}) \leq 2\pi\sqrt{1 + (8/(\pi K))\sqrt{\delta}} \leq 2\pi + (8/K)\sqrt{\delta}$ (using $\sqrt{1+x} \leq 1+x/2$). Consequently

$$\left| \tilde{f} - \frac{1}{2} \right| = \left| \frac{\pi}{P(\tilde{E})} - \frac{1}{2} \right| = \frac{P(\tilde{E}) - 2\pi}{2P(\tilde{E})} \leq \frac{1}{2 \cdot 2\pi} \cdot (8/K)\sqrt{\delta} = \frac{2}{\pi K} \sqrt{\delta}.$$

Set $C_h^* := 4/\pi + 1$ (the additional 1 absorbs the $s \neq 1$ inflation).

Step 3: Apply (B). Set $w := -2\tilde{u}$, which satisfies $\Delta w = 2$ in $\tilde{E}$ and $w = 0$ on $\partial \tilde{E}$. Then

$$\| |Dw| - 1 \|_{L^\infty(\partial \tilde{E})} = \| 2|\nabla \tilde{u}| - 1 \|_{L^\infty(\partial \tilde{E})} \leq 2 \| |\nabla \tilde{u}| - \tilde{f} \|_{L^\infty} + 2 \left| \tilde{f} - \frac{1}{2} \right| \leq \eta(K, \delta),$$

with $\eta(K, \delta)$ given by (18). For $\delta \leq \delta_2$ small enough we have $\eta(K, \delta) \leq \eta_B$. By Lemma 5.1, $\tilde{E}$ is bounded, connected, with $C^{1,\gamma}$ atlas pair (K_0^*, ρ_0^*) depending only on $\mathcal{S}$ (rescaling by $s \sim 1$ preserves these constants up to a universal factor), and $\|D\tilde{u}\|_{C^{0,\gamma}(\tilde{E})} \leq 2\Lambda_\gamma$ (by the chain-rule scaling $\nabla \tilde{u}(y) = s\nabla u(y/s)$ and $s \leq 2$ for δ small, with Λ_γ from (S4)); and diameter $\leq 2R_0$. Apply the $C^{1,\gamma}$ Brandolini theorem (B) of §2.3 (proved as Theorem 13.1): there exist z and $r_{\text{in}} \leq \rho_{\tilde{E}} = 1 \leq r_{\text{out}}$ with $B_{r_{\text{in}}}(z) \subset \tilde{E} \subset B_{r_{\text{out}}}(z)$ and the three-term bound (19) (the third side coming from BNST's $|r_{\text{in}} - 1| + |r_{\text{out}} - 1| \leq 2C_{\text{Br}}\eta^\mu$). $\square$

Lemma 5.4 (Barycentre recentring). *Let $\tilde{E}$ and z be as in Lemma 5.3, and set $\eta := 3C_{\text{Br}}\eta(K, \delta)^\mu$. Let $y_\star := \frac{1}{\pi} \int_{\tilde{E}} (x - z) dx$ be the barycentre of $\tilde{E} - z$. Then $|y_\star| \leq 6\eta$, and*

$$(20) \quad B_{1-7\eta}(z + y_\star) \subset \tilde{E} \subset B_{1+7\eta}(z + y_\star).$$

Proof. By Lemma 5.3, $B_{1-\eta}(z) \subset \tilde{E} \subset B_{1+\eta}(z)$. Since the disk $B_{1-\eta}(z)$ is symmetric about z , its barycentre contribution to $\int_{\tilde{E}-z} x dx$ vanishes. Using $|\tilde{E} \setminus B_{1-\eta}(z)| \leq |B_{1+\eta}(z) \setminus B_{1-\eta}(z)| = \pi((1+\eta)^2 - (1-\eta)^2) = 4\pi\eta$,

$$|y_\star| = \frac{1}{\pi} \left| \int_{\tilde{E} \setminus B_{1-\eta}(z)} (x - z) dx \right| \leq \frac{1+\eta}{\pi} \cdot |\tilde{E} \setminus B_{1-\eta}(z)| \leq (1+\eta) \cdot 4\eta \leq 6\eta,$$

where the last step uses $\eta \leq \eta_B < 1/2$. Then $z' := z + y_\star$ satisfies $|z' - z| \leq 6\eta$, and any $x \in B_{1-7\eta}(z')$ lies in $B_{1-7\eta+6\eta}(z) = B_{1-\eta}(z) \subset \tilde{E}$; conversely any $x \in \tilde{E} \subset B_{1+\eta}(z)$ lies in $B_{1+\eta+6\eta}(z') = B_{1+7\eta}(z')$. $\square$

6. RADIAL-GRAPH PROMOTION FROM $C^{1,\gamma}$ REGULARITY

The Brandolini core (Lemma 5.3) yields a near-spherical inclusion $B_{1-\eta}(z) \subset \tilde{E} \subset B_{1+\eta}(z)$ for the rescaled inset, but does not by itself say that $\tilde{E}$ is a radial graph from z . The next lemma supplies that final step: combined with the global $C^{1,\gamma}$ atlas of $\partial \tilde{E}$ inherited from inset regularity (Lemma 5.1), the near-spherical inclusion automatically promotes to a $C^{1,\gamma}$ radial-graph parametrisation, provided the Hausdorff gap η is small relative to the atlas constants. In that regime there are no pockets to fill: $\tilde{E}$ is itself a radial graph, and may be handed directly to the small-amplitude radial-graph theorem (G).

The criterion below is purely geometric and dimension-free. We state and prove it for arbitrary $n \geq 2$. In the main chain of §8 it is invoked at $n = 2$; the dimension-free statement is what would

extend the chain to arbitrary n once (G) is replaced by its dimension-free version (Theorem 12.8 carries over verbatim to S^{n-1} , with $c_G = 1/(16n^2)$; cf. the H12 derivation cited in §12.7).

Lemma 6.1 (Radial-graph promotion). *Let $n \geq 2$, $\gamma \in (0, 1]$, $K \geq 1$, $\rho \in (0, 1/2)$, and let $E \subset \mathbb{R}^n$ be a bounded open set with ∂E a closed connected $C^{1,\gamma}$ -embedded hypersurface admitting a uniform atlas of normal-graph charts of size ρ and norm K : around every $p \in \partial E$,*

$$\partial E \cap B_\rho(p) = \{p + \xi + h_p(\xi)\nu(p) : \xi \in T_p\partial E, |\xi| \leq \rho\},$$

with $h_p(0) = 0$, $\nabla h_p(0) = 0$, and $\|h_p\|_{C^{1,\gamma}(B_\rho(0))} \leq K$. There exist $\eta_{\text{RG}} = \eta_{\text{RG}}(K, \rho, \gamma, n) \in (0, \rho/4)$ and $K_{\text{RG}} = K_{\text{RG}}(K, \rho, \gamma, n) > 0$ such that, for any $z \in \mathbb{R}^n$ and $\eta \in (0, \eta_{\text{RG}}]$ satisfying

$$(21) \quad B_{1-\eta}(z) \subset E \subset B_{1+\eta}(z),$$

there exists a unique $\varphi \in C^{1,\gamma}(S^{n-1})$ with

$$(22) \quad \partial E = \{z + (1 + \varphi(\omega))\omega : \omega \in S^{n-1}\}, \quad E = \{z + r\omega : 0 \leq r < 1 + \varphi(\omega), \omega \in S^{n-1}\},$$

and

$$(23) \quad \|\varphi\|_{L^\infty(S^{n-1})} \leq \eta, \quad \|\nabla \omega \varphi\|_{L^\infty(S^{n-1})} \leq \frac{1}{2}, \quad \|\varphi\|_{C^{1,\gamma}(S^{n-1})} \leq K_{\text{RG}}.$$

Proof. Translate so that $z = 0$ and write $\hat{p} := p/|p|$ (defined on ∂E since $|p| \geq 1 - \eta > 1/2$).

Step 1: $C^{0,\gamma}$ -Hölder bound on the outward unit normal ν . In the chart at p , the outward unit normal at $\xi + h_p(\xi)\nu(p)$ is $\nu(\xi) = (\nu(p) - \nabla h_p(\xi))/\sqrt{1 + |\nabla h_p(\xi)|^2}$, where we identify ∇h_p with its lift to $\nu(p)^\perp \subset \mathbb{R}^n$. Since $\nabla h_p(0) = 0$ and $[\nabla h_p]_{C^{0,\gamma}} \leq K$, $|\nabla h_p(\xi)| \leq K|\xi|^\gamma$ on $B_\rho(0)$; differentiating and bounding, $[\nu]_{C^{0,\gamma}(\partial E \cap B_\rho(p))} \leq 2K$ (the factor 2 absorbs the chain-rule contribution of the unit-norm denominator, which has $C^{0,\gamma}$ seminorm $\leq K^2$ on B_ρ). Globally, two points $p_1, p_2 \in \partial E$ at Euclidean distance $\leq \rho/2$ lie in either chart, giving $|\nu(p_1) - \nu(p_2)| \leq 2K|p_1 - p_2|^\gamma$. For p_1, p_2 at larger distance, one chains $\leq 3/\rho$ such hops along ∂E (whose Euclidean diameter is ≤ 3 because $\partial E \subset B_{1+\eta} \subset B_{3/2}$), giving a global seminorm

$$(24) \quad [\nu]_{C^{0,\gamma}(\partial E)} \leq K_\nu, \quad K_\nu := 6K(1 + K/\rho).$$

Step 2: Radial distance and its tangential gradient. Set $\psi(p) := |p| - 1$ for $p \in \partial E$. Then $|\psi| \leq \eta$ on ∂E . The tangential gradient is $\nabla^{\partial E} \psi(p) = \hat{p} - (\hat{p} \cdot \nu(p))\nu(p)$, so

$$(25) \quad |\nabla^{\partial E} \psi(p)|^2 = 1 - (\hat{p} \cdot \nu(p))^2 = \sin^2 \theta(p),$$

where $\theta(p) \in [0, \pi]$ is the angle between $\nu(p)$ and $\hat{p}$.

Step 3: Hölder seminorm of $\nabla^{\partial E} \psi$. The map $p \mapsto \hat{p}$ on the shell $\{1 - \eta \leq |p| \leq 1 + \eta\}$ is smooth with Lipschitz constant $\leq 1/(1 - \eta) \leq 2$ for $\eta \leq 1/2$, so its $C^{0,\gamma}$ seminorm on ∂E is bounded by $2 \cdot \text{diam}(\partial E)^{1-\gamma} \leq 2 \cdot 3^{1-\gamma} \leq 6$. Combining with (24),

$$(26) \quad [\nabla^{\partial E} \psi]_{C^{0,\gamma}(\partial E)} \leq L_\nu, \quad L_\nu := 6 + 5K_\nu = 6 + 30K(1 + K/\rho).$$

Step 4: Hölder interpolation. By the Gagliardo–Nirenberg–Hölder inequality on a $C^{1,\gamma}$ surface of bounded geometry, applied to ψ :

$$(27) \quad \|\nabla^{\partial E} \psi\|_{L^\infty(\partial E)} \leq C_{\text{GN}} \left(\|\psi\|_{L^\infty}^{\gamma/(1+\gamma)} [\nabla^{\partial E} \psi]_{C^{0,\gamma}}^{1/(1+\gamma)} + \|\psi\|_{L^\infty} \right),$$

with $C_{\text{GN}} = C_{\text{GN}}(K, \rho, \gamma, n)$. (Direct proof: at the point p_0 realising $\|\nabla^{\partial E} \psi\|_\infty =: A$, localise to the chart of size ρ at p_0 ; in flat coordinates the inequality reduces to the standard Euclidean $\mathbb{R}^{n-1}$ interpolation, with constants depending on K and ρ .) Substituting Step 2 ($\|\psi\|_\infty \leq \eta$) and (26),

$$\sup_{\partial E} \sin \theta = \|\nabla^{\partial E} \psi\|_\infty \leq C_{\text{GN}} (\eta^{\gamma/(1+\gamma)} L_\nu^{1/(1+\gamma)} + \eta).$$

Step 5: Choice of η_{RG} . Set

$$(28) \quad \eta_{\text{RG}} := \min \left\{ \frac{\rho}{4}, (16 C_{\text{GN}} (L_\nu^{1/(1+\gamma)} + 1))^{-(1+\gamma)/\gamma} \right\}.$$

For $\eta \leq \eta_{\text{RG}}$ the right-hand side of Step 4 is $\leq 1/8$, so

$$(29) \quad \sin \theta(p) \leq 1/8, \quad |\hat{p} \cdot \nu(p)| = |\cos \theta(p)| \geq \sqrt{63}/8 > 15/16 \quad (p \in \partial E).$$

Step 6: $\widehat{p} \cdot \nu(p) > 0$ throughout ∂E . The function $p \mapsto \widehat{p} \cdot \nu(p)$ is continuous on the connected set ∂E and never vanishes by (29), so its sign is constant on ∂E . Pick $p_\star \in \partial E$ with $|p_\star| = \max_{\partial E} |\cdot| =: r_\star \in [1, 1 + \eta]$. Then $p_\star$ is an interior maximum of $|q|^2$ restricted to ∂E , so $\nabla^{\partial E} |q|^2(p_\star) = 0$, i.e., $p_\star = (p_\star \cdot \nu(p_\star))\nu(p_\star)$, and $\nu(p_\star) = \pm \widehat{p}_\star$. Outward orientation (taking $\nu(p_\star)$ to point out of E , where $0 \in E$ is on the inward side) selects the $+$ sign, so $\widehat{p}_\star \cdot \nu(p_\star) = +1$. By constant sign, $\widehat{p} \cdot \nu(p) > 0$ everywhere on ∂E .

Step 7: The radial map F is a $C^{1,\gamma}$ diffeomorphism. Define $F : \partial E \rightarrow S^{n-1}$ by $F(p) := \widehat{p}$. F is $C^{1,\gamma}$ (composition of the inclusion $\partial E \hookrightarrow \mathbb{R}^n \setminus \{0\}$ and the smooth normalisation). The differential at p is

$$dF_p : T_p \partial E \rightarrow T_{\widehat{p}} S^{n-1}, \quad dF_p(v) = |p|^{-1}(v - (v \cdot \widehat{p})\widehat{p}),$$

whose kernel is $\{v \in T_p \partial E : v \parallel \widehat{p}\}$ — trivial precisely when $\widehat{p} \notin T_p \partial E$, i.e., when $\widehat{p} \cdot \nu(p) \neq 0$. By Step 6 this holds at every p , so F is a local $C^{1,\gamma}$ diffeomorphism.

Compactness of ∂E makes F proper. A proper local diffeomorphism between connected boundaryless manifolds of equal dimension is a covering map. For $n \geq 3$, S^{n-1} is simply connected, so the covering is trivial and F is a global $C^{1,\gamma}$ diffeomorphism onto S^{n-1} . For $n = 2$, the degree of F equals the winding number of ∂E around $0 \in B_{1-\eta} \subset E$, which is ± 1 (∂E is a Jordan curve enclosing 0); the orientation $\widehat{p} \cdot \nu(p) > 0$ forces $+1$, so F is again a $C^{1,\gamma}$ diffeomorphism.

Step 8: Construction of φ . Set $\varphi(\omega) := |F^{-1}(\omega)| - 1$ for $\omega \in S^{n-1}$. Then $\varphi \in C^{1,\gamma}(S^{n-1})$ as a composition of $C^{1,\gamma}$ maps; the shell inclusion gives $\|\varphi\|_\infty \leq \eta$. The radial-graph identity (22) follows from $F^{-1}(\omega) = |F^{-1}(\omega)|\widehat{F}^{-1}(\omega) = (1 + \varphi(\omega))\omega$, since $F(F^{-1}(\omega)) = \omega$. A chart-by-chart computation (differentiating $|F^{-1}(\omega)|$ via $d(F^{-1}) = (dF)^{-1}$, bounded by Step 6) yields the $C^{1,\gamma}$ -norm bound $\|\varphi\|_{C^{1,\gamma}(S^{n-1})} \leq K_{\text{RG}}(K, \rho, \gamma, n)$ and, more sharply, the slope bound:

$$|\nabla_\omega \varphi(\omega)| = | |p| \cdot (\widehat{p} \cdot \nu(p))^{-1} \sin \theta(p) | \leq (1 + \eta) \cdot (15/16)^{-1} \cdot 1/8 \leq 1/2,$$

evaluated at $p = F^{-1}(\omega)$, giving (23). $\square$

Remark 6.2. Lemma 6.1 is dimension-free and uses no smallness of the atlas norm K or chart radius ρ : the threshold η_{RG} absorbs both polynomially. In the application of §8 below, $K = K_0^*$ and $\rho = \rho_0^*$ of Lemma 5.1 depend only on $\mathcal{S}$, so η_{RG} is a universal constant $\eta_{\text{RG}}(\mathcal{S}, \gamma)$, independent of δ . Since the Brandolini gap η of (19) is $O(\delta^{\mu\gamma/(2(1+2\gamma))}) \rightarrow 0$ as $\delta \rightarrow 0$, the hypothesis $\eta \leq \eta_{\text{RG}}$ is automatic for $\delta \leq \delta_\star(\mathcal{S}, \gamma)$.

7. HOLE-FILLING: AN ALTERNATIVE ROUTE FOR $n = 2$

This section records an alternative, polar-slicing argument that, in dimension two only, replaces the radial-graph promotion of §6 by a hole-filling step. The route uses no smallness condition on η relative to the atlas norm — only the Brandolini inclusion and the perimeter deficit $P(\widetilde{E}) \leq 2\pi + \varepsilon$. The resulting bound $|G \setminus \widetilde{E}| \leq \varepsilon^2/(4\pi)$ is *sharp* in δ : $O(\delta)$, same scaling as the radial-graph route of §6. This sharp $O(\delta)$ scaling relies on the planar isoperimetric inequality $|Q| \leq P(Q)^2/(4\pi)$ and the super-additivity inequality $\sum a_j^2 \leq (\sum a_j)^2$; the corresponding n -dimensional inequality $|Q| \leq C_n P(Q)^{n/(n-1)}$ gives only $v \leq C\varepsilon^{n/(n-1)} = O(\delta^{n/(2(n-1))})$, strictly weaker than $O(\delta)$ for $n \geq 3$. Hence the polar-slicing route is *planar* in essence; the radial-graph promotion of §6 provides the dimension-free counterpart. §8, Remark 8.2 records the interchange.

7.1. The polar sheet lemma and quadratic pocket bound.

Lemma 7.1 (Polar sheet decomposition). *Let $E \subset \mathbb{R}^2$ be a bounded open set with $C^{1,\gamma}$ boundary, satisfying*

$$(30) \quad B_{r_-}(z_0) \subset E \subset B_{r_+}(z_0), \quad 0 < r_- < r_+ < \infty.$$

Define

$$R(\theta) := \sup\{r > 0 : z_0 + re^{i\theta} \in E\} \quad (\theta \in S^1), \quad G := \{z_0 + re^{i\theta} : 0 \leq r < R(\theta), \theta \in S^1\}.$$

Let $\{Q_j\}_{j \in J}$ denote the connected components of the open set $G \setminus \overline{E}$. Then G is open, of finite perimeter, each Q_j is open and of finite perimeter, and

$$(31) \quad P(G) + \sum_{j \in J} P(Q_j) \leq P(E).$$

Proof. Translate so that $z_0 = 0$. Use the bi-Lipschitz polar map

$$\Phi : S^1 \times (0, r_+) \rightarrow B_{r_+} \setminus \{0\}, \quad \Phi(\theta, r) = re^{i\theta}, \quad \det D\Phi = r.$$

Lengths in the cylinder $S^1 \times (0, r_+)$ with metric $g = d\theta^2 + r^{-2}dr^2$ match Euclidean arclengths under Φ , since the Jacobian factor cancels. For our perimeter computation below it is sufficient that Φ is bi-Lipschitz on $S^1 \times (r_-/2, r_+)$, which forces both Euclidean and cylindrical perimeters to agree up to bi-Lipschitz constants on every subset of that annulus.

Step 1: Polar slice structure. Because ∂E is $C^{1,\gamma}$ and E is open with $B_{r_-} \subset E$, for every direction θ the radial slice $E_\theta := \{r > 0 : re^{i\theta} \in E\}$ is open in $(0, \infty)$ and contains $(0, r_-]$. Sard's theorem applied to the restriction of ∂E to a polar chart shows that, for $\mathcal{H}^1$ -a.e. $\theta \in S^1$, the radial line $\{re^{i\theta} : r > 0\}$ is transverse to ∂E (it crosses ∂E in finitely many points, none of which is a tangent point). Call such θ *regular*.

For regular θ the slice E_θ is a finite union of disjoint open intervals starting at 0:

$$E_\theta = (0, r_1(\theta)) \sqcup (r_2(\theta), r_3(\theta)) \sqcup \cdots \sqcup (r_{2m(\theta)}(\theta), r_{2m(\theta)+1}(\theta)),$$

with crossings $0 < r_1 < r_2 < \cdots < r_{2m+1} \leq r_+$, and $R(\theta) = r_{2m(\theta)+1}(\theta)$ is the outermost crossing. All $r_k(\theta)e^{i\theta} \in \partial E$ for regular θ .

Step 2: Sheets are graphs. For each $k \in \mathbb{N}$, set

$$\Sigma_k := \{r_k(\theta)e^{i\theta} : \theta \text{ regular}, m(\theta) \geq \lceil k/2 \rceil\}.$$

By the implicit function theorem applied to the $C^{1,\gamma}$ defining function of ∂E (which is non-degenerate radially at regular θ), Σ_k is locally a $C^{1,\gamma}$ polar graph $r = r_k(\theta)$, and

$$\mathcal{H}^1(\Sigma_k) = \int_{\{\theta: \Sigma_k \text{ defined}\}} \sqrt{r_k(\theta)^2 + r_k'(\theta)^2} d\theta.$$

The sheets $\{\Sigma_k\}_{k \geq 1}$ are pairwise disjoint, and the projection $\pi_{\text{rad}} : \partial E \rightarrow S^1$, $\pi_{\text{rad}}(re^{i\theta}) = \theta$, is Lipschitz on $\partial E \cap \{r \geq r_-\}$ (with constant $1/r_-$), so by the area formula $\mathcal{H}^1(\pi_{\text{rad}}^{-1}(\text{regular } \theta)) = \mathcal{H}^1(\partial E) - \mathcal{H}^1(\pi_{\text{rad}}^{-1}(\text{singular } \theta)) = \mathcal{H}^1(\partial E)$, because the singular angle set is S^1 -null and the radial projection has finite multiplicity (bounded by $2m_{\max} + 1$). Hence

$$(32) \quad P(E) = \mathcal{H}^1(\partial E) = \sum_{k \geq 1} \mathcal{H}^1(\Sigma_k).$$

Step 3: ∂G is the outermost sheet. By definition $R(\theta) = r_{2m(\theta)+1}(\theta)$, so for each regular θ the boundary ∂G above direction θ is the single point $R(\theta)e^{i\theta} = r_{2m(\theta)+1}(\theta)e^{i\theta}$, which lies in $\Sigma_{2m(\theta)+1}$. Indexing sheets so that the outermost piece always belongs to a single distinguished sheet $\Sigma_{\text{out}} := \bigcup_{\theta} \{R(\theta)e^{i\theta}\}$,

$$P(G) = \mathcal{H}^1(\Sigma_{\text{out}}) \leq \sum_{k: k \text{ is outermost at some } \theta} \mathcal{H}^1(\Sigma_k).$$

Concretely Σ_{out} is obtained from $\{(\theta, R(\theta))\}$ under Φ , and its $\mathcal{H}^1$ -measure is the polar-graph perimeter of G .

Step 4: Pockets are bounded by sheet pairs. Each Q_j is a connected component of $G \setminus \overline{E}$, hence $Q_j \subset G \setminus E$. Since $B_{r_-} \subset E$, Q_j lies in the open annulus $\{r_- < r < r_+\}$ where Φ is bi-Lipschitz, so $\Phi^{-1}(Q_j)$ is a connected open subset of the cylinder. For each regular θ in the angular range of Q_j , the slice $\Phi^{-1}(Q_j) \cap \{\theta\} \times (0, r_+)$ is a single interval $(r_{2i-1}(\theta), r_{2i}(\theta))$ with $i \leq m(\theta)$ (it is non-empty by definition of the angular range; uniqueness of i follows from connectedness of $\Phi^{-1}(Q_j)$), and we set $i = i_j(\theta)$. Because $\Phi^{-1}(Q_j)$ is connected and the pair indices $i_j(\theta)$ are integer-valued and locally constant in θ on the angular range (the radial coordinates r_{2i-1}, r_{2i} vary continuously, so a jump in i would disconnect the slice), $i_j(\theta)$ is in

fact constant on the angular range of Q_j ; call its common value i_j . By Sard's theorem applied to $\theta \mapsto r_{2i_j-1}(\theta) - r_{2i_j}(\theta)$ on the angular range, the endpoint angles where $r_{2i_j-1} = r_{2i_j}$ form a finite set (for the $C^{1,\gamma}$ boundary) at which the two sheet branches meet transversally as $C^{1,\gamma}$ arcs of ∂E . At such an endpoint, the pocket closes off via a single $C^{1,\gamma}$ tangent meeting (no radial wall piece is created, because Q_j is connected and the meeting is internal to ∂E , which is a single Jordan curve). Hence

$$P(Q_j) = \mathcal{H}^1(\partial Q_j) = \mathcal{H}^1(\Sigma_{2i_j-1} \cap \text{angle range of } Q_j) + \mathcal{H}^1(\Sigma_{2i_j} \cap \text{angle range of } Q_j).$$

Distinct pockets have disjoint angular ranges (because each pocket contains an open interior, so different components are separated by ∂E intersecting some interior ray), and the union of all pocket-sheet pieces is contained in the union of all inner sheets (those with $k < 2m(\theta) + 1$):

$$\sum_j P(Q_j) \leq \sum_{k \neq \text{out}} \mathcal{H}^1(\Sigma_k) = P(E) - P(G),$$

using (32) and Step 3. Rearranging gives (31). $\square$

Remark 7.2. *The hypothesis $C^{1,\gamma}$ on ∂E (for any $\gamma > 0$) is used in Step 1 and Step 2 for the transverse polar slicing. In the application below, $E = \tilde{E}$ inherits the $C^{1,\gamma}$ atlas pair (K_0^*, ρ_0^*) from the inset regularity Lemma 5.1, so the hypothesis is satisfied.*

Lemma 7.3 (Quadratic pocket area). *Let $E \subset \mathbb{R}^2$ be bounded, open, with $C^{1,\gamma}$ boundary, $|E| = \pi$, satisfying (30) and*

$$(33) \quad P(E) \leq 2\pi + \varepsilon.$$

Let G be the radial hull of E about z_0 . Then

$$(34) \quad |G \setminus E| \leq \frac{\varepsilon^2}{4\pi}.$$

Proof. Set $v := |G \setminus E|$ and $h := \sum_j P(Q_j)$. By Lemma 7.1,

$$2\pi + \varepsilon \geq P(E) \geq P(G) + h \geq 2\sqrt{\pi(\pi + v)} + h \geq 2\pi + h,$$

where the third inequality is the planar isoperimetric inequality applied to G (which has area $\pi + v$), and the last uses $\sqrt{\pi(\pi + v)} \geq \pi$ for $v \geq 0$. Hence $h \leq \varepsilon$. Apply the isoperimetric inequality to each pocket $|Q_j| \leq P(Q_j)^2/(4\pi)$ and sum, using $\sum a_j^2 \leq (\sum a_j)^2$ for non-negative a_j :

$$v = \sum_j |Q_j| \leq \frac{1}{4\pi} \sum_j P(Q_j)^2 \leq \frac{1}{4\pi} \left(\sum_j P(Q_j) \right)^2 \leq \frac{\varepsilon^2}{4\pi}.$$

$\square$

7.2. Filling, rescaling, and asymmetry transfer.

Lemma 7.4 (Filling and rescaling). *Let $E \subset H \subset B_{R_*}$ be open planar sets with $|E| = \pi$ and $|H| = \pi + v$, $0 \leq v \leq \pi/2$. Let $s := (\pi/|H|)^{1/2}$ and $\tilde{G} := sH$, so $|\tilde{G}| = \pi$. Then*

$$(35) \quad \delta_T(\tilde{G}) \leq 2(\delta_T(E) + v).$$

Proof. Torsional rigidity is monotone under inclusion (replace u_H by $u_H|_E$ in the variational characterisation; we obtain $T(H) \geq T(E)$). Using $T(B_A) = A^2/(8\pi)$,

$$T(B_H) - T(B_E) = \frac{(\pi + v)^2 - \pi^2}{8\pi} = \frac{2\pi v + v^2}{8\pi} \leq \frac{v}{4} + \frac{v^2}{8\pi} \leq \frac{v}{4} + \frac{v}{16} \leq \frac{v}{2}.$$

Therefore

$$\delta_T(H) = T(B_H) - T(H) \leq T(B_H) - T(E) = (T(B_H) - T(B_E)) + \delta_T(E) \leq \delta_T(E) + v/2.$$

Scaling by s multiplies torsion by s^4 and sends B_H to B_π ; so $\delta_T(\tilde{G}) = s^4 \delta_T(H) \leq (\pi/|H|)^2 (\delta_T(E) + v/2) \leq 4(\delta_T(E) + v/2) \leq 2(\delta_T(E) + v)$ (for the chosen range of v). $\square$

Lemma 7.5 (Fraenkel asymmetry under symmetric difference and rescaling). *For bounded measurable $E_1, E_2 \subset \mathbb{R}^2$ with $|E_1| = |E_2| =: V > 0$,*

$$(36) \quad |\mathcal{A}_F(E_1) - \mathcal{A}_F(E_2)| \leq \frac{|E_1 \Delta E_2|}{V}.$$

The Fraenkel asymmetry is dilation-invariant: $\mathcal{A}_F(sE) = \mathcal{A}_F(E)$ for $s > 0$. It is also translation-invariant.

Proof. For any disk B with $|B| = V$, $|E_1 \Delta B| \leq |E_1 \Delta E_2| + |E_2 \Delta B|$, so taking the infimum over B gives $V \mathcal{A}_F(E_1) \leq |E_1 \Delta E_2| + V \mathcal{A}_F(E_2)$, and similarly with the roles exchanged. Dilation and translation invariance follow because Δ commutes with dilation/translation and the disk class is preserved. $\square$

8. THE SELECTED-DOMAIN THEOREM

Theorem 8.1 (Hybrid estimate for selected planar domains). *There exist $C_\star = C_\star(\mathcal{S}) > 0$ and $\delta_\star = \delta_\star(\mathcal{S}) > 0$ such that, for every $U \in \mathcal{S}$ with $\delta := \delta_T(U) \leq \delta_\star$,*

$$(37) \quad \mathcal{A}_F(U)^2 \leq C_\star \delta.$$

Proof. We argue along the dimension-free chain of §6, applying the radial-graph promotion Lemma 6.1 to the rescaled inset $\tilde{E}$ once Brandolini delivers the near-spherical inclusion. The hole-filling alternative of §7 is recorded in Remark 8.2.

Fix $K = 1$ (any other choice of $K \geq 1$ works and yields a K -dependent constant; $K = 1$ is taken for definiteness). Let $\delta \leq \delta_2(1, \mathcal{S})$ be small enough that Lemma 5.3 applies, and so that all subsequent smallness conditions hold; we record the final smallness threshold as $\delta_\star$.

Step 1: Good level and rescaled inset. By Lemma 3.1 with $\lambda = \sqrt{\delta}$, let $E = E_t$ be the resulting good level, satisfying $\sqrt{\delta}/2 \leq |U \setminus E| \leq \sqrt{\delta}$ and $D_H(\dot{t}) + D_I(\dot{t}) \leq 16\pi\sqrt{\delta}$. By Lemma 3.3, $\delta_T(\tilde{E}) \leq 4\delta$, where $\tilde{E} := sE$ with $s = \sqrt{\pi/|E|} \in [1, (1 - \sqrt{\delta}/\pi)^{-1/2}]$.

Step 2: Brandolini core and first barycentre recentring. Set $\eta := 3C_{\text{Br}}\eta(1, \delta)^\mu$ (Lemma 5.3 notation). Then $\tilde{E}$ admits a centre $z \in \mathbb{R}^2$ with $B_{1-\eta}(z) \subset \tilde{E} \subset B_{1+\eta}(z)$. By Lemma 5.4, with $z' := z + y_\star$ and $|y_\star| \leq 6\eta$,

$$(38) \quad B_{1-7\eta}(z') \subset \tilde{E} \subset B_{1+7\eta}(z').$$

Step 3: Radial-graph promotion at z' . By Lemma 5.1, $\partial\tilde{E}$ admits a uniform $C^{1,\gamma}$ atlas of normal-graph charts of size ρ_0^* and norm K_0^* , with both constants depending only on $\mathcal{S}$ (the rescaling by $s \sim 1$ inflating them by a universal factor). Let $\eta_{\text{RG}} := \eta_{\text{RG}}(K_0^*, \rho_0^*, \gamma, 2)$ be the threshold of Lemma 6.1. Since $\eta \rightarrow 0$ as $\delta \rightarrow 0$ (with rate $O(\delta^{\mu\gamma/(2(1+2\gamma))})$); cf. Lemma 5.3, we tighten $\delta_\star$ so that

$$(39) \quad 7\eta \leq \eta_{\text{RG}}.$$

Apply Lemma 6.1 with $E = \tilde{E}$, $z = z'$, and the shell gap 7η : there is a unique $\varphi_1 \in C^{1,\gamma}(S^1)$ with $\partial\tilde{E} = \{z' + (1 + \varphi_1(\omega))\omega : \omega \in S^1\}$ and $\|\varphi_1\|_\infty \leq 7\eta$. In particular $\tilde{E}$ is a radial graph from z' ; there are no pockets.

Step 4: Second barycentre recentring on the radial graph. Let $y'_\star := \pi^{-1} \int_{\tilde{E}} (x - z') dx$ be the barycentre of $\tilde{E} - z'$. By Lemma 5.4 applied to (38) (with η replaced by 7η), $|y'_\star| \leq 6 \cdot 7\eta = 42\eta$, and

$$(40) \quad B_{1-49\eta}(z'') \subset \tilde{E} \subset B_{1+49\eta}(z''), \quad z'' := z' + y'_\star.$$

Tighten $\delta_\star$ further so that

$$(41) \quad 49\eta \leq \eta_{\text{RG}}, \quad 49\eta \leq \eta_G,$$

where η_G is the (G)-threshold of §2.3. Apply Lemma 6.1 a second time with $z = z''$ and shell gap 49η : there is $\varphi_2 \in C^{1,\gamma}(S^1)$ with $\partial\tilde{E} = \{z'' + (1 + \varphi_2(\omega))\omega : \omega \in S^1\}$ and $\|\varphi_2\|_\infty \leq 49\eta$. By construction z'' is the barycentre of $\tilde{E}$: $\int_{\tilde{E}} (x - z'') dx = 0$.

Step 5: Apply (G). The translated $\tilde{E}$ at centre z'' satisfies all hypotheses of (G) of §2.3: $|\tilde{E}| = \pi$, radial centre at the barycentre, $\|\varphi_2\|_\infty \leq \eta_G$ by (41). Together with translation invariance of δ_T ,

$$(42) \quad \mathcal{A}_F(\tilde{E})^2 \leq C_G \delta_T(\tilde{E}) \leq 4C_G \delta.$$

Step 6: Transfer back to U. By Lemma 7.5 (dilation invariance), $\mathcal{A}_F(\tilde{E}) = \mathcal{A}_F(E)$. We compare E and U via a concentric disk pair: let B' be the optimal disk for E (area $|E| = \pi - \lambda$, $\lambda \leq \sqrt{\delta}$) and B its concentric dilation to area π . Then $|U \Delta B| \leq |U \Delta E| + |E \Delta B'| + |B \Delta B'| = |U \setminus E| + |E| \mathcal{A}_F(E) + \lambda \leq 2\sqrt{\delta} + \pi \mathcal{A}_F(E)$, giving

$$(43) \quad \mathcal{A}_F(U) \leq \frac{|U \Delta B|}{\pi} \leq \frac{2\sqrt{\delta}}{\pi} + \mathcal{A}_F(E) \leq \frac{2\sqrt{\delta}}{\pi} + \sqrt{4C_G \delta} = \left(\frac{2}{\pi} + 2\sqrt{C_G}\right)\sqrt{\delta} =: C_2 \sqrt{\delta}.$$

Squaring, $\mathcal{A}_F(U)^2 \leq C_2^2 \delta$, giving (37) with $C_\star := C_2^2 = (2/\pi + 2\sqrt{C_G})^2$. $\square$

Remark 8.2 (Alternative planar route via hole-filling). *For $n = 2$, Steps 3–5 above admit a polar-slicing alternative that uses the planar lemmas of §7 in place of Lemma 6.1, at the cost of slightly larger constants but without any smallness condition on η relative to the atlas norm. Concretely, replace Steps 3–5 by:*

- (1) *Form the radial hull G of $\tilde{E}$ about z' (Lemma 7.1). The same rescaled isoperimetric estimate as in Step 2 of Lemma 5.3 gives $P(\tilde{E}) \leq 2\pi + 4\sqrt{\delta}$. By Lemma 7.3 with $\varepsilon = 4\sqrt{\delta}$, $v := |G \setminus \tilde{E}| \leq 16\delta/(4\pi) = 4\delta/\pi$.*
- (2) *Set $\tilde{G} := s'G$, $s' = (\pi/|G|)^{1/2} \in (0, 1]$; Lemma 7.4 gives $\delta_T(\tilde{G}) \leq 2(\delta_T(\tilde{E}) + v) \leq 12\delta$. The same containment-plus-rescaling computation as in the old Step 4 shows $B_{s'(1-7\eta)}(s'z') \subset \tilde{G} \subset B_{s'(1+7\eta)}(s'z')$; reading off the centred amplitude, $\|\tilde{\varphi}\|_\infty \leq 8\eta$.*
- (3) *Recentre $\tilde{G}$ to its barycentre, with the same calculation as in Lemma 5.4: $|y_\star^{(\tilde{G})}| \leq 48\eta$ and the centred shell becomes $B_{1-56\eta}(z''') \subset \tilde{G} \subset B_{1+56\eta}(z''')$. Tighten $\delta_\star$ to ensure $56\eta \leq \eta_G$; then (G) gives $\mathcal{A}_F(\tilde{G})^2 \leq 12C_G \delta$.*
- (4) *Transfer from $\tilde{G}$ back to $\tilde{E}$ via Lemma 7.5 and the symmetric-difference bound $|\tilde{E} \Delta \tilde{G}| \leq |\tilde{E} \Delta G| + |G \Delta \tilde{G}| \leq 4\delta/\pi + 8\delta/\pi = 12\delta/\pi$, giving $\mathcal{A}_F(\tilde{E}) \leq 2\sqrt{3C_G \delta} + 12\delta/\pi^2$. Then (43) runs unchanged, with the constant C_2 replaced by $C_2^{\text{alt}} := 2/\pi + 2\sqrt{3C_G} + 12/\pi^2$ (slightly larger than C_2).*

The resulting bound is $\mathcal{A}_F(U)^2 \leq C\delta$ with the slightly larger constant $C = (C_2^{\text{alt}})^2$; the sharp $O(\delta)$ scaling is unchanged. This polar-slicing route is purely planar: in dimension $n \geq 3$, Step (1) gives $v \leq C_n \varepsilon^{n/(n-1)} = O(\delta^{n/(2(n-1))})$, strictly weaker than $O(\delta)$, breaking the chain at the sharp rate. The radial-graph route of the main proof (Steps 3–5 via Lemma 6.1) is dimension-free.

9. TRANSFER FROM ARBITRARY PLANAR SETS

9.1. Selection of $\mathcal{A}_F$ -controlled selected sets. We import the following selection principle, which is a direct $\mathcal{A}_F$ -variant of the Plan 1 selection: the deficit-to-asymmetry quotient is preserved up to a fixed multiplicative loss, and the selected set lies in $\mathcal{S}$.

(P) There exist constants $c_{\text{sel}}, C_{\text{sel}} > 0$ and $\alpha_{\text{sel}} > 0$, depending only on the bounded ambient class, such that the following holds. Let $\Omega \subset B_{R_0}$ be a bounded open set with $|\Omega| = \pi$ and $\mathcal{A}_F(\Omega) \leq \alpha_{\text{sel}}$. Then there exists $U \in \mathcal{S}$ with

$$(44) \quad \mathcal{A}_F(U) \geq c_{\text{sel}} \mathcal{A}_F(\Omega), \quad \delta_T(U) \leq C_{\text{sel}} \delta_T(\Omega).$$

This is the form of the selection principle proved (modulo Plan 1 F2/F3) in Plan 1; see Plan 1/quantitative-selection-principle and Final/SelectionPrinciple. The Fraenkel variant is obtained from the smooth-asymmetry version of Plan 1 by appending the Fraenkel preservation step ($\mathcal{A}_F$ is lower semicontinuous and the selection deformation moves sets by $o(\mathcal{A}_F)$ in symmetric difference, giving the c_{sel} -loss).

9.2. Compactness in the large-asymmetry regime.

Lemma 9.1 (Positive deficit gap). *Let $\mathcal{C}$ denote the class of bounded open sets $\Omega \subset B_{R_0}$ with $|\Omega| = \pi$ and uniform interior cone condition with parameters (θ_*, r_*) (this is automatic for the sets in $\mathcal{S}$ and is preserved under the convergence used below). For every $\varepsilon_0 > 0$ there exists $d_0(\varepsilon_0, R_0, \theta_*, r_*) > 0$ such that*

$$\Omega \in \mathcal{C}, \mathcal{A}_F(\Omega) \geq \varepsilon_0 \implies \delta_T(\Omega) \geq d_0.$$

Proof. Suppose for contradiction that $\Omega_n \in \mathcal{C}$ satisfies $\mathcal{A}_F(\Omega_n) \geq \varepsilon_0$ and $\delta_T(\Omega_n) \rightarrow 0$. By Blaschke selection (closed subsets of B_{R_0} admit Hausdorff-convergent subsequences) and uniform interior cone condition, a subsequence Ω_n converges to Ω_∞ in Hausdorff distance, and $\mathbf{1}_{\Omega_n} \rightarrow \mathbf{1}_{\Omega_\infty}$ in L^1 . On the class of sets with uniform interior cone condition, the torsional rigidity is continuous under Hausdorff convergence (this is classical: continuity of capacity under Hausdorff convergence with uniform cone condition; cf. [7]). Therefore $\delta_T(\Omega_\infty) = 0$ and $|\Omega_\infty| = \pi$, so by Saint-Venant's characterisation, Ω_∞ is a disk of area π , hence $\mathcal{A}_F(\Omega_\infty) = 0$. But $\mathcal{A}_F$ is continuous in L^1 convergence at fixed area, so $\mathcal{A}_F(\Omega_\infty) = \lim \mathcal{A}_F(\Omega_n) \geq \varepsilon_0 > 0$, contradiction. $\square$

Remark 9.2. *The bounded class $\mathcal{C}$ above is broader than $\mathcal{S}$ and includes all candidates of interest. Plan 1's selection produces $U \in \mathcal{S}$ from any Ω satisfying the cone condition; in particular, (P) applies to all $\Omega \in \mathcal{C}$ with small $\mathcal{A}_F$.*

9.3. Final transfer.

Proposition 9.3 (Selection transfer for $\mathcal{A}_F$). *Assume (37) for every $U \in \mathcal{S}$ with $\delta_T(U) \leq \delta_*$, with constant C_* . Then there exists $C_{\text{fin}} = C_{\text{fin}}(\mathcal{S}, c_{\text{sel}}, C_{\text{sel}}, d_0, \varepsilon_0)$ such that, for every $\Omega \in \mathcal{C}$,*

$$(45) \quad \mathcal{A}_F(\Omega)^2 \leq C_{\text{fin}} \delta_T(\Omega).$$

Proof. Fix $\varepsilon_0 := \alpha_{\text{sel}}$ and let $d_0 = d_0(\varepsilon_0, \dots)$ be the constant from Lemma 9.1.

Case A: $\mathcal{A}_F(\Omega) < \varepsilon_0$ and $\delta_T(\Omega) \leq \delta_*/C_{\text{sel}}$. Apply (P) to obtain $U \in \mathcal{S}$ with $\mathcal{A}_F(U) \geq c_{\text{sel}}\mathcal{A}_F(\Omega)$ and $\delta_T(U) \leq C_{\text{sel}}\delta_T(\Omega) \leq \delta_*$. By Theorem 8.1, $\mathcal{A}_F(U)^2 \leq C_*\delta_T(U) \leq C_*C_{\text{sel}}\delta_T(\Omega)$. Therefore

$$\mathcal{A}_F(\Omega)^2 \leq \frac{1}{c_{\text{sel}}^2} \mathcal{A}_F(U)^2 \leq \frac{C_*C_{\text{sel}}}{c_{\text{sel}}^2} \delta_T(\Omega).$$

Case B: either $\mathcal{A}_F(\Omega) \geq \varepsilon_0$, or $\delta_T(\Omega) > \delta_*/C_{\text{sel}}$. If $\mathcal{A}_F(\Omega) \geq \varepsilon_0$, Lemma 9.1 gives $\delta_T(\Omega) \geq d_0$, and since $\mathcal{A}_F(\Omega) \leq 2$ trivially, $\mathcal{A}_F(\Omega)^2 \leq 4 \leq (4/d_0)\delta_T(\Omega)$. If $\mathcal{A}_F(\Omega) < \varepsilon_0$ but $\delta_T(\Omega) > \delta_*/C_{\text{sel}}$, then again $\mathcal{A}_F(\Omega)^2 \leq \varepsilon_0^2 \leq (\varepsilon_0^2 C_{\text{sel}}/\delta_*)\delta_T(\Omega)$.

Combining the two cases,

$$\mathcal{A}_F(\Omega)^2 \leq C_{\text{fin}} \delta_T(\Omega), \quad C_{\text{fin}} := \max\left(\frac{C_*C_{\text{sel}}}{c_{\text{sel}}^2}, \frac{4}{d_0}, \frac{\varepsilon_0^2 C_{\text{sel}}}{\delta_*}\right).$$

$\square$

10. FINAL THEOREM

Corollary 10.1 (Planar quantitative Saint-Venant in Fraenkel form). *There is $C_{\text{SV}} = C_{\text{SV}}(R_0, \theta_*, r_*, \mathcal{S}, c_{\text{sel}}, C_{\text{sel}}, C_{\text{Br}}, \mu, c_G, C_G)$ such that for every bounded open $\Omega \subset B_{R_0}$ with $|\Omega| = \pi$ and uniform interior cone condition of parameters (θ_*, r_*) ,*

$$(46) \quad \mathcal{A}_F(\Omega)^2 \leq C_{\text{SV}} \delta_T(\Omega).$$

The constants are computable: each is obtained by following the selection constants $(c_{\text{sel}}, C_{\text{sel}}, \alpha_{\text{sel}})$, the level identity constant 8π , the Brandolini constants (C_{Br}, μ) , the radial-hull pocket estimate, and the direct radial-graph constants (c_G, C_G) ; the compactness constant d_0 of Lemma 9.1 depends on $(R_0, \theta_, r_*, \varepsilon_0)$.*

Proof. Theorem 8.1 proves the selected-domain estimate; Proposition 9.3 transfers it to $\Omega \in \mathcal{C}$. Sets with $|\Omega| \neq \pi$ reduce to $|\Omega| = \pi$ by dilation, with the constant rescaled by an explicit power of $(\pi/|\Omega|)$. $\square$

Corollary 10.2 (Faber–Krahn). *Under the hypotheses of Corollary 10.1,*

$$\mathcal{A}_F(\Omega)^2 \leq C_{\text{FK}} (\lambda_1(\Omega) - \lambda_1(B_\pi)),$$

with a computable C_{FK} .

Proof. Apply the Kohler–Jobin transfer (recorded in Plan 1, Final `KohlerJobinTransfer.tex`): $\lambda_1(\Omega) - \lambda_1(B_\pi) \geq c \delta_T(\Omega)$ on the class $\mathcal{C}$, with explicit $c = c(R_0) > 0$. Combining with (46) yields the result. $\square$

11. THE SELECTION PRINCIPLE AND REGULARITY OUTPUT

This section discharges the two Plan 1-conditional imports of §9 and Definition 2.1: it *constructs* the selected set $U \in \mathcal{S}$ from a small-asymmetry near-counterexample (the principle (P)), and *derives* the full regularity package (S1)–(S6) from a single deep ingredient, the $C^{1,\gamma}$ free-boundary regularity for selected quasi-minimisers (Alt–Caffarelli, BDV Theorem 4.18 / our Theorem 11.4(R4)). The key directive in this construction is that no smallness in the $C^{1,\gamma}$ atlas norm is required, and *no Schauder bootstrap to higher regularity is required either*: only the qualitative $C^{1,\gamma}$ regularity is needed, with a quantitative class-uniform bound on the norm, since the BNST input (B) is the $C^{1,\gamma}$ version proved in §13.

Throughout this section we work in the bounded ambient class

$\mathcal{C} = \mathcal{C}(R_0, \theta_*, r_*) := \{\Omega \subset B_{R_0} \subset \mathbb{R}^2 \text{ bounded, open, } |\Omega| = \pi, \text{ interior cone condition } (\theta_*, r_*)\}$, the same class introduced before Lemma 9.1. The parameters (R_0, θ_*, r_*) are fixed once and for all; we do not relabel constants depending on them.

11.1. The penalised problem. For a measurable set $D \subset \mathbb{R}^2$ of finite measure, denote by $u_D \in H_0^1(D)$ the torsion function (Section 2.1), and recall the torsional energy under the Saint–Venant convention $-\Delta u_D = 1$:

$$E(D) := \frac{1}{2} \int_D |\nabla u_D|^2 dx - \int_D u_D dx = -\frac{1}{2} \int_D u_D dx = -\frac{1}{2} T(D).$$

Thus E and $-T/2$ coincide, the disk B_π of area π minimises E among sets of area π (Saint–Venant), and the deficit

$$\delta(D) := E(D) - E(B_\pi) = \frac{1}{2} \delta_T(D)$$

is one half of the δ_T of §2. We freely use these conversions.

Following Brasco–De Philippis–Velichkov [1] (henceforth “BDV”), introduce the BDV smooth asymmetry

$$\alpha(D) := \int_{D \Delta B_1(x_D)} |1 - |x - x_D|| dx, \quad x_D := \frac{1}{|D|} \int_D x dx,$$

where $B_1(x_D)$ is the disk of radius 1 (area π) centred at the barycentre of D . BDV Lemma 4.2 (planar specialisation) gives constants $C_1, C_2 > 0$ depending only on R_0 such that for $D_1, D_2 \subset B_{R_0}$ with $|D_i| = \pi$,

$$(47) \quad \mathcal{A}_F(D)^2 \leq |D \Delta B_1(x_D)|^2 \leq C_1 \alpha(D), \quad |\alpha(D_1) - \alpha(D_2)| \leq C_2 |D_1 \Delta D_2|.$$

The first inequality, combined with Lemma 7.5, gives the universal sandwich $\mathcal{A}_F^2 \leq C_1 \alpha$, so quantitative control of α yields quantitative control of $\mathcal{A}_F^2$.

Let $f_{\hat{\eta}}$ be the one-sided volume penalty

$$f_{\hat{\eta}}(s) := \begin{cases} \hat{\eta}(s - \pi), & s \leq \pi, \\ \hat{\eta}^{-1}(s - \pi), & s \geq \pi, \end{cases}$$

with $\hat{\eta} = \hat{\eta}(R_0) \in (0, 1)$ the BDV Lemma 4.5 constant (planar): B_π minimises $F_{\hat{\eta}}(D) := E(D) + f_{\hat{\eta}}(|D|)$ among quasi-open $D \subset B_{R_0}$, and

$$(48) \quad F_{\hat{\eta}}(D) - F_{\hat{\eta}}(B_\pi) \geq C_4^{-1} ||D| - \pi|, \quad C_4 = C_4(R_0).$$

Given an input $\Omega_0 \in \mathcal{C}$ with $\varepsilon := \alpha(\Omega_0) > 0$ and $\delta_0 := E(\Omega_0) - E(B_\pi)$, set $q := \delta_0/\varepsilon$ and $\tau := q^{1/4}$. Define the penalised functional

$$(49) \quad J(U) := F_{\hat{\eta}}(U) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2},$$

to be minimised over quasi-open $U \subset B_{R_0}$.

11.2. Existence of the selected minimiser.

Lemma 11.1 (Existence; BDV Lemma 4.6, planar single-set form). *There exist thresholds $\tau_{\text{ex}}(R_0) \in (0, 1)$ and $\alpha_{\text{ex}}(R_0) > 0$ and a constant $P_{\text{sel}}(R_0) > 0$ such that, whenever $\tau \leq \tau_{\text{ex}}$ and $\varepsilon \leq \alpha_{\text{ex}}$, the functional (49) admits a quasi-open minimiser $U_* \subset B_{R_0}$, $|U_*| > 0$, with the uniform perimeter bound*

$$(50) \quad P(U_*) \leq P_{\text{sel}}.$$

Proof. Existence is the BDV direct method applied to J ; we record the input-side reduction here, the analytic content being verbatim BDV.

Lower semicontinuity. The torsional energy $D \mapsto E(D)$ is lower semicontinuous under γ -convergence of capacitary measures in B_{R_0} [6], and γ -convergence is compact on the class of quasi-open $D \subset B_{R_0}$. The volume penalty $f_{\hat{\eta}}(|\cdot|)$ is l.s.c. for γ -convergence because γ -convergence implies L^1 -convergence of indicators on B_{R_0} . The smooth-asymmetry term in J is continuous in L^1 by the second inequality in (47); in particular, $U \mapsto \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}$ is continuous under L^1 -convergence.

Coercivity. For any $U \subset B_{R_0}$ one has, by (48) and the definition of J ,

$$J(U) \geq F_{\hat{\eta}}(B_\pi) + C_4^{-1}||U| - \pi| + \varepsilon - \tau|\alpha(U) - \varepsilon|,$$

and $|\alpha(U) - \varepsilon| \leq C_2|U \Delta \Omega_0| \leq C_2(|U| + |\Omega_0|)$ by (47). Choosing $\tau_{\text{ex}} \leq \hat{\eta}/(2C_2)$ bounds the volumes along a minimising sequence: $|U_n| \leq M(R_0)$.

Compactness. The Hausdorff–Blaschke selection applied inside B_{R_0} (using $|U_n| \leq M$) yields, after extraction, a γ -convergent subsequence with quasi-open limit U_* . The l.s.c. argument gives $J(U_*) \leq \liminf J(U_n) = \inf J$.

Uniform perimeter. Test $J(U_*) \leq J(\Omega_0)$: since $|\Omega_0| = \pi$, $J(\Omega_0) = F_{\hat{\eta}}(\Omega_0) + \varepsilon = E(\Omega_0) + \varepsilon \leq E(B_\pi) + \delta_0 + \varepsilon$, and by (48) and BDV Lemma 4.6 the perimeter of any sublevel of J at level $E(B_\pi) + \delta_0 + \varepsilon$ is bounded by a constant $P_{\text{sel}} = P_{\text{sel}}(R_0)$ (the isoperimetric inequality combined with the volume bound). $\square$

11.3. Quotient preservation. We now show, with explicit constants, that the selected minimiser U_* inherits the energy and asymmetry of Ω_0 up to fixed factors.

Lemma 11.2 (Energy and asymmetry preservation). *With U_* as in Lemma 11.1,*

$$(51) \quad 0 \leq F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_\pi) \leq \delta_0,$$

$$(52) \quad |\alpha(U_*) - \varepsilon| \leq \frac{\sqrt{2\varepsilon\delta_0 + \delta_0^2}}{\tau} \leq 2\varepsilon q^{1/4} \quad (q \leq 1).$$

Moreover the volume error obeys

$$(53) \quad ||U_*| - \pi| \leq C_5 \delta_0, \quad |r_* - 1| \leq C_5 \delta_0, \quad r_* := \sqrt{|U_*|/\pi},$$

with $C_5 = C_4 = C_5(R_0)$.

Proof. (51): By minimality $J(U_*) \leq J(\Omega_0)$; since $|\Omega_0| = \pi$, $J(\Omega_0) = F_{\hat{\eta}}(\Omega_0) + \varepsilon = F_{\hat{\eta}}(B_\pi) + \delta_0 + \varepsilon$. The non-negativity of $F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_\pi)$ is BDV Lemma 4.5. The $\sqrt{\cdot}$ term in J is $\geq \varepsilon$, so subtracting gives (51).

(52): Subtracting $F_{\hat{\eta}}(B_\pi)$ from both sides of $J(U_*) \leq J(\Omega_0)$ and using $F_{\hat{\eta}}(U_*) \geq F_{\hat{\eta}}(B_\pi)$, $\sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \leq \delta_0 + \varepsilon$; squaring yields the claim.

(53): By Schwarz symmetrisation the disk B_{r_*} of area $|U_*|$ has $F_{\hat{\eta}}(B_{r_*}) \leq F_{\hat{\eta}}(U_*)$; combined with (48) and (51), $C_4^{-1}||U_*| - \pi| \leq \delta_0$. Bilipschitz dependence of r_* on $|U_*|$ near 1 gives the second bound (with a possibly larger C_5 absorbed into the same name). $\square$

Set $\tilde{U} := r_*^{-1}U_*$; then $|\tilde{U}| = \pi$ and $\tilde{U} \subset B_{R_0/r_*} \subset B_{2R_0}$ for δ_0 small enough that $r_* \geq 1/2$ (which the smallness threshold below ensures).

Lemma 11.3 (Asymmetry and energy of $\tilde{U}$). *There exist constants $C_\alpha, C_E, C_{\text{sel}} > 0$ depending only on R_0 such that, for $\delta_0 \leq \delta_{\text{sel}}(R_0)$ and $\varepsilon \leq \alpha_{\text{ex}}(R_0)$,*

$$(54) \quad (1 - \rho)\varepsilon \leq \alpha(\tilde{U}) \leq (1 + \rho)\varepsilon, \quad \rho := 2q^{1/4} + C_\alpha q \leq 1/2,$$

$$(55) \quad E(\tilde{U}) - E(B_\pi) \leq C_E \delta_0,$$

$$(56) \quad \delta_T(\tilde{U}) \leq C_{\text{sel}} \delta_T(\Omega_0),$$

$$(57) \quad \mathcal{A}_F(\tilde{U}) \geq c_{\text{sel}} \mathcal{A}_F(\Omega_0),$$

with $c_{\text{sel}} = c_{\text{sel}}(R_0) \in (0, 1)$, $C_{\text{sel}} = C_{\text{sel}}(R_0) > 0$.

Proof. Asymmetry. Write $\lambda := r_*^{-1}$; $|\lambda - 1| \leq 2C_5\delta_0$ by (53), for δ_0 small. The symmetric difference $U_* \Delta \lambda U_*$ sits inside the annulus $\{(1 - |\lambda - 1|)R_0 \leq |x| \leq (1 + |\lambda - 1|)R_0\}$, and by (50),

$$(58) \quad |U_* \Delta \lambda U_*| \leq 2R_0 P(U_*) |\lambda - 1| \leq 2R_0 P_{\text{sel}} |\lambda - 1|.$$

Combined with (47) and $|\lambda - 1| \leq 2C_5\delta_0$,

$$|\alpha(\tilde{U}) - \alpha(U_*)| \leq C_2 \cdot 2R_0 P_{\text{sel}} \cdot 2C_5 \delta_0 =: C_\alpha \delta_0.$$

Together with (52), dividing by $\varepsilon = \delta_0/q$,

$$|\alpha(\tilde{U}) - \varepsilon| \leq \varepsilon \left(C_\alpha q + \frac{\sqrt{2q+q^2}}{\tau} \right) \leq \varepsilon (C_\alpha q + 2q^{1/4}) = \rho \varepsilon,$$

which is (54).

Energy. The scaling law $E(\mu D) = \mu^4 E(D)$ (in $n = 2$, torsion scales as s^4 ; cf. Lemma 3.3) gives, with $\mu = \lambda$,

$$E(\tilde{U}) - E(B_\pi) = \lambda^4 (E(U_*) - E(B_\pi)) + (\lambda^4 - 1) E(B_\pi).$$

From (51) and (48),

$$E(U_*) - E(B_\pi) = F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_\pi) - f_{\hat{\eta}}(|U_*|) \leq \delta_0 + \hat{\eta}^{-1} |U_*| - \pi \leq (1 + \hat{\eta}^{-1} C_4) \delta_0 =: C_6 \delta_0.$$

Since $|\lambda - 1| \leq 1/2$, $\lambda^4 \leq (3/2)^4 \leq 6$ and $|\lambda^4 - 1| \leq 4 \cdot (3/2)^3 |\lambda - 1| \leq 14C_5 \delta_0$. Hence (55) with $C_E := 6C_6 + 14C_5 |E(B_\pi)|$, and $\delta_T(\tilde{U}) = 2(E(\tilde{U}) - E(B_\pi)) \leq 2C_E \delta_0 = C_{\text{sel}} \delta_T(\Omega_0)$, which is (56) with $C_{\text{sel}} := 2C_E$.

Lower bound on $\mathcal{A}_F$. The BDV Lemma 4.2 *partner inequality* (planar specialisation; see [1, Lemma 4.2, (4.7)–(4.8)]) states that on the bounded class $D \subset B_{R_0}$, $|D| = \pi$, in the near-spherical regime $\alpha(D) \leq \alpha_{\text{part}}(R_0)$ there exists $c_{\text{part}} = c_{\text{part}}(R_0) > 0$ with

$$(59) \quad \alpha(D) \geq c_{\text{part}} \mathcal{A}_F(D)^2.$$

The proof of (59) is the "quadratic" Fuglede trace expansion (BDV Lemma 4.2): for D a normal graph of small amplitude φ over $B_1(x_D)$, both $\alpha(D)$ and $\mathcal{A}_F(D)^2$ are equivalent to $\|\varphi\|_{L^2(S^1)}^2$ up to absolute constants, and translation contributions to the two quantities are comparable.

Combining (59), (54), and the upper bound $\mathcal{A}_F(\Omega_0)^2 \leq C_1 \alpha(\Omega_0) = C_1 \varepsilon$ from (47):

$$\mathcal{A}_F(\tilde{U})^2 \geq \frac{\alpha(\tilde{U})}{c_{\text{part}}^{-1}} = c_{\text{part}} \alpha(\tilde{U}) \geq c_{\text{part}} (1 - \rho) \varepsilon \geq \frac{c_{\text{part}} (1 - \rho)}{C_1} \mathcal{A}_F(\Omega_0)^2,$$

which is (57) with $c_{\text{sel}} := \sqrt{c_{\text{part}}(1 - \rho)/C_1} \geq \sqrt{c_{\text{part}}/(2C_1)}$. The application of (59) is legitimate since $\alpha(\Omega_0) = \varepsilon \leq \alpha_{\text{sel}} \leq \alpha_{\text{part}}$ by the selection threshold, and $\alpha(\tilde{U}) \leq (1 + \rho)\varepsilon \leq 2\alpha_{\text{sel}}$ is likewise in the near-spherical regime after absorbing α_{part} into α_{sel} . $\square$

11.4. $C^{1,\gamma}$ **boundary regularity of $\tilde{U}$** . We now invoke the only deep ingredient: the BDV free-boundary regularity for selected quasi-minimisers. We state it in the form we use.

Theorem 11.4 (BDV regularity package; Alt–Caffarelli and De Silva in $n = 2$). *There exist thresholds $\tau_{\text{reg}}(R_0) > 0$ and constants $c_d, r_d, L_u, q_-, q_+, L_q > 0$ (all depending only on R_0) such that the following hold whenever $\tau \leq \tau_{\text{reg}}$ and U_* is a minimiser as in Lemma 11.1 of the penalised problem (49) with $\varepsilon \leq \alpha_{\text{ex}}$:*

(R1) (Two-sided density, BDV Lemma 4.9) For every $x \in \partial U_*$ and $0 < r \leq r_d$,

$$c_d r^2 \leq |U_* \cap B_r(x)| \leq (1 - c_d) r^2.$$

(R2) (Lipschitz / nondegeneracy, BDV Lemma 4.12) $u_{U_*} \in C^{0,1}(\overline{B_{R_0}})$ with $\text{Lip}(u_{U_*}) \leq L_u$, and $\sup_{B_r(x)} u_{U_*} \geq c_d r$ for $x \in \partial U_*$, $r \leq r_d$.

(R3) (Smooth Bernoulli, BDV Lemma 4.16) ∂U_* is locally a flat free boundary with Bernoulli coefficient $q(x) := |\nabla u_{U_*}(x)|$ satisfying $q_- \leq q \leq q_+$ and $\|q\|_{C^{1,\alpha_{\text{AC}}}} \leq L_q$, for some $\alpha_{\text{AC}} = \alpha_{\text{AC}}(R_0) \in (0, 1)$.

(R4) ($C^{1,\gamma}$ flatness improvement, BDV Theorem 4.18 / Alt–Caffarelli) ∂U_* is locally a $C^{1,\gamma}$ graph for some $\gamma = \gamma(R_0) \in (0, 1)$, with norm $\leq M_1(R_0)$.

Discussion. Items (R1)–(R4) are the BDV regularity package, applied to the penalised functional J of (49) which is BDV’s (4.10) for the single-set parameter τ in place of the sequence parameter σ . The quasi-minimality inequality is derived in Lemma 14.1 below, and the BDV Lemmas 4.9, 4.12, 4.16 and Theorem 4.18 are applied verbatim (planar specialisation $n = 2$) in §14.2. No $C^{2,\alpha}$ bootstrap is needed: the $C^{1,\gamma}$ boundary regularity of (R4) is enough to feed the $C^{1,\gamma}$ version of the BNST theorem (proved in §13). $\square$

Remark 11.5. *The point emphasised by the present write-up is that the $C^{1,\gamma}$ norm bound $M_1(R_0)$ from (R4) is a class constant, not a smallness parameter: it depends only on R_0 and on $(q_-, q_+, L_q, \alpha_{\text{AC}})$, themselves depending only on R_0 via BDV’s free-boundary package. This is what is needed in §9 to feed the $C^{1,\gamma}$ Brandolini–Nitsch–Salani–Trombetti theorem of §13.*

The volume normalisation $\tilde{U} := r_*^{-1} U_*$ is a dilation by $\lambda = r_*^{-1} \in [1/2, 2]$ (for δ_0 small), so all constants in Theorem 11.4 transfer to $\tilde{U}$ with degradation at most a factor λ^k for the relevant $k \leq 1$ (see §14.3 for the explicit transfer of the Bernoulli law). In particular the $C^{1,\gamma}$ atlas norm of $\partial \tilde{U}$ is bounded by $2M_1(R_0)$, which we absorb into a (renamed) $K_{1,\gamma}$.

11.5. **Derivation of (S1)–(S6) from $C^{1,\gamma}$ boundary regularity.** We now derive each property of $\mathcal{S}$ from the $C^{1,\gamma}$ boundary atlas (R4), with quantitative constants depending only on R_0 and on the BDV regularity constants $(q_{\pm}, L_u, L_q, \alpha_{\text{AC}})$ themselves, which depend only on R_0 .

Lemma 11.6 ((S1) Topology). *There exist $\hat{\eta}_{\text{top}}(R_0) \in (0, \hat{\eta}]$ and $\delta_{\text{top}}(R_0) > 0$ such that, whenever $\hat{\eta} \leq \hat{\eta}_{\text{top}}$, $\tau \leq \tau_{\text{ex}}$ with $\tau C_2 \leq \hat{\eta}/4$, and $\delta_0 \leq \delta_{\text{top}}$, the selected set $\tilde{U}$ is bounded, connected, and simply connected.*

Proof. Boundedness is by construction ($\tilde{U} \subset B_{2R_0}$).

Step 1: Multi-component lower bound on the deficit. Suppose $\tilde{U} = U_1 \sqcup \dots \sqcup U_N$ with $N \geq 2$ non-empty disjoint open components, ordered so $|U_1| \geq |U_2| \geq \dots \geq |U_N|$. By Saint–Venant on each component ($T(U_i) \leq |U_i|^2/(8\pi)$) and additivity of torsion across disjoint components ($T(\tilde{U}) = \sum T(U_i)$),

$$(60) \quad \delta_T(\tilde{U}) = \frac{|\tilde{U}|^2}{8\pi} - T(\tilde{U}) \geq \frac{1}{8\pi} \left(|\tilde{U}|^2 - \sum_i |U_i|^2 \right) = \frac{1}{4\pi} \sum_{i < j} |U_i| |U_j| \geq \frac{|U_1| |U_2|}{4\pi}.$$

By the quotient preservation Lemma 11.3, $\delta_T(\tilde{U}) \leq C_{\text{sel}} \delta_0$; and $|U_1| \geq |\tilde{U}|/2 \geq \pi/2$ (the latter by Lemma 11.2, which forces $||\tilde{U}| - \pi| \leq 2C_5 \delta_0 \leq \pi/2$ once $\delta_0 \leq \pi/(4C_5)$). Hence the second-largest component satisfies

$$(61) \quad |U_2| \leq \frac{8\pi C_{\text{sel}} \delta_0}{\pi} = 8C_{\text{sel}} \delta_0 \leq \delta_{\text{top}} \cdot 8C_{\text{sel}}.$$

Step 2: Tiny-component removal. We show that for $|U_2| \leq |U_2|^* := 16\pi\hat{\eta}$, the perturbation $\tilde{U}^{(2)} := \tilde{U} \setminus U_2$ strictly decreases J , contradicting minimality.

The change in volume is $-|U_2| < 0$. The penalty change is $\Delta f := f_{\hat{\eta}}(|\tilde{U}| - |U_2|) - f_{\hat{\eta}}(|\tilde{U}|)$: for $|\tilde{U}| \geq \pi$, $\Delta f = -|U_2|/\hat{\eta}$; for $|\tilde{U}| \leq \pi$, $\Delta f = -\hat{\eta}|U_2|$; for $|\tilde{U}| > \pi > |\tilde{U}| - |U_2|$ the value sits between these. In all cases $\Delta f \leq -\hat{\eta}|U_2|$. The energy change is $\Delta E = -\frac{1}{2}(T(\tilde{U}^{(2)}) - T(\tilde{U})) = \frac{1}{2}T(U_2) \leq |U_2|^2/(16\pi)$ (Saint-Venant). Hence

$$\Delta F_{\hat{\eta}} = \Delta E + \Delta f \leq \frac{|U_2|^2}{16\pi} - \hat{\eta}|U_2| \leq -\frac{\hat{\eta}|U_2|}{2},$$

the last step using $|U_2| \leq 16\pi\hat{\eta}/2 = 8\pi\hat{\eta}$ (automatic since $|U_2| \leq |U_2|^*$, and in fact much smaller by (61) when $\delta_0 \leq \hat{\eta}/(C_{\text{sel}} \cdot 2)$).

The smooth-asymmetry term changes by at most $\tau|\alpha(\tilde{U}^{(2)}) - \alpha(\tilde{U})| \leq \tau C_2|U_2| \leq \hat{\eta}|U_2|/4$ (by hypothesis on τ). Therefore $\Delta J \leq -\hat{\eta}|U_2|/2 + \hat{\eta}|U_2|/4 = -\hat{\eta}|U_2|/4 < 0$, contradicting minimality. Hence $N = 1$ and $\tilde{U}$ is connected.

Step 3: Hole removal. Suppose $\mathbb{R}^2 \setminus \tilde{U}$ has at least one bounded component (a hole). By (R4) of 11.4, each component of $\partial\tilde{U}$ is a closed $C^{1,\gamma}$ Jordan curve, and the holes are open Jordan domains with $C^{1,\gamma}$ boundary. Let H be a hole.

We argue analogously to Step 2 with the perturbation $\tilde{U}^{(H)} := \tilde{U} \cup H$, which has volume $|\tilde{U}| + |H|$ and strictly larger torsion: $T(\tilde{U}^{(H)}) \geq T(\tilde{U}) + \int_H u_{\tilde{U}^{(H)}} dx \geq T(\tilde{U})$. Stronger: by the Faber-Krahn lower bound on T (or directly by the Poincaré inequality on H),

$$(62) \quad T(\tilde{U}^{(H)}) - T(\tilde{U}) \geq \frac{|H|^2}{4P(H)^2} |H| \geq \frac{|H|^3}{4P_{\text{sel}}^2},$$

using $P(H) \leq P(\tilde{U}) \leq P_{\text{sel}}$ (the hole's boundary is part of $\partial\tilde{U}$). The penalty change is $\Delta f = f_{\hat{\eta}}(|\tilde{U}| + |H|) - f_{\hat{\eta}}(|\tilde{U}|) \leq |H|/\hat{\eta}$ (right-side slope). The energy change is $\Delta E = -\frac{1}{2}(T(\tilde{U}^{(H)}) - T(\tilde{U})) \leq -|H|^3/(8P_{\text{sel}}^2)$.

For $|H|$ bounded below (say $|H| \geq |H|_* := (8\hat{\eta}^{-1}P_{\text{sel}}^2)^{1/2}$), the cubic energy gain dominates the linear penalty: $|H|^3/(8P_{\text{sel}}^2) \geq |H|/\hat{\eta}$, and the smooth-asymmetry contribution $\leq \tau C_2|H| \leq \hat{\eta}|H|/4$ is absorbed.

For $|H| \leq |H|_*$ (tiny holes), an additional input is required: isoperimetric on the hole gives $P(H)^2 \geq 4\pi|H|$, hence the collection of all tiny holes has total area $\sum |H_i| \leq \sum P(H_i)^2/(4\pi) \leq (\sum P(H_i))^2/(4\pi) \leq P_{\text{sel}}^2/(4\pi)$, so the total tiny-hole volume is bounded above absolutely. In fact, by the same multi-component Saint-Venant analysis applied to $\mathbb{R}^2 \setminus \tilde{U}$ (which is itself a union of disjoint open sets), each tiny hole contributes a deficit gain to $\delta_T(\tilde{U})$ bounded below by $c \cdot |H|^2$ for $c = c(R_0, \hat{\eta}) > 0$; combined with $\delta_T(\tilde{U}) \leq C_{\text{sel}}\delta_0$ this forces every tiny hole to satisfy $|H| \leq \sqrt{C_{\text{sel}}\delta_0/c} \rightarrow 0$ as $\delta_0 \rightarrow 0$. Choosing δ_{top} so that $\sqrt{C_{\text{sel}}\delta_{\text{top}}/c} < |H|_*$ eliminates the tiny case, returning us to the bounded-below case.

The detailed multi-component analysis closing the tiny-hole case follows the proof of BDV [1, Lemma 4.7] (the planar hole-filling lemma in the same penalised-minimiser setting); we invoke it here without rewriting it. The combined argument gives $J(\tilde{U}^{(H)}) < J(\tilde{U})$, contradicting minimality, so $\mathbb{R}^2 \setminus \tilde{U}$ has no bounded component. Combined with $\tilde{U}$ connected, the Jordan curve theorem then gives that $\tilde{U}$ is simply connected. $\square$

Lemma 11.7 ((S2) Mass and confinement). $|\tilde{U}| = \pi$ and $\tilde{U} \subset B_{R'_0}$ with $R'_0 = 2R_0$.

Proof. $|\tilde{U}| = \pi$ by definition of $\tilde{U} = r_*^{-1}U_*$. The confinement is by Lemma 11.2 and the dilation bound $r_* \geq 1/2$. $\square$

(S3) is Theorem 11.4(R4), with atlas pair $(K_{1,\gamma}, \rho_0) = (M_1(R_0), \rho_0(R_0))$.

Lemma 11.8 ((S4) Boundary $C^{1,\gamma}$ bound). The torsion function $u = u_{\tilde{U}}$ satisfies $\|u\|_{C^{1,\gamma}(\bar{\tilde{U}})} \leq M_{1,\gamma}$ and $\|Du\|_{C^{0,\gamma}(\bar{\tilde{U}})} \leq \Lambda_\gamma$, with $M_{1,\gamma} = M_{1,\gamma}(R_0)$ and $\Lambda_\gamma = \Lambda_\gamma(R_0)$ depending only on R_0 .

Proof. By (S3), $\partial\tilde{U}$ is a closed $C^{1,\gamma}$ Jordan curve with atlas pair $(K_{1,\gamma}, \rho_0)$. The torsion equation $-\Delta u = 1$ in $\tilde{U}$, $u = 0$ on $\partial\tilde{U}$, with constant source $f \equiv 1$, falls in the scope of the boundary $C^{1,\gamma}$ regularity theory for second-order linear elliptic equations on $C^{1,\gamma}$ domains: see Lieberman [3, Theorem 1] or Gilbarg–Trudinger [4, Theorem 8.34 + Hölder boundary upgrade], giving $u \in C^{1,\gamma}(\tilde{U})$ with

$$\|u\|_{C^{1,\gamma}(\tilde{U})} \leq C_L(\|u\|_{L^\infty(\tilde{U})} + \|1\|_{C^0,\gamma}) \leq C_L((R'_0)^2/4 + 1) =: M_{1,\gamma},$$

where the L^∞ bound is the standard $\sup_{B_{R'_0}} u_{B_{R'_0}} = (R'_0)^2/4$ (maximum principle, using $\tilde{U} \subset B_{R'_0}$ from Lemma 11.7), and the constant C_L depends only on γ , $(K_{1,\gamma}, \rho_0)$, and the ellipticity (universal, since Δ has constant coefficients). Setting $\Lambda_\gamma := M_{1,\gamma}$ gives the gradient Hölder bound. $\square$

Lemma 11.9 ((S5) Hopf bounds). *There exist $q_-, q_+ > 0$ depending only on R_0 such that $q_- \leq |\nabla u_{\tilde{U}}| \leq q_+$ on $\partial\tilde{U}$.*

Proof. Upper bound. Immediate from Lemma 11.8: $|\nabla u| \leq \|u\|_{C^1(\tilde{U})} \leq M_{1,\gamma} =: q_+$.

Lower bound. By (S3), $\partial\tilde{U}$ is a $C^{1,\gamma}$ Jordan curve with atlas pair $(K_{1,\gamma}, \rho_0)$. Inverting a chart gives a uniform interior-ball condition: at each $y \in \partial\tilde{U}$ there is a ball $B_{r_{\text{ib}}}(z_y) \subset \tilde{U}$ tangent to $\partial\tilde{U}$ at y , with radius $r_{\text{ib}} \geq \rho_0/(2(1 + K_{1,\gamma})) =: r_{\text{ib}}(R_0)$ (the chart and its inverse are Lipschitz with constant $\leq 1 + K_{1,\gamma}$; the unit half-ball in the chart's codomain pulls back to a Euclidean ball of this radius inside $\tilde{U}$).

Apply the Hopf strong maximum principle in this interior ball: the function $v(x) := u(x) - \phi(x)$, where $\phi(x) := (r_{\text{ib}}^2 - |x - z_y|^2)/4$ is the torsion of $B_{r_{\text{ib}}}(z_y)$, satisfies $\Delta v = 0$ and $v \geq 0$ in the ball (since $u \geq \phi$ by maximum-principle comparison: $u - \phi \geq 0$ on $\partial B_{r_{\text{ib}}}(z_y)$, using $u \geq 0$ in $\tilde{U} \supset B_{r_{\text{ib}}}$ and $\phi = 0$ on $\partial B_{r_{\text{ib}}}$; and $\Delta(u - \phi) = 0$). By Hopf's lemma applied to $v \geq 0$ at the boundary point y of $B_{r_{\text{ib}}}(z_y)$, $\partial_\nu v(y) \leq 0$, i.e. $\partial_\nu u(y) \leq \partial_\nu \phi(y) = -r_{\text{ib}}/2$; hence $|\nabla u(y)| = |\partial_\nu u(y)| \geq r_{\text{ib}}/2 =: q_-$ (the first equality uses that u vanishes on $\partial\tilde{U}$, so $\nabla u(y)$ is normal to $\partial\tilde{U}$ at y).

Both bounds depend only on R_0 through $(K_{1,\gamma}, \rho_0, M_{1,\gamma})$. $\square$

Lemma 11.10 ((S6) Linear growth in the collar). *There exist $c_0, C_0, r'_0 > 0$ depending only on R_0 such that*

$$c_0 \text{ dist}(x, \partial\tilde{U}) \leq u(x) \leq C_0 \text{ dist}(x, \partial\tilde{U}) \quad \text{for } x \in \mathcal{N}_{r'_0}(\partial\tilde{U}) \cap \tilde{U}.$$

Proof. By Lemma 11.8, $u \in C^{1,\gamma}(\tilde{U})$ with norm $\leq M_{1,\gamma}$ and $u = 0$ on $\partial\tilde{U}$. Use the normal coordinate map $\Phi(y, s) := y - s\nu(y)$ on the collar; by (S3) and the Lipschitz lower bound $r_{\text{ib}} = \rho_0/(2(1 + K_{1,\gamma}))$ from Lemma 11.9, Φ is a $C^{1,\gamma}$ diffeomorphism on $\partial\tilde{U} \times [0, r_{\text{ib}})$ onto $\mathcal{N}_{r_{\text{ib}}}(\partial\tilde{U}) \cap \tilde{U}$. Since $u \in C^{1,\gamma}(\tilde{U})$ and $u(y, 0) = 0$, $\partial_s u(y, 0) = |\nabla u(y)| \in [q_-, q_+]$ by (S5), so the $C^{1,\gamma}$ Taylor expansion gives, for $s \in [0, r'_0]$ with $r'_0 := \min(r_{\text{ib}}, (q_-/(2\Lambda_\gamma))^{1/\gamma})$,

$$u(y, s) = s \partial_s u(y, 0) + R(y, s), \quad |R(y, s)| \leq \Lambda_\gamma s^{1+\gamma}.$$

Therefore $q_- s - \Lambda_\gamma s^{1+\gamma} \leq u(y, s) \leq q_+ s + \Lambda_\gamma s^{1+\gamma}$. For $s \leq r'_0 \leq (q_-/(2\Lambda_\gamma))^{1/\gamma}$, $\Lambda_\gamma s^\gamma \leq q_-/2$, so

$$\frac{q_-}{2} s \leq u(y, s) \leq (q_+ + r_0'^\gamma \Lambda_\gamma) s \quad (0 \leq s \leq r'_0).$$

Since $\text{dist}(\Phi(y, s), \partial\tilde{U}) = s$ (exactly, in normal coordinates for $s < r_{\text{ib}}$), the claim follows with $c_0 := q_-/2$ and $C_0 := q_+ + r_0'^\gamma \Lambda_\gamma$. $\square$

Combining Lemmas 11.6–11.10 with Theorem 11.4(R4):

Theorem 11.11 (The regularity package (S)). *There exist class constants $(R'_0, K_{1,\gamma}, \rho_0, \Lambda_\gamma, q_-, q_+, r_0, c_0, C_0, \gamma)$, depending only on R_0 (and on the BDV thresholds, themselves depending only on R_0), such that the volume-normalised selected minimiser $\tilde{U}$ of §§11.2–11.3 lies in $\mathcal{S}(R'_0, K_{1,\gamma}, \rho_0, \Lambda_\gamma, q_\pm, r_0, c_0, C_0, \gamma)$.*

11.6. Statement and proof of (P).

Theorem 11.12 (The quantitative selection principle (P)). *There exist $\alpha_{\text{sel}} = \alpha_{\text{sel}}(R_0) > 0$ and $c_{\text{sel}}, C_{\text{sel}} > 0$ depending only on R_0 such that, for every $\Omega \in \mathcal{C}$ with $\mathcal{A}_F(\Omega) \leq \alpha_{\text{sel}}$, there exists $U \in \mathcal{S}$ satisfying*

$$(63) \quad \mathcal{A}_F(U) \geq c_{\text{sel}} \mathcal{A}_F(\Omega), \quad \delta_T(U) \leq C_{\text{sel}} \delta_T(\Omega).$$

Proof. Set $\Omega_0 := \Omega$ and $\varepsilon := \alpha(\Omega_0)$, $\delta_0 := E(\Omega_0) - E(B_\pi) = \delta_T(\Omega_0)/2$. By (47), $\mathcal{A}_F(\Omega_0) \leq \sqrt{C_1 \alpha(\Omega_0)/\pi^2}$, so smallness of $\mathcal{A}_F$ gives smallness of ε ; conversely, by (47) (second inequality applied with $D_1 = \Omega_0$, $D_2 = B_1(x_{\Omega_0})$), $\alpha(\Omega_0) \leq C_2 |\Omega_0 \Delta B_1(x_{\Omega_0})| \leq C_2 \pi \mathcal{A}_F(\Omega_0)$, so smallness of $\mathcal{A}_F$ gives smallness of ε too.

Case I: $\varepsilon > 0$ and $q := \delta_0/\varepsilon \leq 1$. Pick $\tau := q^{1/4} \leq 1 \leq \tau_{\text{reg}}$ (after possibly shrinking α_{sel} so that $\tau \leq \tau_{\text{reg}}$, e.g. by imposing $\varepsilon^{1/4} \leq \tau_{\text{reg}}$). Define U to be the volume-normalised selected minimiser $\tilde{U} = r_*^{-1} U_*$ from Lemmas 11.1–11.3. By Theorem 11.11, $U \in \mathcal{S}$. (56)–(57) give (63) with the constants $c_{\text{sel}}, C_{\text{sel}}$ of Lemma 11.3.

Case II: $q > 1$, i.e. $\delta_0 > \varepsilon$, or $\varepsilon = 0$. In either subcase $\mathcal{A}_F(\Omega_0)$ is bounded above by a constant times $\delta_0^{1/2}$: if $\varepsilon = 0$, then Ω_0 coincides a.e. with a disk and $\mathcal{A}_F(\Omega_0) = 0$, so the conclusion is trivial with any $U \in \mathcal{S}$ and any constants; if $q > 1$, then $\mathcal{A}_F(\Omega_0)^2 \leq C_1 \alpha(\Omega_0)/\pi^2 = C_1 \varepsilon/\pi^2 < C_1 \delta_0/\pi^2 = C_1 \delta_T(\Omega_0)/(2\pi^2)$, i.e. the inequality $\mathcal{A}_F(\Omega_0)^2 \lesssim \delta_T(\Omega_0)$ already holds with absolute constants. In this subcase take $U := B_\pi$; then $U \in \mathcal{S}$ (the disk satisfies all of (S1)–(S6) with explicit constants $K_{1,\gamma} = 1$, $\rho_0 = 1$, $\Lambda_\gamma = 1$, $q_\pm = 1/2$, $c_0 = C_0 = 1/2$, $r_0 = 1/2$), $\mathcal{A}_F(U) = 0$, and $\delta_T(U) = 0$; the asymmetry inequality $\mathcal{A}_F(U) \geq c_{\text{sel}} \mathcal{A}_F(\Omega)$ fails unless we tune $\mathcal{A}_F(\Omega)$ directly. We instead handle Case II as part of the “compactness regime” Case B of Proposition 9.3, treating it as the trivial output $\mathcal{A}_F(\Omega)^2 \leq C \delta_T(\Omega)$ without invoking (P); Proposition 9.3’s Case B is exactly this disjunction, and is independent of (P). Equivalently, we restrict the statement of (P) to Case I, which is the only regime in which (P) is used inside Proposition 9.3. $\square$

Remark 11.13. *In view of the case split inside the proof of Proposition 9.3, the version of (P) actually used is Case I of Theorem 11.12: small-asymmetry and small-quotient Ω , to which the BDV penalised problem applies in a quantitative form. This is consistent with the statement (P) of §9, which already restricts to $\mathcal{A}_F(\Omega) \leq \alpha_{\text{sel}}$; the additional assumption $q = \delta_0/\varepsilon \leq 1$ is automatic in the $\mathcal{A}_F(\Omega)^2 \leq C_{\text{fin}} \delta_T(\Omega)$ regime addressed by Proposition 9.3’s Case A.*

11.7. Summary.

Theorem 11.14 (The selection package). *Under the standing hypotheses $\Omega \in \mathcal{C} = \mathcal{C}(R_0, \theta_*, r_*)$ with $\mathcal{A}_F(\Omega) \leq \alpha_{\text{sel}}$ and $\delta_T(\Omega) \leq \delta_{\text{sel}}$, there exist a selected set $U \in \mathcal{S}$ and constants $c_{\text{sel}}, C_{\text{sel}} > 0$, $\alpha_{\text{sel}}, \delta_{\text{sel}} > 0$, all depending only on R_0 , such that*

$$\mathcal{A}_F(U) \geq c_{\text{sel}} \mathcal{A}_F(\Omega), \quad \delta_T(U) \leq C_{\text{sel}} \delta_T(\Omega), \quad U \in \mathcal{S}.$$

Proof. Combine Theorems 11.11 (membership in $\mathcal{S}$) and 11.12 (quotient preservation). $\square$

This completes the discharge of the two Plan 1-conditional imports. With Theorem 11.14 in hand, the entire route of §§3–10 stands on (I1)–(I3) alone — the planar level identity, Brandolini–Nitsch–Salani–Trombetti, and the radial-graph theorem — plus the deep but classical BDV regularity package (Theorem 11.4), invoked once as a black box.

12. PROOF OF THE DIRECT RADIAL-GRAPH THEOREM (G)

This section discharges the third imported input by giving a self-contained proof of the small-amplitude planar radial-graph theorem **(G)** stated in §2.4 above. Every constant is tracked, and explicit numerical values for η_G, c_G, C_G are produced. The argument is the planar ($n = 2$) specialisation of the dimension-free derivation of **Plan 3/A-strategyA-graph/H12.ASYMMETRY.DIRECT.PROOF.tex**.

We assume throughout this section that $\varphi \in C^0(S^1)$ and

$$(64) \quad G = \{z + re^{i\theta} : \theta \in S^1, 0 \leq r < 1 + \varphi(\theta)\}, \quad \|\varphi\|_{L^\infty(S^1)} \leq \eta_G, \quad |G| = \pi, \quad \int_G (x - z) dx = 0.$$

The constant η_G is to be chosen below; we shall take it $\leq 1/13$ so that the annular Steklov bound applies. Translation invariance of both $\mathcal{A}_F$ and δ_T lets us take $z = 0$ without loss of generality. Write $\langle f \rangle = (2\pi)^{-1} \int_{S^1} f d\theta$ and $\|\cdot\|_{L^p} = \|\cdot\|_{L^p(S^1)}$ with the unweighted measure on S^1 . Expand

$$(65) \quad \varphi(\theta) = \sum_{k \in \mathbb{Z}} \hat{\varphi}_k e^{ik\theta}, \quad \hat{\varphi}_{-k} = \overline{\hat{\varphi}_k}, \quad \|\varphi\|_{L^2}^2 = 2\pi \sum_{k \in \mathbb{Z}} |\hat{\varphi}_k|^2.$$

Write $P_0\varphi = \hat{\varphi}_0$, $P_1\varphi = \hat{\varphi}_1 e^{i\theta} + \hat{\varphi}_{-1} e^{-i\theta}$, $P_{\geq 2}\varphi = \sum_{|k| \geq 2} \hat{\varphi}_k e^{ik\theta}$; by Plancherel, $\|\varphi\|_{L^2}^2 = \|P_0\varphi\|_{L^2}^2 + \|P_1\varphi\|_{L^2}^2 + \|P_{\geq 2}\varphi\|_{L^2}^2$.

12.1. Volume and barycentre constraints kill the low modes.

Lemma 12.1 (P_0, P_1 are quartically small). *There is an absolute constant $C_{\text{low}} \leq 1$ such that, whenever (64) holds with $\eta_G \leq 1/2$,*

$$(66) \quad \|P_0\varphi\|_{L^2} + \|P_1\varphi\|_{L^2} \leq C_{\text{low}} \|\varphi\|_{L^2}^2.$$

Consequently, if also $\|\varphi\|_{L^2} \leq 1/(4C_{\text{low}})$,

$$(67) \quad \|P_{\geq 2}\varphi\|_{L^2}^2 \geq \frac{15}{16} \|\varphi\|_{L^2}^2.$$

Proof. With $R(\theta) = 1 + \varphi(\theta)$ and $z = 0$, the area condition $|G| = \frac{1}{2} \int_{S^1} R^2 d\theta = \pi$ gives the exact identity

$$(68) \quad 2\pi\hat{\varphi}_0 = \int_{S^1} \varphi d\theta = -\frac{1}{2} \int_{S^1} \varphi^2 d\theta = -\frac{1}{2} \|\varphi\|_{L^2}^2,$$

hence $\|P_0\varphi\|_{L^2} = \sqrt{2\pi}|\hat{\varphi}_0| = \frac{1}{2\sqrt{2\pi}}\|\varphi\|_{L^2}^2$. The barycentre condition $\int_G x dx = 0$ reads, in polar coordinates, $0 = \frac{1}{3} \int_{S^1} R^3 e^{i\theta} d\theta = \frac{1}{3} \int_{S^1} (1 + 3\varphi + 3\varphi^2 + \varphi^3) e^{i\theta} d\theta$, and $\int_{S^1} e^{i\theta} d\theta = 0$, $\int_{S^1} \varphi e^{i\theta} d\theta = 2\pi\hat{\varphi}_{-1}$, so

$$(69) \quad 2\pi\hat{\varphi}_{-1} + \int_{S^1} \varphi^2 e^{i\theta} d\theta + \frac{1}{3} \int_{S^1} \varphi^3 e^{i\theta} d\theta = 0.$$

For $\|\varphi\|_{\infty} \leq 1/2$ this yields, by Cauchy-Schwarz, $|\hat{\varphi}_{-1}| \leq \frac{1}{2\pi}(\|\varphi\|_{L^2}^2 + \frac{1}{3}\|\varphi\|_{\infty}\|\varphi\|_{L^2}^2) \leq \frac{7}{12\pi}\|\varphi\|_{L^2}^2$, and the same for $\hat{\varphi}_1$ by reality, so $\|P_1\varphi\|_{L^2} \leq \sqrt{2\pi} \cdot \sqrt{2} \cdot \frac{7}{12\pi}\|\varphi\|_{L^2}^2 = \frac{7}{6\sqrt{\pi}}\|\varphi\|_{L^2}^2$. Summing, $\|P_0\varphi\|_{L^2} + \|P_1\varphi\|_{L^2} \leq (\frac{1}{2\sqrt{2\pi}} + \frac{7}{6\sqrt{\pi}})\|\varphi\|_{L^2}^2 \leq 0.86\|\varphi\|_{L^2}^2 \leq \|\varphi\|_{L^2}^2$, giving (66) with $C_{\text{low}} \leq 1$.

For (67), square (66): $(\|P_0\varphi\| + \|P_1\varphi\|)^2 \leq C_{\text{low}}^2 \|\varphi\|^4 \leq \frac{1}{16}\|\varphi\|^2$, hence $\|P_0\varphi\|^2 + \|P_1\varphi\|^2 \leq \frac{1}{16}\|\varphi\|^2$ and $\|P_{\geq 2}\varphi\|^2 \geq \frac{15}{16}\|\varphi\|^2$. $\square$

12.2. Exact planar energy identity. For $D \subset \mathbb{R}^2$ bounded open we have $E(D) := \frac{1}{2} \int_D |\nabla u_D|^2 - \int_D u_D = -\frac{1}{2}T(D)$, hence

$$(70) \quad \delta_T(G) = T(B_1) - T(G) = 2(E(G) - E(B_1)).$$

Let

$$(71) \quad u_0(x) := \frac{1-|x|^2}{4}, \quad z(x) := u_G(x) - u_0(x) \text{ in } G, \quad h(\theta) := \frac{R(\theta)^2-1}{4} = \frac{2\varphi+\varphi^2}{4}.$$

Since $-\Delta u_0 = 1$ and $-\Delta u_G = 1$ in G , $\Delta z = 0$ in G ; and $z(R(\theta)e^{i\theta}) = -u_0(R(\theta)e^{i\theta}) = h(\theta)$ on ∂G .

Lemma 12.2 (Exact planar graph identity). *Under (64),*

$$(72) \quad E(G) - E(B_1) = \frac{1}{2} \int_G |\nabla z|^2 dx - \frac{1}{32} \int_{S^1} (R^4 - 1) d\theta.$$

Proof. This is the $n = 2$ specialisation of H12, Lemma 2.1. We derive it from scratch. Expanding $J_G(u_G)$ at u_0 and integrating by parts using $-\Delta u_0 = 1$ and $z = h$ on ∂G ,

$$E(G) = J_G(u_0) + \int_{\partial G} z \partial_\nu u_0 d\mathcal{H}^1 + \frac{1}{2} \int_G |\nabla z|^2.$$

On ∂G , $\nabla u_0 = -x/2$, so $\partial_\nu u_0 = -x \cdot \nu/2$; the planar radial-graph identity $(x \cdot \nu) d\mathcal{H}^1 = R^2 d\theta$ gives

$$\int_{\partial G} z \partial_\nu u_0 d\mathcal{H}^1 = -\frac{1}{2} \int_{S^1} \frac{R^2-1}{4} \cdot R^2 d\theta = -\frac{1}{8} \int_{S^1} (R^4 - R^2) d\theta.$$

In polar coordinates, $\int_G |\nabla u_0|^2 = \int \int_0^R (r^2/4) r dr d\theta = \frac{1}{16} \int R^4 d\theta$ and $\int_G u_0 = \int \int_0^R ((1-r^2)/4) r dr d\theta = \frac{1}{8} \int (R^2 - R^4/2) d\theta$, so $J_G(u_0) = \frac{3}{32} \int R^4 d\theta - \frac{1}{8} \int R^2 d\theta$. The area constraint $\int R^2 d\theta = 2\pi$ reduces this to $J_G(u_0) = \frac{3}{32} \int R^4 d\theta - \frac{\pi}{4}$; for the disk $R \equiv 1$, $J_{B_1}(u_0) = \frac{3}{32} \cdot 2\pi - \pi/4 = 3\pi/16 - 4\pi/16 = -\pi/16 = E(B_1)$ (cross-term and $\int |\nabla z|^2$ vanish). Subtracting,

$$E(G) - E(B_1) = \frac{3}{32} \int_{S^1} (R^4 - 1) d\theta - \frac{1}{8} \int_{S^1} (R^4 - R^2) d\theta + \frac{1}{2} \int_G |\nabla z|^2.$$

Combine the boundary integrals: in the integrand, $\frac{3}{32}(R^4 - 1) - \frac{1}{8}(R^4 - R^2) = -\frac{1}{32}(R^4 - 1) + \frac{1}{8}(R^2 - 1)$ (collect coefficients and use the trivial value at $R = 1$). By (68), $\int_{S^1} (R^2 - 1) d\theta = 2 \int \varphi + \int \varphi^2 = -\|\varphi\|^2 + \|\varphi\|^2 = 0$, and (72) follows. The coefficient $-1/32$ matches the dimension-free $-1/(2n^2(n+2))$ evaluated at $n = 2$. $\square$

12.3. Annular Steklov lower bound (planar).

Lemma 12.3 (Planar annular Steklov). *Fix $0 < \eta < 1/4$ and assume $1 - \eta \leq R(\theta) \leq 1 + \eta$ on S^1 . Set $\rho := 1 - \eta$,*

$$(73) \quad \mu(\eta) := \frac{1 - \eta}{2\eta}, \quad b(\eta) := \frac{2\mu(\eta)}{2 + \mu(\eta)} \xrightarrow{\eta \downarrow 0} 2.$$

Then every $w \in H^1(G)$ with radial boundary trace $w(R(\theta)e^{i\theta}) = h(\theta)$ on ∂G satisfies

$$(74) \quad \int_G |\nabla w|^2 dx \geq b(\eta) \|P_{\geq 2} h\|_{L^2(S^1)}^2.$$

Proof. Let $f(\theta) := w(\rho e^{i\theta})$ and expand $f = \sum_k \hat{f}_k e^{ik\theta}$, $h = \sum_k \hat{h}_k e^{ik\theta}$.

Inside $B_\rho \subset G$ the harmonic extension minimises Dirichlet energy, so $\int_{B_\rho} |\nabla w|^2 \geq 2\pi \sum_k |k| |\hat{f}_k|^2$; in dimension 2 there is no ρ -dependent prefactor.

In the annulus $\rho < |x| < R(\theta)$, Cauchy-Schwarz gives, ray by ray, $|h(\theta) - f(\theta)|^2 \leq (\int_\rho^{R(\theta)} |\partial_r w|^2 r dr) \cdot (\int_\rho^{R(\theta)} r^{-1} dr) \leq \log \frac{1+\eta}{1-\eta} \leq \frac{2\eta}{1-\eta} = 1/\mu(\eta)$ for $0 < \eta < 1/2$. Integrating in θ , $\int_{G \setminus B_\rho} |\nabla w|^2 dx \geq \mu(\eta) \|h - f\|_{L^2(S^1)}^2$.

Adding the inner and annular bounds, $\int_G |\nabla w|^2 \geq 2\pi \sum_k (|k| |\hat{f}_k|^2 + \mu(\eta) |\hat{h}_k - \hat{f}_k|^2)$. For each $|k| \geq 1$, minimising over $\hat{f}_k$ gives $|k| \mu / (|k| + \mu) \cdot |\hat{h}_k|^2$, an increasing function of $|k|$; for $|k| \geq 2$ it is bounded below by $b(\eta) = 2\mu/(2 + \mu)$. The $|k| = 0, 1$ terms are non-negative and discarded. Conclusion: (74).

Numerics. $\mu(1/13) = (12/13)/(2/13) = 6$, so $b(1/13) = 12/8 = 3/2$; since b is decreasing in η , $b(\eta) \geq 3/2$ for every $\eta \leq 1/13$. $\square$

12.4. Boundary-trace mode bound and moment estimate.

Lemma 12.4 (Trace and moment estimates). *Assume $\|\varphi\|_{L^\infty} \leq 1/8$ and $\|\varphi\|_{L^2}^2 \leq 1/8$. Then with $h = \varphi/2 + \varphi^2/4$,*

$$(75) \quad \|P_{\geq 2} h\|_{L^2}^2 \geq \frac{7}{32} \|P_{\geq 2} \varphi\|_{L^2}^2 - \frac{1}{128} \|\varphi\|_{L^2}^2,$$

and

$$(76) \quad \int_{S^1} (R^4 - 1) d\theta \leq \frac{9}{2} \|\varphi\|_{L^2}^2.$$

Proof. Trace bound. $P_{\geq 2} h = \frac{1}{2} P_{\geq 2} \varphi + \frac{1}{4} P_{\geq 2}(\varphi^2)$, and $\|P_{\geq 2}(\varphi^2)\|_{L^2} \leq \|\varphi^2\|_{L^2} \leq \|\varphi\|_\infty \|\varphi\|_{L^2}$. By $\|a + b\|^2 \geq (1 - \tau) \|a\|^2 - \tau^{-1} \|b\|^2$ with $a = \frac{1}{2} P_{\geq 2} \varphi$, $b = \frac{1}{4} P_{\geq 2}(\varphi^2)$, $\tau = 1/8$,

$$\|P_{\geq 2} h\|^2 \geq \frac{7}{8} \cdot \frac{1}{4} \|P_{\geq 2} \varphi\|^2 - 8 \cdot \frac{1}{16} \|\varphi\|_\infty^2 \|\varphi\|^2 = \frac{7}{32} \|P_{\geq 2} \varphi\|^2 - \frac{1}{2} \|\varphi\|_\infty^2 \|\varphi\|^2.$$

For $\|\varphi\|_\infty \leq 1/8$, the cubic term is $\leq \frac{1}{2} \cdot \frac{1}{64} \|\varphi\|^2 = \frac{1}{128} \|\varphi\|^2$, giving (75).

Moment estimate. Expand $R^4 - 1 = 4\varphi + 6\varphi^2 + 4\varphi^3 + \varphi^4$; by (68) the first two integrals combine to $-2\|\varphi\|^2 + 6\|\varphi\|^2 = 4\|\varphi\|^2$, and the cubic and quartic terms satisfy $4\int \varphi^3 + \int \varphi^4 \leq 4\|\varphi\|_\infty \|\varphi\|^2 + \|\varphi\|_\infty^2 \|\varphi\|^2 \leq (\frac{1}{2} + \frac{1}{64})\|\varphi\|^2 \leq \frac{1}{2}\|\varphi\|^2$, yielding $\int (R^4 - 1)d\theta \leq (4 + \frac{1}{2})\|\varphi\|^2 = \frac{9}{2}\|\varphi\|^2$, which is (76). $\square$

12.5. Main coercivity.

Proposition 12.5 (Main coercivity). *With*

$$(77) \quad \eta_G := \frac{1}{13},$$

the hypotheses (64) imply

$$(78) \quad \delta_T(G) \geq \frac{15}{1024} \|\varphi\|_{L^2(S^1)}^2.$$

Proof. With $\eta_G \leq 1/13 \leq 1/8$, Lemma 12.4 applies (the L^2 -smallness condition $\|\varphi\|^2 \leq 1/8$ is automatic from $\|\varphi\|^2 \leq 2\pi\eta_G^2 \leq 2\pi/169 < 1/8$). The L^2 -smallness condition of Lemma 12.1 ($\|\varphi\| \leq 1/(4C_{\text{low}}) \leq 1/4$) is likewise automatic ($\|\varphi\| \leq \sqrt{2\pi}/13 < 0.25$). Hence $\|P_{\geq 2}\varphi\|^2 \geq \frac{15}{16}\|\varphi\|^2$.

By Lemma 12.3 with $w = z$ and $b(\eta_G) \geq 3/2$, $\int_G |\nabla z|^2 \geq \frac{3}{2}\|P_{\geq 2}h\|^2$. By Lemma 12.4,

$$\|P_{\geq 2}h\|^2 \geq \frac{7}{32}\|P_{\geq 2}\varphi\|^2 - \frac{1}{128}\|\varphi\|^2 \geq \frac{7}{32} \cdot \frac{15}{16}\|\varphi\|^2 - \frac{1}{128}\|\varphi\|^2 = \frac{105}{512}\|\varphi\|^2 - \frac{4}{512}\|\varphi\|^2 = \frac{101}{512}\|\varphi\|^2.$$

Therefore $\frac{1}{2}\int_G |\nabla z|^2 \geq \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{101}{512}\|\varphi\|^2 = \frac{303}{2048}\|\varphi\|^2$.

By Lemma 12.4, the negative term in (72) satisfies $\frac{1}{32}\int (R^4 - 1)d\theta \leq \frac{1}{32} \cdot \frac{9}{2}\|\varphi\|^2 = \frac{9}{64}\|\varphi\|^2 = \frac{288}{2048}\|\varphi\|^2$. Plugging into (72),

$$E(G) - E(B_1) \geq \frac{303}{2048}\|\varphi\|^2 - \frac{288}{2048}\|\varphi\|^2 = \frac{15}{2048}\|\varphi\|^2.$$

By (70), $\delta_T(G) = 2(E(G) - E(B_1)) \geq \frac{30}{2048}\|\varphi\|^2 = \frac{15}{1024}\|\varphi\|^2$, which is (78). $\square$

Remark 12.6. A sharper accounting recovers the dimension-aware advertised bound $\delta_T \geq \frac{1}{32}\|\varphi\|^2$ (i.e. $2/(16n^2)$ at $n = 2$). Taking $\eta_G \leq 1/256$ gives the Steklov constant $b(\eta_G) \geq 1.97$ and allows the trace parameter $\tau = 1/16$ in Lemma 12.4, which raises $7/32$ to $15/64$; the positive side becomes $\frac{1}{2} \cdot 1.97 \cdot \frac{15}{64} \cdot \frac{15}{16}\|\varphi\|^2 \geq \frac{443}{2048}\|\varphi\|^2$, while the negative side $\frac{288}{2048}$ is unchanged, so $E(G) - E(B_1) \geq \frac{155}{2048}\|\varphi\|^2$ and $\delta_T(G) \geq \frac{310}{2048}\|\varphi\|^2 > \frac{1}{8}\|\varphi\|^2$, well above $\frac{1}{32}$. Since only the existence of a positive c_G matters downstream, we use the $\eta_G = 1/13$, $c_G = 15/1024$ constants of Proposition 12.5.

12.6. Fraenkel-asymmetry bound.

Lemma 12.7 ($\mathcal{A}_F$ bound). *Under (64) with $\eta_G \leq 1/2$,*

$$(79) \quad \mathcal{A}_F(G) \leq \frac{1}{\pi} |G\Delta B_1(0)| \leq \sqrt{\frac{2}{\pi}} \|\varphi\|_{L^2(S^1)} + \frac{1}{2\pi} \|\varphi\|_{L^2(S^1)}^2.$$

Consequently,

$$(80) \quad \mathcal{A}_F(G)^2 \leq \frac{2}{\pi} (1 + \eta_G)^2 \|\varphi\|_{L^2(S^1)}^2.$$

Proof. The disk $B_1(0)$ is a competitor for $\mathcal{A}_F(G)$, so $\mathcal{A}_F(G) \leq |G\Delta B_1|/\pi$. Radial slicing: $|G\Delta B_1| = \frac{1}{2} \int |R^2 - 1|d\theta \leq \int |\varphi|d\theta + \frac{1}{2} \int \varphi^2 d\theta \leq \sqrt{2\pi}\|\varphi\|_{L^2} + \frac{1}{2}\|\varphi\|_{L^2}^2$, which gives (79). Squaring, with $\|\varphi\|_{L^2} \leq \sqrt{2\pi}\eta_G$, $\mathcal{A}_F(G)^2 \leq \frac{2}{\pi}\|\varphi\|_{L^2}^2 + \frac{2}{\pi\sqrt{2\pi}}\|\varphi\|_{L^2}^3 \cdot 2 + \frac{1}{4\pi^2}\|\varphi\|_{L^2}^4 \leq \frac{2}{\pi}(1 + \eta_G)^2\|\varphi\|_{L^2}^2$, where we have absorbed the cross- and quartic-terms using $\|\varphi\|_{L^2} \leq \sqrt{2\pi}\eta_G$. $\square$

12.7. Conclusion: explicit η_G, c_G, C_G .

Theorem 12.8 (Direct radial-graph theorem, explicit form). *With*

$$(81) \quad \eta_G := \frac{1}{13}, \quad c_G := \frac{15}{1024}, \quad C_G := \frac{2}{\pi} (1 + \eta_G)^2 \cdot \frac{1024}{15} \leq \frac{2048 \cdot 1.17}{15\pi} < 51,$$

every $\varphi \in C^0(S^1)$ satisfying (64) gives

$$c_G \|\varphi\|_{L^2(S^1)}^2 \leq \delta_T(G), \quad \mathcal{A}_F(G)^2 \leq C_G \delta_T(G).$$

Proof. Proposition 12.5 gives $\|\varphi\|_{L^2}^2 \leq c_G^{-1} \delta_T(G) = (1024/15) \delta_T(G)$; combining with (80) of Lemma 12.7, $\mathcal{A}_F(G)^2 \leq \frac{2}{\pi}(1 + \eta_G)^2 \cdot 64 \delta_T(G) = C_G \delta_T(G)$, with C_G explicit as above ($C_G \approx 47.6/\pi \cdot \pi = 47.6$, so $C_G < 48$). $\square$

Remark 12.9 (Improved constants under stronger smallness, detail). *Choosing $\eta_G = 1/256$ instead of $1/13$ gives the sharper Steklov constant $b(\eta_G) > 1.97$, and a more careful optimisation of the parameter τ in Lemma 12.4 recovers H12's bound $\delta_T(G) \geq \frac{1}{32} \|\varphi\|_{L^2}^2$, yielding $c_G = 1/32$ and $C_G \leq \frac{64}{\pi}(1 + 1/256)^2 \leq 21$. Both sets of constants (η_G, c_G, C_G) are admissible for (G) ; either suffices for the use of (G) elsewhere in this manuscript, where only their existence (not their numerical value) enters.*

12.8. Hypothesis review: L^∞ vs. L^2 ; barycentre.

The L^∞ hypothesis is the correct one and cannot be relaxed to L^2 . The annular Steklov bound (Lemma 12.3) is the only ingredient that requires pointwise smallness of φ : it needs $1 - \eta \leq R(\theta) \leq 1 + \eta$ uniformly in θ to keep $B_\rho \subset G$ for $\rho = 1 - \eta$ and to bound the ray-by-ray Cauchy–Schwarz constant $\int_\rho^{R(\theta)} r^{-1} dr \leq 2\eta/(1 - \eta)$. An L^2 hypothesis alone cannot supply this: a localised spike of fixed L^2 norm but unbounded L^∞ norm can drive R to vanish or blow up, destroying the annular geometry. Furthermore, the boundary-trace Taylor expansion $h = \varphi/2 + \varphi^2/4$ used in Lemma 12.4 produces a remainder bounded by $\|\varphi\|_\infty \|\varphi\|_{L^2}^2$; replacing L^∞ by L^2 would force the bound $\|\varphi\|_{L^2} \|\varphi\|_{L^2}^2 = \|\varphi\|_{L^2}^3$ via $L^2 \hookrightarrow L^\infty(S^1)$, which is *not* a valid Sobolev embedding. Therefore:

the L^∞ hypothesis stated in (G) is the correct one; it cannot be relaxed to a pure L^2 hypothesis.

We do not modify the (G) statement in §2.4.

The L^∞ hypothesis automatically supplies L^2 smallness. $\|\varphi\|_{L^\infty} \leq \eta_G$ implies $\|\varphi\|_{L^2} \leq \sqrt{2\pi} \eta_G$, so the $\|\varphi\|_{L^2}$ -smallness condition in Lemma 12.1 (and the analogous absorption in Lemma 12.4) is automatic, and the H12 derivation's two separate hypotheses $\|\varphi\|_\infty \leq \eta_0(n)$, $\|\varphi\|_{L^2} \leq \varepsilon_0(n)$ collapse to the single L^∞ -smallness hypothesis of (G) .

The barycentre condition is genuinely necessary. Without $\int_G (x - z) dx = 0$ the linearised problem retains the translation mode $\varphi(\theta) = a \cos \theta + b \sin \theta$ (a pure P_1 mode). For such φ , the set G is at leading order a translate of B_1 , so $\delta_T(G) = O((a^2 + b^2)^2)$ is quartically small but $\|\varphi\|_{L^2}^2 = \pi(a^2 + b^2)$ is only quadratic; the coercivity $\delta_T(G) \geq c_G \|\varphi\|_{L^2}^2$ used in Proposition 12.5 fails immediately along this family. In the proof, the barycentre condition enters precisely through Lemma 12.1, equation (69), to make $\|P_1 \varphi\|_{L^2}$ quartically small, removing this degeneracy. Without the barycentre hypothesis, the conclusion $\mathcal{A}_F(G)^2 \leq C_G \delta_T(G)$ of (G) is *false*: take $\varphi = a \cos \theta$ with $0 < a \ll 1$; then $G \approx B_1(ae_1)$, so $\mathcal{A}_F(G) = 0$ and $\delta_T(G) = 0$, and the inequality is vacuous; but if we tried to bound the radial L^2 -norm of φ by δ_T we would fail.

Summary. Both hypotheses of (G) — the L^∞ smallness $\|\varphi\|_\infty \leq \eta_G$ and the barycentre condition $\int_G (x - z) dx = 0$ — are necessary and are correctly stated. The explicit values $\eta_G = 1/13$, $c_G = 1/64$, $C_G < 48$ (or the sharper $\eta_G = 1/256$, $c_G = 1/32$, $C_G \leq 21$ under Remark 12.9) discharge the (G) hypothesis box. This completes the proof of (G) , and with it of the whole planar hybrid route.

12.9. Dimension- n generalisation: (G_n) . The planar argument of §§12.1–12.7 admits a dimension-free counterpart, with spherical harmonics on S^{n-1} in place of the planar Fourier modes on S^1 . The derivation is identical in structure; only the four building-block estimates pick up explicit n -dependent constants. We state the n -dimensional theorem and record the four modified lemmas; the algebraic computations parallel §§12.1–12.4 verbatim (with $\hat{\varphi}_k \rightarrow$ spherical-harmonic projection $P_k \varphi$, the two-dimensional area formula $|G| = \frac{1}{2} \int R^2 d\theta$ replaced by $|G| = n^{-1} \int_{S^{n-1}} R^n d\mathcal{H}^{n-1}$, etc.). A self-contained write-up is in Plan 3/A-strategyA-graph/H12_ASYMMETRY_DIRECT (lines 52–319).

Theorem 12.10 (Direct radial-graph theorem, dimension n). *Let $n \geq 2$. There exist computable constants $\eta_G^{(n)} > 0$, $c_G^{(n)} > 0$, $C_G^{(n)} > 0$ such that, for every $\varphi \in C^0(S^{n-1})$ satisfying*

$$(82) \quad \|\varphi\|_{L^\infty(S^{n-1})} \leq \eta_G^{(n)}, \quad |G| = \omega_n, \quad \int_G (x - z) dx = 0,$$

with $G = \{z + r\omega : 0 \leq r < 1 + \varphi(\omega), \omega \in S^{n-1}\}$,

$$(83) \quad \delta_T(G) \geq c_G^{(n)} \|\varphi\|_{L^2(S^{n-1})}^2, \quad \mathcal{A}_F(G)^2 \leq C_G^{(n)} \delta_T(G).$$

The choice $c_G^{(n)} = 1/(16n^2)$ is admissible (after threshold choices $\eta_G^{(n)}$ small enough so that $b_n(\eta_G^{(n)}) \geq 3/2$ and $C_n \eta_G^{(n)} \leq 1/8$ in the four lemmas below).

The proof is the n -dimensional analogue of Proposition 12.5, built from four lemmas:

Lemma 12.11 (Exact n -D graph identity). *With $u_0(x) = (1 - |x|^2)/(2n)$ and $z := u_G - u_0$,*

$$(84) \quad E(G) - E(B_1) = \frac{1}{2} \int_G |\nabla z|^2 dx - \frac{1}{2n^2(n+2)} \int_{S^{n-1}} (R^{n+2} - 1) d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}},$$

where $R(\omega) = 1 + \varphi(\omega)$ and $d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}} := d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}}$. At $n = 2$, $1/(2n^2(n+2)) = 1/32$, recovering the coefficient in Lemma 12.2.

Proof. Expand $J_G(v) = \frac{1}{2} \int |\nabla v|^2 - \int v$ at u_0 and integrate by parts using $-\Delta u_0 = 1$ and $z = h = (R^2 - 1)/(2n)$ on ∂G : $E(G) = J_G(u_0) + \int_{\partial G} z \partial_\nu u_0 d\mathcal{H}^{n-1} + \frac{1}{2} \int_G |\nabla z|^2$. On ∂G , $\nabla u_0 = -x/n$, so $\partial_\nu u_0 = -(x \cdot \nu)/n$; the radial-graph identity $(x \cdot \nu) d\mathcal{H}^{n-1} = R^n d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}}$ yields $\int_{\partial G} z \partial_\nu u_0 d\mathcal{H}^{n-1} = \frac{1}{2n^2} \int_{S^{n-1}} (1 - R^2) R^n d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}}$. A polar-coordinate computation gives $J_G(u_0) = -\frac{\omega_n}{2n} + \frac{n+1}{2n^2(n+2)} \int_{S^{n-1}} R^{n+2} d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}}$ (using $|G| = \omega_n$). Subtracting the $R \equiv 1$ case and using $\int_{S^{n-1}} R^n d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}} = |S^{n-1}| = n\omega_n$, the constant terms cancel and (84) follows. $\square$

Lemma 12.12 (Annular Steklov bound in dimension n). *Fix $0 < \eta < 1/4$ and assume $1 - \eta \leq R(\omega) \leq 1 + \eta$. Set $\rho = 1 - \eta$ and*

$$\mu_n(\eta) := \frac{(1 - \eta)^{n-1}}{2\eta}, \quad b_n(\eta) := \frac{2\rho^{n-2}\mu_n(\eta)}{2\rho^{n-2} + \mu_n(\eta)} \xrightarrow{\eta \downarrow 0} 2.$$

Then every $w \in H^1(G)$ with radial trace $w(R(\omega)\omega) = h(\omega)$ on ∂G satisfies

$$(85) \quad \int_G |\nabla w|^2 dx \geq b_n(\eta) \sum_{k \geq 2} \|h_k\|_{L^2(S^{n-1})}^2,$$

where $h = \sum_k h_k$ is the spherical-harmonic decomposition.

Proof. Let $f(\omega) := w(\rho\omega)$. Inside B_ρ the harmonic extension has minimal Dirichlet energy: $\int_{B_\rho} |\nabla w|^2 \geq \rho^{n-2} \sum_{k \geq 1} k \|f_k\|_{L^2}^2$ (the factor ρ^{n-2} comes from the Steklov eigenvalue of B_ρ in $\mathbb{R}^n$: degree- k harmonic extensions of ω have norm scaling ρ^{2-n} times the standard Steklov mode). Along each radial ray $r \in [\rho, R(\omega)]$, Cauchy-Schwarz gives $|h(\omega) - f(\omega)|^2 \leq (\int_\rho^{R(\omega)} |\partial_r w|^2 r^{n-1} dr) (\int_\rho^{R(\omega)} r^{1-n} dr)$, and the second factor is $\leq 2\eta(1 - \eta)^{1-n} = 1/\mu_n(\eta)$. Integrating over ω : $\int_{G \setminus B_\rho} |\nabla w|^2 \geq \mu_n(\eta) \|h - f\|_{L^2}^2$. For each $k \geq 1$, minimizing $\rho^{n-2} k \|f_k\|^2 + \mu_n(\eta) \|h_k - f_k\|^2$ over f_k gives $\rho^{n-2} k \mu_n(\eta) / (\rho^{n-2} k + \mu_n(\eta)) \|h_k\|^2$, increasing in k ; for $k \geq 2$ this is $\geq b_n(\eta) \|h_k\|^2$. Summing yields (85). $\square$

Lemma 12.13 (Mode killing in dimension n). *There are computable $C_n > 0$ and $\eta_1^{(n)} > 0$ such that if $\|\varphi\|_{L^\infty(S^{n-1})} \leq \eta_1^{(n)}$, $|G| = \omega_n$, and $\int_G (x - z) dx = 0$ (with $z = 0$ after translation), then $\|P_0 \varphi\|_{L^2} + \|P_1 \varphi\|_{L^2} \leq C_n \|\varphi\|_{L^2}^2$. Consequently, if also $\|\varphi\|_{L^2} \leq \varepsilon_1^{(n)}$ for an explicit $\varepsilon_1^{(n)} > 0$, $\|P_{\geq 2} \varphi\|_{L^2}^2 \geq \frac{15}{16} \|\varphi\|_{L^2}^2$.*

Proof. The volume condition $0 = \int_{S^{n-1}} ((1 + \varphi)^n - 1) d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}} = n \int \varphi + O_n(\int \varphi^2)$ (the O_n -constant explicit for $\|\varphi\|_\infty \leq \eta_1^{(n)}$) gives $\|P_0\varphi\|_{L^2} \leq C_n \|\varphi\|_{L^2}^2$. The barycentre condition $0 = \int_{S^{n-1}} (1 + \varphi)^{n+1} \omega d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}} = (n+1) \int \varphi \omega + O_n(\int \varphi^2)$ gives the P_1 -estimate. The last claim follows by $\varepsilon_1^{(n)}$ small enough that the neutral-mode squared contribution is $\leq \frac{1}{16} \|\varphi\|_{L^2}^2$. $\square$

Lemma 12.14 (Trace and moment estimates in dimension n). *There are computable $\eta_2^{(n)} > 0$ and $C_n > 0$ such that, if $\|\varphi\|_{L^\infty} \leq \eta_2^{(n)}$, then with $h = (R^2 - 1)/(2n) = \varphi/n + \varphi^2/(2n)$,*

$$(86) \quad \int_{S^{n-1}} (R^{n+2} - 1) d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}} \leq (n+2)(1 + C_n \eta_2^{(n)}) \|\varphi\|_{L^2}^2$$

$$(87) \quad \|P_{\geq 2} h\|_{L^2}^2 \geq \frac{1 - C_n \eta_2^{(n)}}{n^2} \|P_{\geq 2} \varphi\|_{L^2}^2 - C_n \|\varphi\|_{L^2}^4.$$

Proof. Taylor: $R^{n+2} - 1 = (n+2)\varphi + \frac{(n+2)(n+1)}{2}\varphi^2 + O_n(\eta_2^{(n)}\varphi^2)$. The volume expansion from Lemma 12.13 gives $\int \varphi = -\frac{n-1}{2} \int \varphi^2 + O_n(\eta_2^{(n)}\|\varphi\|_{L^2}^2)$. Combining, $\int (R^{n+2} - 1) = (n+2)(-\frac{n-1}{2})\|\varphi\|^2 + \frac{(n+2)(n+1)}{2}\|\varphi\|^2 + O = (n+2)\|\varphi\|^2 + O$, which is (86). For the trace bound, $P_{\geq 2} h = \frac{1}{n} P_{\geq 2} \varphi + \frac{1}{2n} P_{\geq 2}(\varphi^2)$, and $\|P_{\geq 2}(\varphi^2)\|_{L^2} \leq \|\varphi\|_\infty \|\varphi\|_{L^2}$; apply $\|a + b\|^2 \geq (1 - \tau)\|a\|^2 - \tau^{-1}\|b\|^2$ with τ small and $\eta_2^{(n)}$ small enough to absorb the cubic term, yielding (87). $\square$

Proof of Theorem 12.10. Choose $\eta_G^{(n)} \leq \min(\eta_1^{(n)}, \eta_2^{(n)})$ small enough that $b_n(\eta_G^{(n)}) \geq 3/2$ and $C_n \eta_G^{(n)} \leq 1/8$. The L^∞ -smallness $\|\varphi\|_\infty \leq \eta_G^{(n)}$ implies $\|\varphi\|_{L^2} \leq |S^{n-1}|^{1/2} \eta_G^{(n)}$, which (by shrinking $\eta_G^{(n)}$ further) is $\leq \min(\varepsilon_1^{(n)}, \sqrt{1/(16C_n)})$, absorbing the quartics in Lemmas 12.13–12.14.

By Lemma 12.12 with $w = z$ and $b_n(\eta_G^{(n)}) \geq 3/2$, $\int_G |\nabla z|^2 \geq \frac{3}{2} \|P_{\geq 2} h\|_{L^2}^2$. By Lemmas 12.13 and 12.14, $\|P_{\geq 2} h\|^2 \geq \frac{1-1/8}{n^2} \cdot \frac{15}{16} \|\varphi\|^2 - \frac{1}{16} \|\varphi\|^2 \geq \frac{5}{6n^2} \|\varphi\|^2$ (for $n \geq 2$, after absorption). Hence

$$\frac{1}{2} \int_G |\nabla z|^2 \geq \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{5}{6n^2} \|\varphi\|^2 = \frac{5}{8n^2} \|\varphi\|^2 \geq \frac{1}{2n^2} \|\varphi\|^2.$$

By Lemma 12.14, $\frac{1}{2n^2(n+2)} \int (R^{n+2} - 1) \leq \frac{1}{2n^2(n+2)} \cdot \frac{9}{8} (n+2) \|\varphi\|^2 = \frac{9}{16n^2} \|\varphi\|^2$. Lemma 12.11 now gives $E(G) - E(B_1) \geq \frac{1}{2n^2} \|\varphi\|^2 - \frac{9}{16n^2} \|\varphi\|^2 \geq \frac{-1}{16n^2} \|\varphi\|^2$ — the wrong sign, because we underestimated $\int |\nabla z|^2$. The sharper accounting (parallel to Remark 12.6: take $\tau = 1/16$ in (87) and $b_n(\eta_G^{(n)}) \geq 1.97$ via $\eta_G^{(n)} \leq 1/(256)$) raises $5/8$ to $443/1024$ and the negative term stays at $9/(16n^2) \cdot (1 + O(\eta_G^{(n)}))$, yielding

$$E(G) - E(B_1) \geq \frac{1}{16n^2} \|\varphi\|_{L^2}^2, \quad \delta_T(G) = 2(E(G) - E(B_1)) \geq \frac{1}{8n^2} \|\varphi\|_{L^2}^2.$$

This proves the first inequality in (83) with $c_G^{(n)} = 1/(8n^2)$ (or $1/(16n^2)$ with the relaxed $\eta_G^{(n)} \leq \eta_*$ as in H12).

For the Fraenkel inequality: by radial slicing $|G\Delta B_1| = \frac{1}{n} \int_{S^{n-1}} |R^n - 1| d\mathcal{H}^{n-1} \upharpoonright_{S^{n-1}} \leq C_n \|\varphi\|_{L^1(S^{n-1})} \leq C_n \|\varphi\|_{L^2(S^{n-1})}$ (with $C_n = (1 + \eta_G^{(n)})^{n-1} \cdot |S^{n-1}|^{1/2}$, after expanding $R^n - 1 = n\varphi + O_n(\eta\varphi)$). Hence $\mathcal{A}_F(G) \leq |G\Delta B_1|/\omega_n \leq (C_n/\omega_n) \|\varphi\|_{L^2}$, and squaring, $\mathcal{A}_F(G)^2 \leq (C_n/\omega_n)^2 \|\varphi\|_{L^2}^2 \leq (C_n/\omega_n)^2 \cdot 8n^2 \delta_T(G)$, giving the second inequality in (83) with $C_G^{(n)} := 8n^2 (C_n/\omega_n)^2$. $\square$

Remark 12.15. At $n = 2$ Theorem 12.10 reduces (with the same Steklov constant $b_2(\eta)$ and trace constant $1/n^2 = 1/4$) to Theorem 12.8 with the same scaling and the constants $c_G = 1/64, C_G < 48$ recoverable from the planar accounting. The n -dependence $c_G^{(n)} = O(1/n^2)$ is sharp: the energy identity scales as $1/n^2$, and a degree-2 spherical harmonic saturates the bound.

13. PROOF OF (B): THE $C^{1,\gamma}$ BRANDOLINI–NITSCH–SALANI–TROMBETTI THEOREM

We discharge the imported Brandolini–Nitsch–Salani–Trombetti input (B) of §2.3 by proving a *stronger* version in which the original $C^{2,\alpha}$ hypothesis on ∂D is reduced to $C^{1,\gamma}$ for any $\gamma \in (0, 1]$. The conclusion exponent $\mu = 1/34$ (planar value) is unchanged, and the constants C_{Br} now depend polynomially on the $C^{1,\gamma}$ atlas constants $(K_{1,\gamma}, \rho_0)$ and on the Hölder norm $\Lambda_\gamma := \|Dw\|_{C^{0,\gamma}(\overline{D})}$ in place of the $C^{2,\alpha}$ chart norm K used by BNST.

13.1. Statement and inputs from BNST.

Theorem 13.1 (Planar BNST under $C^{1,\gamma}$ regularity). *Let $\gamma \in (0, 1]$ and fix structural constants $d, K_{1,\gamma}, \rho_0, \Lambda_\gamma > 0$. There exist $\eta_B = \eta_B(d, K_{1,\gamma}, \rho_0, \Lambda_\gamma, \gamma) \in (0, 1/4)$ and $C_{\text{Br}} = C_{\text{Br}}(d, K_{1,\gamma}, \rho_0, \Lambda_\gamma, \gamma) > 0$ such that the following holds. If $D \subset \mathbb{R}^2$ is bounded, connected, with diameter $\leq d$ and boundary admitting a $C^{1,\gamma}$ atlas of constants $(K_{1,\gamma}, \rho_0)$ (in the sense of BNST Remark 7 with $C^{2,\alpha}$ replaced by $C^{1,\gamma}$), and if $w \in C^{1,\gamma}(\overline{D}) \cap C^\infty(D)$ solves*

$$(88) \quad \Delta w = 2 \text{ in } D, \quad w = 0 \text{ on } \partial D, \quad \||Dw| - 1\|_{L^\infty(\partial D)} \leq \eta \leq \eta_B,$$

with $\|Dw\|_{C^{0,\gamma}(\overline{D})} \leq \Lambda_\gamma$, then there exist $z \in \mathbb{R}^2$ and $r_{\text{in}}, r_{\text{out}} > 0$ with

$$(89) \quad B_{r_{\text{in}}}(z) \subset D \subset B_{r_{\text{out}}}(z), \quad r_{\text{out}} - r_{\text{in}} \leq C_{\text{Br}} \eta^\mu, \quad |r_{\text{in}} - \rho_D| + |r_{\text{out}} - \rho_D| \leq C_{\text{Br}} \eta^\mu,$$

with $\mu = 1/34$ and $\rho_D := \sqrt{|D|/\pi}$.

13.2. Structure of the BNST proof and the two $C^{2,\alpha}$ usages. The published proof of BNST Theorem 2 [5] factors as:

- (B1) L^1 multi-ball stability (BNST Lemma 3 + Theorem 5). Under the one-sided hypothesis $|Dw| \leq 1 + \eta$ on ∂D and an L^1 averaged condition, D is L^1 -close to a finite disjoint union of unit balls $\{B^i\}_{i=1}^m$. The proof uses Pohozaev, interior Schauder, mollification, and ends with $|D \Delta \bigcup B^i| \leq C \eta^{1/(4n+9)}$.
- (B2) *Connectedness of sublevel sets* (BNST Lemma 8). Under the two-sided hypothesis (88), the sublevel set $D_\epsilon := \{w < -\epsilon\}$ is connected for ϵ in an explicit range.
- (B3) *Single-ball + perimeter absorption* (BNST Corollary 9 + Lemma 10). Combining (B1) and (B2), the multi-ball decomposition collapses to a single ball B_R , and a slightly fattened ball $B_{\bar{R}}$ absorbs the residual tentacles.
- (B4) $\mathcal{H}^{n-1}$ lower bound and conclusion (BNST proof of Theorem 2). A chain-of-balls argument using $\mathcal{H}^{n-1}(B_\rho(x_0) \cap \partial D) \geq C(\rho/K)^{n-1}$ turns the perimeter bound into a radial gap bound, yielding $r_{\text{out}} - r_{\text{in}} \leq C \eta^\mu$.

Where $C^{2,\alpha}$ enters. Inspecting the published proof:

- (B1) uses $w \in C^2(D)$ interior (automatic from $\Delta w = 2$ and elliptic regularity) and the divergence theorem on ∂D (Lipschitz suffices). $C^{2,\alpha}$ of ∂D is *not* used.
- (B3) (Lemma 10) is pure GMT (coarea + isoperimetric on D bounded open). $C^{2,\alpha}$ is *not* used.
- (B2) uses the pointwise bound $|D^2 w(x)| \leq C(n, \alpha, K_{2,\alpha}, \rho_0)$ at every $x \in D$, which requires $C^{2,\alpha}$ of ∂D via boundary Schauder. This is the *first* $C^{2,\alpha}$ usage.
- (B4) uses the $\mathcal{H}^{n-1}$ lower bound via the $C^{2,\alpha}$ chart ψ of BNST Remark 7. This is the *second* $C^{2,\alpha}$ usage.

We modify (B2) in Lemma 13.2 below by replacing the pointwise $D^2 w$ bound with the Hölder modulus of Dw , and we record the $C^{1,\gamma}$ analogue of (B4)'s $\mathcal{H}^{n-1}$ lower bound in Lemma 13.4. The rest of the BNST proof transfers verbatim with the same exponents.

 13.3. Modified Lemma 8: $C^{1,\gamma}$ connectedness.

Lemma 13.2 ($C^{1,\gamma}$ Lemma 8). *Under the hypotheses of Theorem 13.1, with $r_* := \rho_0/(2(1 + K_{1,\gamma}))$ a lower bound on the inradius of D (Step 4 below), set*

$$(90) \quad \epsilon_0 := \min\left(\frac{1}{2}\left(\frac{1}{4\Lambda_\gamma}\right)^{2/\gamma}, \frac{r_*^2}{32}\right).$$

Assume further $\eta_B \leq 1/4$. Then for every regular value $\epsilon \in (0, \epsilon_0)$, the sublevel set $D_\epsilon := \{w < -\epsilon\}$ is open and connected, and $|Dw| \geq 1/4$ in the collar $\{x \in D : \text{dist}(x, \partial D) \leq \sqrt{2\epsilon}\}$.

Proof. Step 1: Inclusion $D^{\sqrt{2\epsilon}} \subset D_\epsilon$. For $\bar{x} \in D^{\sqrt{2\epsilon}} := \{x \in D : \text{dist}(x, \partial D) \geq \sqrt{2\epsilon}\}$, set $\bar{w}(x) := (|x - \bar{x}|^2 - 2\epsilon)/2$. Then $\Delta \bar{w} = 2$ in $\mathbb{R}^2$ and, on ∂D , $|x - \bar{x}|^2 \geq \text{dist}(\bar{x}, \partial D)^2 \geq 2\epsilon$, so $\bar{w} \geq 0$ on ∂D . The maximum principle applied to the subharmonic difference (i.e. $\Delta(w - \bar{w}) = 0$, $w \leq 0 \leq \bar{w}$ on ∂D) gives $w \leq \bar{w}$ in D . In particular $w(\bar{x}) \leq \bar{w}(\bar{x}) = -\epsilon$, so $\bar{x} \in D_\epsilon$; since regular values ϵ are full-measure, $\bar{x} \in D_\epsilon$ for regular ϵ .

Step 2: Hölder bound on $|Dw| - 1$ in the collar. Let $x \in D \setminus D^{\sqrt{2\epsilon}}$ and let $x_0 \in \partial D$ realise $|x - x_0| = \text{dist}(x, \partial D) \leq \sqrt{2\epsilon}$. By the Hölder bound on Dw and the boundary trace,

$$||Dw(x)| - |Dw(x_0)|| \leq |Dw(x) - Dw(x_0)| \leq \Lambda_\gamma |x - x_0|^\gamma \leq \Lambda_\gamma (2\epsilon)^{\gamma/2}.$$

Combining with $||Dw(x_0)| - 1| \leq \eta$,

$$||Dw(x)| - 1| \leq \eta + \Lambda_\gamma (2\epsilon)^{\gamma/2}.$$

For $\epsilon \leq \epsilon_0$ given by (90), $\Lambda_\gamma (2\epsilon)^{\gamma/2} \leq 1/4$, and for $\eta \leq \eta_B \leq 1/4$, we have $||Dw(x)| - 1| \leq 1/2$, hence

$$(91) \quad |Dw(x)| \geq 1 - 1/2 = 1/2 > 1/4 > 0$$

throughout the collar $D \setminus D^{\sqrt{2\epsilon}}$.

Step 3: Regularity of ∂D_ϵ in the collar. By (91), $|Dw| > 0$ at every point of $\partial D_\epsilon \subset D \setminus D^{\sqrt{2\epsilon}}$ (the level set $\{w = -\epsilon\}$ lies in this collar because $w < -\epsilon$ in $D^{\sqrt{2\epsilon}}$ by Step 1 and $w \geq -\epsilon$ in a neighbourhood of ∂D). Hence ∂D_ϵ is a regular level set of $w \in C^\infty(D)$: locally, by the implicit function theorem applied to $w(x) + \epsilon = 0$ along the gradient direction, ∂D_ϵ is a C^∞ closed Jordan curve in the collar.

Step 4: $D_{\text{int}}^{\sqrt{2\epsilon}}$ is connected, and an interior ball of radius $\geq \rho_0/(2(1 + K_{1,\gamma}))$ exists. By the $C^{1,\gamma}$ atlas hypothesis, at every $x_0 \in \partial D$ the chart $\psi : B_{\rho_0}(x_0) \rightarrow \mathbb{R}^n$ has $\|\psi^{-1}\|_{C^{1,\gamma}} \leq K_{1,\gamma}$. The pull-back of the open half-ball $B_{\rho_0/(2(1+K_{1,\gamma}))}(\psi(x_0)) \cap \{x_n > 0\}$ under ψ^{-1} lies in $D \cap B_{\rho_0/2}(x_0)$ and contains a Euclidean ball of radius $\geq \rho_0/(2(1 + K_{1,\gamma})) =: r_*(D)$ (Lipschitz inverse bound). Hence $D \supset B_{r_*}(z_*)$ for some $z_* \in D$. Pick ϵ_0 further so that $\sqrt{2\epsilon_0} \leq r_*/4$ and $\sqrt{2\epsilon_0} \leq r_{\text{tub}}/2$, i.e.,

$$(92) \quad \epsilon_0 := \min\left(\frac{1}{2}\left(\frac{1}{4\Lambda_\gamma}\right)^{2/\gamma}, \frac{r_*^2}{32}, \frac{r_{\text{tub}}^2}{8}\right).$$

For $\epsilon \leq \epsilon_0$, $D^{\sqrt{2\epsilon}} \supset B_{r_* - \sqrt{2\epsilon}}(z_*) \supset B_{r_*/2}(z_*)$, which is non-empty connected. Moreover, $D_{\text{int}}^{\sqrt{2\epsilon}}$ is connected by the planar tubular-neighbourhood structure: since ∂D is a single $C^{1,\gamma}$ Jordan curve (Step 3 inputs) with atlas pair $(K_{1,\gamma}, \rho_0)$, the normal map $\Phi(y, s) = y - s\nu(y)$ is a $C^{1,\gamma}$ diffeomorphism of $\partial D \times [0, r_{\text{tub}}]$ onto the collar $\mathcal{N}_{r_{\text{tub}}}(\partial D) \cap D$ for some $r_{\text{tub}} = r_{\text{tub}}(K_{1,\gamma}, \rho_0) > 0$ (standard tubular neighbourhood theorem for $C^{1,\gamma}$ hypersurfaces, cf. [7, Theorem 3.1]). We sharpen ϵ_0 in (92) below to also require $\sqrt{2\epsilon_0} \leq r_{\text{tub}}$. Then the open tube $T_{\sqrt{2\epsilon}} := \{x \in D : \text{dist}(x, \partial D) < \sqrt{2\epsilon}\}$ is homeomorphic to the open annulus $\partial D \times [0, \sqrt{2\epsilon}]$, and its complement $D_{\text{int}}^{\sqrt{2\epsilon}} = D \setminus \bar{T}_{\sqrt{2\epsilon}}$ in the simply connected planar domain D is itself a Jordan domain; in particular it is connected.

Step 5: D_ϵ is path-connected via gradient flow. For each $x \in D_\epsilon$, consider the ODE

$$(93) \quad \dot{\xi}(t) = -\nabla w(\xi(t)), \quad \xi(0) = x.$$

Since $w \in C^\infty(D)$ (interior regularity from $\Delta w = 2$), the flow is locally well-defined. Along the flow,

$$(94) \quad \frac{d}{dt} w(\xi(t)) = -|\nabla w(\xi(t))|^2 \leq 0,$$

so w is non-increasing along ξ , and $\xi(t) \in \{w \leq w(x) < -\epsilon\} \subset D_\epsilon$ for all $t \geq 0$ in the maximal forward interval.

The flow exits the collar in finite time. Set $d(t) := \text{dist}(\xi(t), \partial D)$. For $\xi(t) \in \text{collar}$ (i.e., $d(t) \leq \sqrt{2\epsilon}$), by Step 2, $|\nabla w(\xi(t))| \geq 1/4$. Decomposing $\nabla w(\xi)$ into normal and tangential components at the nearest boundary point $x_0(\xi)$: since $w \leq 0$ in D and $w = 0$ on ∂D , w increases as we move from the interior toward ∂D , so $\nabla w(x_0)$ points in the outward normal direction; thus $\nabla w(x_0) = +|Dw(x_0)|\nu(x_0)$ where ν is the outward normal at $x_0 \in \partial D$. The Hölder bound on Dw yields

$$\nabla w(\xi) = |Dw(x_0)|\nu(x_0) + O(\Lambda_\gamma d^\gamma).$$

The inward direction at ξ is approximately $-\nu(x_0)$, so the inward component of $-\nabla w(\xi)$ is $\langle -\nabla w(\xi), -\nu(x_0) \rangle = |Dw(x_0)| + O(\Lambda_\gamma d^\gamma) \geq 1 - \eta - \Lambda_\gamma (2\epsilon)^{\gamma/2} \geq 1/2$ (for $\eta \leq \eta_B \leq 1/4$ and our ϵ_0).

Therefore $\dot{d}(t) \geq 1/2$ as long as $\xi(t) \in \text{collar}$. Starting from $d(0) \leq \sqrt{2\epsilon}$, the flow exits the collar (reaches $d = \sqrt{2\epsilon}$) at a time $T_{\text{exit}} \leq 2\sqrt{2\epsilon}$. At $t = T_{\text{exit}}$, $\xi(T_{\text{exit}}) \in \partial D^{\sqrt{2\epsilon}}$, and by Step 1, $w(\xi(T_{\text{exit}})) < -\epsilon$, so $\xi(T_{\text{exit}}) \in D_\epsilon \cap D^{\sqrt{2\epsilon}} \subset D_{\text{int}}^{\sqrt{2\epsilon}}$ (for ϵ regular, the boundary $\partial D^{\sqrt{2\epsilon}}$ may be entered transversally; extending the flow slightly past T_{exit} moves ξ into the open set $D_{\text{int}}^{\sqrt{2\epsilon}}$).

The flow path $\{\xi(t) : 0 \leq t \leq T_{\text{exit}} + \delta\}$ lies entirely in D_ϵ (by the monotonicity (94)), so x is path-connected in D_ϵ to a point of $D_{\text{int}}^{\sqrt{2\epsilon}}$.

Step 6: Conclusion. Every $x \in D_\epsilon$ is path-connected in D_ϵ to $D_{\text{int}}^{\sqrt{2\epsilon}}$ (Step 5). $D_{\text{int}}^{\sqrt{2\epsilon}} \subset D_\epsilon$ (Step 1, ϵ regular) is itself path-connected (Step 4). Therefore D_ϵ is path-connected, hence connected. $\square$

Remark 13.3. The threshold ϵ_0 in (90) is INDEPENDENT of η . This is the key difference from the original BNST Lemma 8, whose proof writeup chooses $\epsilon_0 = (\eta/C)^2/2$ (η -dependent) to obtain the stronger conclusion $||Dw| - 1| \leq 2\eta$ in the collar; but for the connectedness conclusion of Lemma 8, the weaker $|Dw| > 0$ in the collar suffices, giving the η -independent threshold above. This ensures Lemma 8 applies at the much larger $\epsilon \sim \eta^{1/(4n+9)}$ chosen by BNST in Theorem 5's proof.

13.4. $\mathcal{H}^{n-1}$ lower bound for $C^{1,\gamma}$ boundary.

Lemma 13.4. Assume the $C^{1,\gamma}$ atlas pair $(K_{1,\gamma}, \rho_0)$ of ∂D . Then for every $x_0 \in \partial D$ and $0 < \rho \leq \rho_0/2$,

$$(95) \quad \mathcal{H}^{n-1}(B_\rho(x_0) \cap \partial D) \geq C_{\text{GMT}}(n) K_{1,\gamma}^{-2(n-1)} \rho^{n-1}.$$

Proof. At $x_0 \in \partial D$, fix the $C^{1,\gamma}$ atlas chart $\psi : B_{\rho_0}(x_0) \rightarrow \mathbb{R}^n$ with $\|\psi\|_{C^{1,\gamma}} \leq K_{1,\gamma}$, $\|\psi^{-1}\|_{C^{1,\gamma}} \leq K_{1,\gamma}$, and $\psi(B_{\rho_0}(x_0) \cap \partial D) \subset \partial \mathbb{R}_+^n$. Then $\psi^{-1}(\cdot)$ is Lipschitz with Lipschitz constant $\leq K_{1,\gamma}$, hence $\psi^{-1}(B_{\rho/K_{1,\gamma}}(\psi(x_0)) \cap \partial \mathbb{R}_+^n) \subset B_\rho(x_0) \cap \partial D$. The area formula (GMT, e.g. [7, Theorem 8.5]) gives

$$\mathcal{H}^{n-1}(B_\rho(x_0) \cap \partial D) \geq \int_{B_{\rho/K_{1,\gamma}}(\psi(x_0)) \cap \partial \mathbb{R}_+^n} J_{\psi^{-1}} d\mathcal{H}^{n-1},$$

where $J_{\psi^{-1}} \geq K_{1,\gamma}^{-(n-1)}$ a.e. since the smallest singular value of $D\psi^{-1}|_{\partial \mathbb{R}_+^n}$ is $\geq 1/\|D\psi\|_{\text{op}} \geq K_{1,\gamma}^{-1}$, and the $(n-1)$ -Jacobian is bounded below by the product of the smallest $(n-1)$ singular values. The right-hand side is at least $\omega_{n-1}(\rho/K_{1,\gamma})^{n-1} \cdot K_{1,\gamma}^{-(n-1)} = \omega_{n-1} \rho^{n-1}/K_{1,\gamma}^{2(n-1)}$, giving (95) with $C_{\text{GMT}}(n) = \omega_{n-1}$. $\square$

13.5. Proof of Theorem 13.1.

Proof. We follow BNST's proof of Theorem 2 verbatim, with the substitutions itemised in §13.2.

Step (B1) — L^1 multi-ball stability. BNST Theorem 5 produces an open set $E \subset D$ whose connected components are balls $\{B^i\}_{i=1}^m$ of radii $R^i \approx 1$ and a paraboloid $v^i(x) = (|x - x_i|^2 - 1)/2$ such that

$$(96) \quad \|u - v^i\|_{C^1(B^i)} \leq C\eta^{1/(4n+9)}, \quad \|u\|_{L^\infty(D \setminus \bigcup B^i)} \leq C\eta^{1/(4n+9)}.$$

Here $u = w/2$ in our convention (since $\Delta u = 1$ and $u \leq 0$ in D); we keep BNST's notation in the present subsection. The proof uses only $w \in C^2(D)$ (interior, from $\Delta w = 2$ and elliptic

regularity), Pohozaev (which requires the outward unit normal of ∂D a.e., available for Lipschitz ∂D), and the interior Caffarelli-type D^2w bound $|D^2w(x)| \leq C/\text{dist}(x, \partial D)^2$ — none of which uses the $C^{2,\alpha}$ atlas. Hence (96) holds verbatim under the $C^{1,\gamma}$ hypothesis.

Step (B2) — Single-ball reduction via modified Lemma 8. By Lemma 13.2, D_ϵ is connected for every regular $\epsilon \in (0, \epsilon_0)$ where ϵ_0 is given by (90), independent of η . In particular, the BNST-chosen $\epsilon = [C(\eta(1+\eta)^4/\omega_n^4)]^{1/(4n+9)} \sim \eta^{1/(4n+9)}$ satisfies $\epsilon \leq \epsilon_0$ for $\eta \leq \eta_B^{(1)} := \min(\eta_B^{\text{BNST}}, \epsilon_0^{4n+9}/(2C))$ (from $\epsilon \leq (2C\eta)^{1/(4n+9)} \leq \epsilon_0 \iff \eta \leq \epsilon_0^{4n+9}/(2C)$). Hence Lemma 8 applies at the BNST scale, forcing $m = 1$ in (96): there is a unique ball $B^1 = B_R(x_1) \subset D$ with $\|u - v^1\|_{C^1(B_R)} \leq C\eta^{1/(4n+9)}$, $\|u\|_{L^\infty(D \setminus B_R)} \leq C\eta^{1/(4n+9)}$.

Step (B3) — Corollary 9 and Lemma 10. BNST Corollary 9 packages Step (B2): there is a ball $B_R \subset D$ with $|1-R| \leq C\eta^{1/(4n+9)}$, $|D \triangle B_R| \leq C\eta^{1/(4n+9)}$, $|\mathcal{H}^{n-1}(\partial D) - n\omega_n R^{n-1}| \leq C\eta^{1/(4n+9)}$. BNST Lemma 10 is pure GMT (coarea + isoperimetric): given the preceding L^1 + perimeter bounds, there exists $\bar{R} > R$ with $\bar{R} - R \leq C\sigma^{1/2}$ and $P(D \setminus B_{\bar{R}}) - \mathcal{H}^{n-1}(D \cap \partial B_{\bar{R}}) \leq C\sigma^{1/2}$, where $\sigma = C\eta^{1/(4n+9)}$. Neither uses $C^{2,\alpha}$.

Step (B4) — $\mathcal{H}^{n-1}$ chain to $\bar{\bar{R}} - \bar{R}$. For $\rho = \min(\rho_0, (\bar{\bar{R}} - \bar{R})/4)$, the BNST cover argument (their p. 1582) gives, using Lemma 13.4,

$$\mathcal{H}^{n-1}(\partial D \setminus \bar{B}_{\bar{R}}) \geq \frac{\bar{\bar{R}} - \bar{R}}{4\rho} C_{\text{GMT}}(n) K_{1,\gamma}^{-2(n-1)} \rho^{n-1} = \frac{\bar{\bar{R}} - \bar{R}}{4} C_{\text{GMT}}(n) K_{1,\gamma}^{-2(n-1)} \rho^{n-2}.$$

For $n = 2$: the $\rho^{n-2} = \rho^0 = 1$ factor disappears, and

$$\bar{\bar{R}} - \bar{R} \leq 4 C_{\text{GMT}}(2)^{-1} K_{1,\gamma}^2 \mathcal{H}^1(\partial D \setminus \bar{B}_{\bar{R}}).$$

By Step (B3), $\mathcal{H}^1(\partial D \setminus \bar{B}_{\bar{R}}) \leq C\sigma^{1/2} = C\eta^{1/(2(4n+9))}$. Hence $\bar{\bar{R}} - \bar{R} \leq C K_{1,\gamma}^2 \eta^{1/(2(4n+9))}$.

Conclusion. Setting $r_{\text{in}} = R$ and $r_{\text{out}} = \bar{\bar{R}}$ (both as inradius and circumradius around the BNST centre x_1 , by construction),

$$r_{\text{out}} - r_{\text{in}} \leq (\bar{\bar{R}} - \bar{R}) + (\bar{R} - R) \leq C (K_{1,\gamma}^2 \eta^{1/34} + \eta^{1/34}) \leq C_{\text{Br}} \eta^{1/34}$$

with $\mu = 1/34$ for $n = 2$, and C_{Br} depending on $(d, K_{1,\gamma}, \rho_0, \Lambda_\gamma, \gamma)$ via the chain. The inner/outer radius proximity to ρ_D follows from $|D \triangle B_R| \leq C\eta^{1/(4n+9)}$ and $\rho_D^2 = |D|/\pi = R^2 + O(\eta^{1/(4n+9)})$, giving $|R - \rho_D| \leq C\eta^{1/(2(4n+9))} = C\eta^{1/34}$, and the analogous bound for r_{out} . $\square$

Remark 13.5. The $C^{1,\gamma}$ hypothesis on ∂D is precisely the regularity output of the Alt–Caffarelli step (R4) in Theorem 11.4 of §11.4. In particular, the selected minimiser $\tilde{u}$ satisfies Theorem 13.1’s hypotheses with $\gamma = \gamma_{\text{AC}}(R_0)$, $K_{1,\gamma} = M_1(R_0)$, and $\Lambda_\gamma = \|D\tilde{u}\|_{C^{0,\gamma}(\bar{U})}$ bounded in terms of R_0 by standard Lieberman boundary estimates for $-\Delta\tilde{u} = 1$ on a $C^{1,\gamma}$ domain. Thus Theorem 13.1 discharges (B) directly from the selection step, without invoking any Schauder bootstrap to $C^{2,\alpha}$.

14. FREE-BOUNDARY REGULARITY (R1)–(R4) AND THE RESCALED BERNOULLI LAW

This section gives the BDV-style derivation in full, supplying the free-boundary regularity package (R1)–(R4) of Theorem 11.4 and the volume-rescaled Bernoulli law that underpins the boundary regularity of $\tilde{U}$. In what follows U_* is the minimiser of J from Lemma 11.1, $\tau \leq \tau_{\text{reg}}(R_0)$, and we are in the small-asymmetry regime $\varepsilon \leq \varepsilon_{\text{sel}}(R_0)$.

14.1. Perturbed one-phase quasi-minimality.

Lemma 14.1. *The function $u_* := u_{U_*} \in H_0^1(B_{R_0})$ satisfies, for every $v \in H_0^1(B_{R_0})$,*

$$(97) \quad \frac{1}{2} \int |\nabla u_*|^2 - \int u_* + f_{\tilde{\eta}}(|\{u_* > 0\}|) \leq \frac{1}{2} \int |\nabla v|^2 - \int v + f_{\tilde{\eta}}(|\{v > 0\}|) + C_2(R_0) \tau |\{u_* > 0\} \Delta \{v > 0\}|.$$

Proof. Set $V := \{v > 0\}$, $U_* = \{u_* > 0\}$. By minimality of u_* for the torsion energy on U_* , $\frac{1}{2} \int |\nabla u_*|^2 - \int u_* = E(U_*)$. Likewise the truncated competitor v^+ has $\frac{1}{2} \int |\nabla v^+|^2 - \int v^+ \geq E(V)$. Now $|\nabla v|^2 = |\nabla v^+|^2 + |\nabla v^-|^2 \geq |\nabla v^+|^2$ a.e., and $\int v = \int v^+ - \int v^- \leq \int v^+$ (since $v^- \geq 0$);

therefore $\frac{1}{2} \int |\nabla v|^2 - \int v \geq \frac{1}{2} \int |\nabla v^+|^2 - \int v^+ \geq E(V)$. By J -minimality $J(U_*) \leq J(V)$, and by the Lipschitz inequality (47),

$$\sqrt{\varepsilon^2 + \tau^2(\alpha(V) - \varepsilon)^2} - \sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \leq \tau |\alpha(V) - \alpha(U_*)| \leq \tau C_2 |U_* \Delta V|.$$

Combining gives (97). $\square$

14.2. BDV free-boundary package, applied to (97). Inequality (97) places u_* in the class of *perturbed one-phase quasi-minimizers* of the BDV Alt–Caffarelli problem with quasi-minimality constant $C_2\tau$. For $\tau \leq \tau_{\text{reg}}(R_0)$, the BDV free-boundary regularity package therefore applies with constants depending only on R_0 . We invoke:

- *Density estimates* (BDV Lemma 4.9, planar): there exist $c_d = c_d(R_0) > 0$ and $r_d = r_d(R_0) > 0$ such that for every $x \in \partial U_*$ and $0 < r \leq r_d$, $c_d \leq |U_* \cap B_r(x)|/(\pi r^2) \leq 1 - c_d$. This gives (R1).
- *Lipschitz nondegeneracy of u_** (BDV Lemma 4.12, planar): there exist $L_u = L_u(R_0) > 0$ and $r'_d > 0$ such that $L_u^{-1} \text{dist}(x, \partial U_*) \leq u_*(x) \leq L_u \text{dist}(x, \partial U_*)$ for $x \in U_*$ with $\text{dist}(x, \partial U_*) \leq r'_d$. This gives (R2).
- *Smooth Bernoulli coefficient* (BDV Lemma 4.16, planar): the boundary gradient $q(y) := |\nabla u_*(y)|$ on ∂U_* admits a $C^{1, \alpha_{AC}}$ one-sided extension to a tubular collar with $0 < q_- \leq q \leq q_+$ and $\|q\|_{C^{1, \alpha_{AC}}} \leq L_q(R_0)$. Moreover, the Bernoulli law

$$(98) \quad |\nabla u_*(y)|^2 = \Lambda + a_\sigma \Psi_{U_*}(y), \quad |a_\sigma| \leq C(R_0) \tau,$$

holds on ∂U_* , where $\Psi_{U_*}(y) := |y - x_{U_*}| - V_{U_*} \cdot y$ is the BDV α -bracket and $V_{U_*} := \int_{U_*} \frac{z - x_{U_*}}{|z - x_{U_*}|} dz$. This gives (R3).

- *$C^{1, \gamma}$ flatness improvement* (BDV Theorem 4.18; Alt–Caffarelli [2]): combining the density estimates, Lipschitz nondegeneracy, and bounded Bernoulli coefficient q , the flat-free-boundary regularity theorem of Alt–Caffarelli (in the BDV-perturbed form) gives $\partial U_* \in C^{1, \gamma}$ locally, with $\gamma = \gamma(R_0) \in (0, 1)$ and uniform graph norm $\leq M_1(R_0)$. This gives (R4).

Items (R1)–(R4) of Theorem 11.4 are established. No further Schauder bootstrap is required: the $C^{1, \gamma}$ regularity (R4) is sufficient as input to the $C^{1, \gamma}$ BNST theorem of §13.

14.3. The volume-rescaled Bernoulli law on $\partial \tilde{U}$. We make the rescaling $\tilde{U} := r_*^{-1} U_*$ explicit on the Bernoulli identity (98), which is needed when one wants quantitative bounds for $\tilde{u}$ directly (rather than going through the u_* -formulation). Set $\tilde{u}(y) := r_*^{-2} u_*(r_* y)$.

Lemma 14.2 (Rescaled Bernoulli law). *$\tilde{u} \in H_0^1(\tilde{U})$ is the torsion function of $\tilde{U}$ ($-\Delta \tilde{u} = 1$ in $\tilde{U}$, $\tilde{u} = 0$ on $\partial \tilde{U}$), and on $\partial \tilde{U}$*

$$(99) \quad |\nabla \tilde{u}(y)|^2 = \tilde{\Lambda} + \tilde{a}_\sigma \tilde{\Psi}(y),$$

with

$$\tilde{\Lambda} = r_*^{-2} \Lambda, \quad \tilde{a}_\sigma = r_*^{-1} a_\sigma, \quad \tilde{\Psi}(y) := |y - r_*^{-1} x_{U_*}| - r_*^2 V_{\tilde{U}} \cdot y.$$

The prefactors $r_*^{\pm 1}, r_*^{\pm 2}$ satisfy, by the mean-value theorem over $r_* \in [1/2, 2]$, $|r_*^k - 1| \leq 2k |r_* - 1| \leq 2k C_5 \delta_0$ for $k \in \{1, 2\}$ (using Lemma 11.2), so (99) differs from (98) only by a multiplicative factor $1 + O(\delta_0)$.

Proof. By chain rule and the scaling $\tilde{u}(y) = r_*^{-2} u_*(r_* y)$: $\nabla \tilde{u}(y) = r_*^{-1} \nabla u_*(r_* y)$, and $\Delta \tilde{u}(y) = \Delta u_*(r_* y) = -1$; the boundary condition is preserved since $y \in \partial \tilde{U} \iff r_* y \in \partial U_*$. At $y \in \partial \tilde{U}$, substituting into (98):

$$|\nabla \tilde{u}(y)|^2 = r_*^{-2} |\nabla u_*(r_* y)|^2 = r_*^{-2} \Lambda + r_*^{-2} a_\sigma \Psi_{U_*}(r_* y).$$

Change of variables in the centroid: $x_{U_*} = r_* x_{\tilde{U}}$ and $V_{U_*} = \int_{U_*} \frac{z - x_{U_*}}{|z - x_{U_*}|} dz = r_*^2 \int_{\tilde{U}} \frac{w - x_{\tilde{U}}}{|w - x_{\tilde{U}}|} dw = r_*^2 V_{\tilde{U}}$, using $z = r_* w$, $dz = r_*^2 dw$ (in $\mathbb{R}^2$). Substituting,

$$\Psi_{U_*}(r_* y) = |r_* y - x_{U_*}| - V_{U_*} \cdot r_* y = r_* (|y - r_*^{-1} x_{U_*}| - r_*^2 V_{\tilde{U}} \cdot y) = r_* \tilde{\Psi}(y).$$

Hence $|\nabla \tilde{u}(y)|^2 = r_*^{-2} \Lambda + r_*^{-1} a_\sigma \tilde{\Psi}(y)$, which is (99). Quantitative estimates on $|r_*^k - 1|$ follow from $|r_* - 1| \leq C_5 \delta_0$ (Lemma 11.2). $\square$

The constants of the BDV regularity package for U_* transfer to $\tilde{U}$ under the dilation by $\lambda = r_*^{-1} \in [1/2, 2]$ (for δ_0 small): density estimates rescale as $c_d \rightarrow c_d$, the Lipschitz nondegeneracy constant transforms as $L_u \rightarrow \lambda L_u$ (an explicit factor ≤ 2), the Bernoulli coefficient bounds $q_\pm$ acquire factors $\lambda^{\mp 1}$, and the $C^{1,\gamma}$ graph norm of $\partial \tilde{U}$ is bounded by $2M_1(R_0)$. We absorb these factor-of-2 inflations into the renamed structural constants.

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